UM-3 field equations

The complete set of coupled field equations of Einstein-Lovelock gravity, dirac16complex, the Higgs doublet with the ad hoc vacuum, the gauge fields of SU(3), SU(2) and U(1), and three generations of quarks and leptons with three right-handed neutrinos

Underlying theory: Patrick L. Nash Derived and verified by Claude (Anthropic)

Field equations of UM-3, 2026-10-24

Repository once-ere/Dirac_claude, folder universe_model_3

Contents

1 How to read this document **2**
- 1.1 What this document contains . . . 2
- 1.2 The status labels . . . 3
- 1.3 Every equation is followed by its symbols . . . 3
- 1.4 Coordinates, signature and units . . . 3
- 1.5 The fields of UM-3 and where they live . . . 3
- 1.6 Conventions . . . 5

2 The action **5**
- 2.1 The sum of all terms . . . 5
- 2.2 Gravity . . . 6
- 2.3 The field dirac16complex . . . 8
- 2.4 The stabiliser of the bulk . . . 10
- 2.5 The fold . . . 10
- 2.6 The Higgs doublet with the ad hoc vacuum . . . 12
- 2.7 The gauge fields . . . 14
- 2.8 The quarks and leptons . . . 15
- 2.9 The portals of dirac16complex . . . 16

3 Gravity **18**
- 3.1 The Einstein-Lovelock equations in the bulk . . . 18
- 3.2 The junction conditions at the fold . . . 20
- 3.3 The energy-momentum tensors . . . 21
- 3.4 The conservation laws . . . 24

4 The matter fields **25**
- 4.1 dirac16complex . . . 25
- 4.2 The Higgs doublet . . . 28
- 4.3 The gauge fields . . . 29
- 4.4 The quarks and leptons . . . 32
- 4.5 The ad hoc Higgs vacuum . . . 36

5 The background **36**
- 5.1 The author's metric and the proper chart . . . 36
- 5.2 The bulk equations of the background . . . 39
- 5.3 The fold equations of the background . . . 40
- 5.4 The author's vacuum . . . 41

5.5 The static block and the frozen extra times . . . 42
5.6 The general warped metric and the power series of the metric . . . 44

6 dirac16complex: modes and Kohn-Sham equations 46
6.1 The equation of the hidden direction . . . 46
6.2 The fold and the states bound to it . . . 48
6.3 The Kohn-Sham equations . . . 50

7 The Standard Model on the fold 52
7.1 The four-dimensional couplings . . . 52
7.2 The electroweak vacuum with the ad hoc component . . . 53
7.3 The masses and the mixing of quarks and leptons . . . 54
7.4 The four-dimensional couplings of the bound states of dirac16complex . . . 55

8 The three neutrino flavours 56
8.1 The equations of motion of the flavour content . . . 56
8.2 Matter-enhanced conversion . . . 60
8.3 The fast flavour instability of dense neutrino gases . . . 60
8.4 Neutrinos in the expanding universe . . . 62

9 The cosmological perturbations 63

10 The thermal history 69

11 The coupled system 73
11.1 Which field appears in which equation . . . 73
11.2 How the system is solved . . . 73

12 What was verified 74

1 How to read this document

1.1 What this document contains

This document states the complete set of coupled field equations of the universe model UM-3, derives every one of them from one action, and explains every term. It is self-contained: every equation it uses is written in it, and every symbol of every equation is defined right after the equation. The sections are:

- **Where the fields live** (this section): the author's metric has a fold, the seven-dimensional surface $z = \pi/2$, that carries energy; the Standard Model lives on the fold, dirac16complex and gravity in the bulk.
- **The action** (IN-1 to IN-16): every term of the action, written out and explained.
- **Gravity** (FE-1 to FE-11): the Einstein-Lovelock equations in the bulk, the junction conditions at the fold, the energy-momentum tensor of every field and the conservation laws.
- **The matter fields** (FE-12 to FE-26): the field equations of dirac16complex, of the Higgs doublet, of the gauge fields and of the quarks and leptons, and the equations of the ad hoc Higgs vacuum.
- **The background** (FE-27 to FE-35): the author's metric, its exact curvature, the exact background equations in the bulk and at the fold, the author's vacuum, the static late-time state, the frozen extra times and the power series of the metric.
- **dirac16complex: modes and Kohn-Sham equations** (FE-36 to FE-43): the equation of the hidden direction, the bulk continuum, the states bound to the fold, the Kohn-Sham equations with the mean fields of every other field, the Krein norm.
- **The Standard Model on the fold** (FE-44 to FE-47): the four-dimensional couplings, the masses with the ad hoc vacuum, the mixing matrices and the four-dimensional couplings of the bound states of dirac16complex.
- **The three neutrino flavours** (FE-48 to FE-53): the flavour evolution in vacuum, in matter, in dense neutrino gases (supernovae) and in the early Universe.
- **The cosmological perturbations** (FE-54 to FE-58): the metric as a power series and the Einstein-Boltzmann equations of the cosmic microwave background (CMB), the cosmic neutrino background (CNB) and the baryon acoustic oscillations (BAO).

- **The thermal history** (FE-59 to FE-62): the expansion of the fold, the degrees of freedom and the Boltzmann equations of the bound states of dirac16complex and of the right-handed neutrinos.
- **The coupled system**: which field appears in which equation, and how the system is solved.
- **What was verified**: the checks of the computations behind every derived statement.

1.2 The status labels

Every equation and every result carries one of the labels

- **AUTHOR**: from the author (his notebooks `4+4.nb` and `Dirac_Einstein-Lovelock_4+4_Nash.nb`, and his order of 2026-10-24: the coordinates, the metric, the Lovelock weights, the Dirac matrices, the chiral projectors, the list of fields, the Higgs potential and the ad hoc vacuum);
- **RECORD**: from the record of this repository (the Lagrangian of dirac16complex with its canonical spin connection and its contact self-interaction, the Kohn-Sham treatment);
- **STANDARD**: textbook physics;
- **ASSUMED**: a choice made in UM-3 without derivation;
- **INVENTED**: a new interaction of UM-3 (the portals of dirac16complex);
- **DERIVED**: computed and verified in UM-3 (the verification is stated);
- **ESTIMATE**: an order-of-magnitude evaluation;
- **OPEN**: not settled.

1.3 Every equation is followed by its symbols

Every display equation is followed by the list "Symbols in (FE-n)" (or "Symbols in this equation" for a display without a tag) of EVERY symbol that appears in it: every quantity, field, coupling, constant and operator, every index letter with the values it takes, every mark (prime, dagger, transpose, star), the imaginary unit, Euler's number and the differential. Each item gives the meaning, the unit (natural units $\hbar = c = 1$), the value where one exists, and the status. A field equation carries the tag FE-1 to FE-62 and a term of the action the tag IN-1 to IN-16; the numbering is that of the inventory of UM-3 and is the same in every UM-3 document. Displays without a tag are steps of a derivation.

1.4 Coordinates, signature and units

The author's coordinates are $X = \{x_0, x_1, \ldots, x_7\}$, always with the hidden direction first: x_0 is the hidden space-like coordinate, on which the metric depends periodically with the period $\pi/(6H)$ through the angle $z = 6Hx_0$; x_1, x_2, x_3 are ordinary space, x_4 is the time, and x_5, x_6, x_7 are three extra time-like coordinates. The signature is $(+,+,+,+,-,-,-,-)$ in this order; the four observable coordinates $x_1, \ldots, x_4$ have the signature $(+,+,+,-)$. H is the author's constant (an inverse length); this document keeps it as a free parameter. Units are natural, $\hbar = c = 1$; in eight dimensions the field dirac16complex has the dimension $\mathrm{GeV}^{7/2}$ and the gravitational coupling $\kappa = 8\pi G_8$ the dimension GeV^{-6}. A prime on a function of the time x_4 alone is its derivative $\mathrm{d}/\mathrm{d}x_4$ (in the cosmological perturbation equations, and only there, a dot is the derivative with respect to the conformal time). The proper coordinate of the hidden direction is σ (defined in FE-27): $\sigma = 0$ is the fold $z = \pi/2$.

1.5 The fields of UM-3 and where they live

The author orders a model universe with the following coupled fields (his order of 2026-10-24, items 1 to 10); UM-3 adds the right-handed quark singlets, the Yukawa couplings and three right-handed neutrinos (accepted by the author on 2026-10-24 at 17:32), the stabiliser of the bulk, the fold fluid and the induced gravity of the fold (ASSUMED, explained below).

field	symbol	what it is	lives in	it couples to	field equations
Einstein-Lovelock gravity	g_{MN}	the metric of eight dimensions	bulk and fold	every field	FE-1 to FE-11, FE-27 to FE-35, FE-54, FE-59

field	symbol	what it is	lives in	it couples to	field equations
dirac16complex	Ψ	the author's 16-component complex spinor field; its left- and right-handed halves Ψ_{L}, Ψ_{R}	the bulk	gravity (spin connection), itself, every Standard-Model field at the fold (IN-16)	FE-6, FE-12 to FE-14, FE-36 to FE-43, FE-47, FE-55, FE-61
Higgs doublet	Φ	the scalar doublet with $V = \mu^2\Phi^\dagger\Phi + \lambda(\Phi^\dagger\Phi)^2$, expanded about the ad hoc vacuum $\Phi = (\mathcal{A}, v+h)/\sqrt{2}$	the fold	gravity, W, B, the fermions (Yukawa), Ψ_{L} only	FE-7, FE-15, FE-26, FE-45, FE-49
three left-handed lepton doublets	ℓ_i	$(\nu_{\mathrm{L}i}, e_{\mathrm{L}i})$, $i = 1, 2, 3$	the fold	gravity, W, B, Φ, Ψ_{L} and Ψ_{R}	FE-9, FE-22, FE-48 to FE-53
right-handed electron, muon, tau	$e_{\mathrm{R}i}$	the charged-lepton singlets	the fold	gravity, B, Φ, Ψ_{L}, Ψ_{R}	FE-9, FE-23
three right-handed neutrinos	$\nu_{\mathrm{R}i}$	gauge singlets; Dirac neutrino masses	the fold	gravity, Φ, Ψ_{L}, Ψ_{R}	FE-9, FE-24, FE-62
gauge fields of $SU(2)_L$ and $U(1)_Y$	W^1, W^2, W^3, B	the electroweak gauge fields	the fold	gravity, Φ, the fermions, Ψ_{L}, Ψ_{R}	FE-8, FE-17, FE-18, FE-45
three quark generations	Q_i, $u_{\mathrm{R}i}$, $d_{\mathrm{R}i}$	the quark doublets and singlets	the fold	gravity, gluons, W, B, Φ, Ψ_{L}, Ψ_{R}	FE-9, FE-19 to FE-21, FE-46
gluons	G^A	the gauge field of $SU(3)_C$	the fold	gravity, quarks, Ψ_{L}, Ψ_{R}	FE-8, FE-16
stabiliser	$\mathcal{T}_{\mathrm{stab}}$	a medium of the bulk that keeps the late-time bulk static	the bulk	gravity	FE-3, FE-32
fold fluid	$\mathcal{T}_{\mathrm{ff}}$	the part of the thin source that the kink of the metric requires at the fold not supplied by the other fields	the fold	gravity	FE-4, FE-5, FE-31 to FE-33, FE-59
induced gravity of the fold	κ_4	an Einstein-Hilbert term of the fold	the fold	gravity	FE-5, FE-59

Why the Standard Model lives on the fold (DERIVED). In the proper coordinate σ of the hidden direction (FE-27), one period of x_0 is the whole line $-\infty < \sigma < \infty$ and the author's metric is $\mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\big(\mathrm{e}^{2a_4}\,\mathrm{d}\vec{x}^2 - \mathrm{e}^{-2a_4}b_4^2\,\mathrm{d}\vec{x}_5^2\big) - \mathrm{d}x_4^2$: ordinary space is warped by $\mathrm{e}^{-H|\sigma|}$, the time x_4 is not. A field confined to the fold $\sigma = 0$ has the light cone of the fold; a field that spreads into the bulk moves along ordinary space with a speed set by the σ-average of $\mathrm{e}^{2H|\sigma|}$ over its profile. With the author's metric a bulk photon moves $\sqrt{2}$ times faster than light on the fold, a bulk scalar $\sqrt{3/2}$ times, a bulk fermion with a profile of width $1/M$ $2M/(2M-H)$ times (DERIVED, `um3_bulk.py`, checks B1 to B3). Species with different profiles would have different limiting speeds (excluded at the level 10^{-15} to 10^{-20} for photons, electrons and nucleons), and three neutrino flavours with three profiles would oscillate with energy-proportional phases that observation excludes. The author's request for correct flavour conversion therefore fixes the place of the Standard Model: all of it lives on the fold, where it has one light cone, and its flavour physics is that of the Standard Model with three Dirac neutrinos. dirac16complex is the author's field of eight dimensions and lives in the bulk; it couples to the Standard Model at the fold.

Why the fold carries a source (DERIVED). The warp $\mathrm{e}^{-2H|\sigma|}$ has a kink at the fold. The Einstein-Lovelock equations, which are linear in the second derivatives of the metric along σ, then contain a delta function at $\sigma = 0$: the metric is a solution only if the fold carries a thin source $\mathcal{S}^a{}_b$ (FE-4). At the author's vacuum $\kappa\mathcal{S} = \mathrm{diag}(-10H, -6H, -10H)$ for (ordinary space, time, extra times) (DERIVED, `um3_fold.py`, check J5). This source is the fold fluid together with the Standard Model and the portals.

Why the fold has its own Einstein-Hilbert term (ASSUMED). The transverse traceless gravitons of the bulk, with the author's Lovelock weights, propagate with $c_{\mathrm{g}}^2 = 15/2$ times the light cone of the fold at the static late-time state, and they have no time-kinetic term at all at the author's vacuum (DERIVED, `um3_bulk.py`, B5). Gravitational waves move with the speed of light to 10^{-15} (GW170817). UM-3 therefore gives the fold an induced Einstein-Hilbert term (IN-9) that dominates four-dimensional gravity; the bulk contributes a fraction $\epsilon_{\mathrm{r}} \lesssim 10^{-15}$ (FE-59).

1.6 Conventions

- Indices. M, N, P, Q (with numbered copies $M_1, M_2, \ldots$): coordinate indices of eight dimensions; a, b, c, d: frame indices of eight dimensions (the frame direction (0) is the σ direction); μ, ν, ρ: coordinate indices of the four observable directions $x_1, \ldots, x_4$; $A, B, C = 1, \ldots, 8$: colour (adjoint) indices; $I, J, K = 1, 2, 3$: weak-isospin (adjoint) indices; $i, j = 1, 2, 3$: generations (and the neutrino mass states); $\alpha, \beta = e, \mu, \tau$: lepton flavours; $p = 1, 2, 3$: the Lovelock order; $l = 0, 1, 2, \ldots$: the multipole index of the cosmological perturbations; $X = \mathrm{L}, \mathrm{R}$: the two halves of dirac16complex; $f \in \{Q, u, d, \ell, e, \nu\}$: the six kinds of Standard-Model multiplets. A repeated index is summed over its range.
- Two kinds of Dirac matrices. The author's real 16×16 matrices T_a (IN-5) act on dirac16complex; the four-dimensional complex 4×4 matrices $\hat{\gamma}^\mu$ (IN-14) act on the Standard-Model fermions. They are never multiplied with each other; the couplings between dirac16complex and the Standard Model are products of Lorentz tensors.
- Normalisation of the fold. No field depends on the extra times x_5, x_6, x_7 (ASSUMED: a dependence on them gives tachyonic four-dimensional modes, FE-38). Integrated over them, the extra times give their physical volume on the fold, V_*; the Standard-Model fields are normalised as four-dimensional fields, the couplings are their measured four-dimensional values, and the fold action carries the factor $1/V_*$ (IN-8).
- A word or a particle name in a subscript is upright (m_{W}, ν_{R}, Ψ_{L}); an italic letter in a subscript is an index. e is Euler's number, i the imaginary unit, d the differential; δ is the Kronecker delta and δ_{D} the Dirac delta function.

2 The action

All field equations of UM-3 follow from one action by varying it with respect to every field. This section writes out every term of the action (IN-1 to IN-16) and says in plain words what it couples. The Standard-Model part has exactly the gauge group, the fermion content (three generations, each with a right-handed neutrino), the Higgs potential and the gauge couplings of the Standard Model; it lives on the fold. dirac16complex is a singlet of the Standard-Model gauge group; its left-handed and right-handed halves couple to every Standard-Model field at the fold through gauge-invariant portals with their own coupling constants, and only its left-handed half couples to the Higgs doublet.

2.1 The sum of all terms

$$\begin{aligned} S_{\mathrm{tot}} &= S_{\mathrm{EL}} + S_{\mathrm{stab}} + S_\Psi + S_{\mathrm{fold}}, \\ S_{\mathrm{fold}} &= \int \mathrm{d}^8X \, \sqrt{|g|} \, \delta_{\mathrm{D}}(\sigma) \, \frac{1}{V_*} \left(\mathcal{L}_{\mathrm{IG}} + \mathcal{L}_{\mathrm{ff}} + \mathcal{L}_{\mathrm{H}} + \mathcal{L}_{\mathrm{gauge}} + \mathcal{L}_{\mathrm{f}} + \mathcal{L}_{\mathrm{Y}} + \mathcal{L}_{\mathrm{P}} \right) \end{aligned} \tag{IN-1}$$

Symbols in (IN-1).

- S_{tot}: The action of UM-3 (the integral of the Lagrangian density over spacetime); its variation with respect to every field gives the field equations. Unit: dimensionless ($\hbar = 1$). Status: ASSUMED.
- S_{EL}: The Einstein-Lovelock action of gravity in eight dimensions. Unit: dimensionless. Status: AUTHOR.
- S_{stab}: The action of the stabiliser of the bulk (a medium specified by its energy-momentum tensor $\mathcal{T}_{\mathrm{stab}}$). Unit: dimensionless. Status: ASSUMED.

- S_Ψ: The action of dirac16complex in the bulk (kinetic term with the canonical spin connection, mass term, contact self-interaction). Unit: dimensionless. Status: RECORD.
- S_{fold}: The action of the fold: the Standard Model, the portals of dirac16complex, the fold fluid and the induced gravity of the fold, all at $\sigma = 0$. Unit: dimensionless. Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- δ_{D}: The Dirac delta function: $\int \mathrm{d}\sigma\, \delta_{\text{D}}(\sigma)F(\sigma) = F(0)$. Unit: 1/length (for $\delta_{\text{D}}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \operatorname{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\text{t}}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\mathcal{L}_{\text{IG}}$: The Lagrangian of the induced gravity of the fold. Unit: GeV^4. Status: ASSUMED.
- $\mathcal{L}_{\text{ff}}$: The Lagrangian of the fold fluid (only its energy-momentum tensor $\mathcal{T}_{\text{ff}}$ enters the equations). Unit: GeV^4. Status: ASSUMED.
- $\mathcal{L}_{\text{H}}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\mathcal{L}_{\text{gauge}}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\text{f}}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\text{Y}}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\text{P}}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

The action has four parts: Einstein-Lovelock gravity in eight dimensions (S_{EL}, IN-2, IN-3), the stabiliser of the bulk (S_{stab}, IN-7), the field dirac16complex in the bulk (S_Ψ, IN-4 to IN-6) and the fold (S_{fold}, IN-8 to IN-16). The Dirac delta $\delta_{\text{D}}(\sigma)$ of the proper hidden coordinate σ confines the terms of the fold to $\sigma = 0$; the factor $1/V_*$ turns the integral over the extra times into the normalisation of four-dimensional fields (IN-8). On the fold live: the induced gravity of the fold ($\mathcal{L}_{\text{IG}}$), the fold fluid ($\mathcal{L}_{\text{ff}}$), the Higgs doublet ($\mathcal{L}_{\text{H}}$), the gauge fields ($\mathcal{L}_{\text{gauge}}$), the quarks and leptons ($\mathcal{L}_{\text{f}}$), their Yukawa couplings ($\mathcal{L}_{\text{Y}}$) and the portals of dirac16complex ($\mathcal{L}_{\text{P}}$). Varying S_{tot} with respect to the metric gives the gravitational equations (FE-1, FE-4); with respect to $\bar\Psi$ the equation of dirac16complex (FE-12, FE-13); with respect to $\Phi^\dagger$ the Higgs equation (FE-15); with respect to the gauge fields the Yang-Mills equations (FE-16 to FE-18); with respect to every fermion its Dirac equation (FE-19 to FE-24). Status: ASSUMED (the choice and the place of the terms; the content of each term is stated with its own label).

2.2 Gravity

$$S_{\text{EL}} = \frac{1}{2\kappa}\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big(\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{L}_{(p)} - 2\Lambda\Big), \qquad \kappa = 8\pi G_8,$$
$$\alpha_{(p)} = w_{(p)}\,H^{2-2p}, \qquad w_{(1)} = 1, \quad w_{(2)} = \frac{1}{32}, \quad w_{(3)} = \frac{1}{1152}, \qquad \Lambda = -10H^2 \tag{IN-2}$$

Symbols in (IN-2).

- S_{EL}: The Einstein-Lovelock action of gravity in eight dimensions. Unit: dimensionless. Status: AUTHOR.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate

with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$ (the author writes the pure number $\Lambda_\mathrm{N} = 10$ in $0 = -\Lambda_\mathrm{N} + \sum_p H^{-2p}w_{(p)}\mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_8: The eight-dimensional Newton constant ($\kappa = 8\pi G_8$). Unit: GeV^{-6}. Status: AUTHOR.
- $w_{(p)}$: The author's Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ (Dirac_Einstein-Lovelock_4+4_Nash.nb, In[134]). Unit: dimensionless. Value: $(1, 1/32, 1/1152)$. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

Gravity is the Einstein-Lovelock theory in eight dimensions with the author's weights. $\mathcal{L}_{(1)}$ is the Einstein-Hilbert density (the Ricci scalar), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density and $\mathcal{L}_{(3)}$ the cubic Lovelock density; in eight dimensions the series ends with $p = 3$ (the density of order 4 is a total derivative in eight dimensions). The author writes the vacuum equations as $0 = -\Lambda_\mathrm{N} + H^{-2}w_{(1)}\mathcal{E}_{(1)} + H^{-4}w_{(2)}\mathcal{E}_{(2)} + H^{-6}w_{(3)}\mathcal{E}_{(3)}$ with the pure number $\Lambda_\mathrm{N} = 10$ (`Dirac_Einstein-Lovelock_4+4_Nash.nb`, In[134]); multiplied by H^2 this is FE-1 without sources and with $\Lambda = -\Lambda_\mathrm{N}H^2 = -10H^2$. Gravity couples to every field through the metric that appears in every other term of the action (in $\sqrt{|g|}$, in the contractions of indices and in the spin connection). Status: AUTHOR.

$$\begin{aligned}\mathcal{L}_{(p)} &= \frac{1}{2^p}\,\delta^{M_1N_1\cdots M_pN_p}_{P_1Q_1\cdots P_pQ_p}\,R^{P_1Q_1}{}_{M_1N_1}\cdots R^{P_pQ_p}{}_{M_pN_p},\\ \mathcal{L}_{(1)} &= R, \qquad \mathcal{L}_{(2)} = R^2 - 4R_{MN}R^{MN} + R_{MNPQ}R^{MNPQ}\end{aligned} \tag{IN-3}$$

Symbols in (IN-3).

- $\mathcal{L}_{(p)}$: The Lovelock density of order p (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}} = 1$ for $X = \mathrm{L}$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor (convention $[\nabla_P, \nabla_Q]V^M = R^M{}_{NPQ}V^N$), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\mathrm{length}^2$. Status: STANDARD.

The Lovelock densities are built with the generalised Kronecker delta from p Riemann tensors. Their key property

(Lovelock's theorem, STANDARD) is that their metric variation contains no derivative of the metric higher than the second, so that the field equations FE-1 are second-order differential equations, and that it is linear in the second derivatives along any one direction (quasi-linearity): a thin source at the fold produces a finite jump of first derivatives, the junction conditions FE-4. Status: STANDARD.

2.3 The field dirac16complex

$$S_\Psi = \int \mathrm{d}^8X\,\sqrt{|g|}\,\Big[\frac{1}{2}\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m_\Psi\,\bar\Psi\Psi - \frac{\lambda_\Psi}{2}\big(\bar\Psi\Psi\big)^2\Big] \tag{IN-4}$$

Symbols in (IN-4).

- S_Ψ: The action of dirac16complex in the bulk (kinetic term with the canonical spin connection, mass term, contact self-interaction). Unit: dimensionless. Status: RECORD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0,\dots,x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0,x_1,\dots,x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\dots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \dots$): it runs over $x_0, x_1, \dots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- λ_Ψ: The coupling of the contact self-interaction of dirac16complex. Unit: GeV^{-6}. Status: RECORD.

dirac16complex Ψ is the author's complex spinor field of eight dimensions with 16 components. Its Lagrangian is that of the record: the kinetic term with the canonical spin connection (in the symmetric form, which is real without a factor i for the author's real matrices, IN-5), the mass term and a contact self-interaction. It couples to gravity through $\sqrt{|g|}$, γ^M and the spin connection Ω_M inside D_M (IN-6), and to itself through λ_Ψ. $\bar\Psi\Psi = \bar\Psi_\mathrm{L}\Psi_\mathrm{L} + \bar\Psi_\mathrm{R}\Psi_\mathrm{R}$ (the mass term does not mix the halves, IN-5), while the kinetic term mixes them (it contains one Dirac matrix). Status: RECORD.

$$\begin{aligned}
&\{T_a,T_b\} = 2\eta_{ab}\mathbf{1}, \qquad \eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1), \qquad T^a = \eta^{ab}T_b, \qquad \gamma^M = \mathbf{e}_a{}^M\,T^a,\\
&\bar\Psi = \Psi^\dagger\sigma_{16}, \qquad \sigma_{16} = T_0T_1T_2T_3, \qquad \Gamma_8 = T_0T_1T_2T_3T_4T_5T_6T_7,\\
&P_\mathrm{L} = \frac{\mathbf{1}-\Gamma_8}{2}, \qquad P_\mathrm{R} = \frac{\mathbf{1}+\Gamma_8}{2}, \qquad \Psi_\mathrm{L} = P_\mathrm{L}\Psi, \qquad \Psi_\mathrm{R} = P_\mathrm{R}\Psi
\end{aligned} \tag{IN-5}$$

Symbols in (IN-5).

- T_a, T^a: The author's eight real 16×16 Dirac matrices T16^A[a] of 4+4.nb ($a = 0,\dots,7$; $T_0,\dots,T_7$), with $\{T_a,T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab}T_b$. Unit: dimensionless. Status: AUTHOR.
- a: A frame (orthonormal) index running over $0,1,\dots,7$ (frame direction (0) along σ, (1),(2),(3) along ordinary space, (4) along the time, (5),(6),(7) along the extra times); raised and lowered with η_{ab}; on the fold ($a,b\in\{1,\dots,7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- η_{ab}: The flat metric of the frame, $\eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$ (eta4488 of 4+4.nb); η^{ab} is its inverse (the same matrix). Unit: dimensionless. Status: AUTHOR.

- **1**: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M \mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- σ_{16}: The matrix of the Dirac adjoint, $\sigma_{16} = T_0T_1T_2T_3$, real and symmetric; $\sigma_{16}T_a$ is real and antisymmetric. Unit: dimensionless. Status: AUTHOR.
- Γ_8: The chirality matrix of eight dimensions, $\Gamma_8 = T_0T_1\cdots T_7$ (the author's T16^A[8]) $= \mathrm{diag}(-\mathbf{1}_8, \mathbf{1}_8)$; $\Gamma_8^2 = \mathbf{1}$; it anticommutes with every T_a. Unit: dimensionless. Status: AUTHOR.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- P_R: The author's right-handed chiral projector, $P_\mathrm{R} = (\mathbf{1} + \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- Ψ_L: The left-handed half of dirac16complex, $\Psi_\mathrm{L} = P_\mathrm{L}\Psi$ (the upper eight components in the basis of 4+4.nb). Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- Ψ_R: The right-handed half of dirac16complex, $\Psi_\mathrm{R} = P_\mathrm{R}\Psi$ (the lower eight components). Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.

T_a ($a = 0, \ldots, 7$) are the author's eight real 16×16 Dirac matrices `T16^A[a]` of `4+4.nb`, and Γ_8 is the author's chirality matrix `T16^A[8]` $= \mathrm{diag}(-\mathbf{1}_8, \mathbf{1}_8)$; P_L and P_R are the author's chiral projectors (the provenance document `dirac matrices.md` of UM-3 proves the Clifford relations, the reality and the projectors; the facts used here are restated in this paragraph). The Dirac adjoint uses $\sigma_{16} = T_0T_1T_2T_3$: σ_{16} is real and symmetric, $\sigma_{16}T_a$ is real and antisymmetric, and σ_{16} commutes with Γ_8 (it is a product of four Dirac matrices), so $\overline{P_X\Psi} = \bar\Psi P_X$ and every bilinear with an even number of Dirac matrices joins a half with itself, every bilinear with an odd number joins Ψ_L with Ψ_R (DERIVED, `um3_fields.py`, checks F1 and F2). Status: AUTHOR (matrices, projectors), RECORD (adjoint).

$$D_M\Psi = \partial_M\Psi + \Omega_M\Psi, \qquad D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\,\Omega_M, \qquad \Omega_M = \frac{1}{2}\,\varpi_{Mab}\,\Sigma^{ab}, \qquad \Sigma^{ab} = \frac{1}{4}\big[T^a, T^b\big],$$
$$\varpi_{Mab} = \mathbf{e}_a{}^N\big(\partial_M \mathbf{e}_{bN} - \Gamma^P{}_{MN}\,\mathbf{e}_{bP}\big) \tag{IN-6}$$

Symbols in (IN-6).

- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- Ω_M: The spin connection acting on dirac16complex, $\Omega_M = \frac{1}{2}\varpi_{Mab}\Sigma^{ab}$. Unit: 1/length. Status: RECORD.
- ϖ_{Mab}: The canonical (Levi-Civita) spin connection of the vielbein, antisymmetric in a, b. Unit: 1/length. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.

- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu \mathbf{e}_b{}^\nu \Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- T_a, T^a: The author's eight real 16×16 Dirac matrices T16ˆA[a] of 4+4.nb ($a = 0, \ldots, 7$; $T_0, \ldots, T_7$), with $\{T_a, T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab} T_b$. Unit: dimensionless. Status: AUTHOR.
- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\, \mathbf{e}^a{}_M \mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\, \mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Gamma^P{}_{MN}$: The Christoffel symbols of the metric g_{MN} (the Levi-Civita connection). Unit: 1/length. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

The canonical spin connection ϖ_{Mab} of the vielbein (antisymmetric in a, b) acts on the spinor through the generators Σ^{ab} of Spin(4,4). With it the Dirac matrices are covariantly constant ($D_M \gamma^N = 0$), the kinetic term of IN-4 is a scalar and the field equation FE-12 is covariant. For the author's metric $\gamma^M \Omega_M$ is never zero (FE-14). Status: RECORD (the canonical construction), DERIVED (the vielbein postulate for the author's metric, `um3_geometry.py`, G5).

2.4 The stabiliser of the bulk

$$\mathcal{T}_{\mathrm{stab}}{}^a{}_b = \mathrm{diag}\big(p_\mathrm{s}, p_\mathrm{s}, p_\mathrm{s}, p_\mathrm{s}, -\rho_\mathrm{s}, p_\mathrm{s}, p_\mathrm{s}, p_\mathrm{s}\big), \qquad \kappa\rho_\mathrm{s} = -\frac{H^2}{16}, \qquad \kappa p_\mathrm{s} = -\frac{5H^2}{16} \tag{IN-7}$$

Symbols in (IN-7).

- $\mathcal{T}_{\mathrm{stab}}{}^a{}_b$: The energy-momentum tensor of the stabiliser of the bulk (frame components). Unit: GeV^8. Status: ASSUMED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- p_s: The pressure of the stabiliser (equal along σ, ordinary space and the extra times). Unit: GeV^8. Value: $\kappa p_\mathrm{s} = -5H^2/16$. Status: ASSUMED.
- ρ_s: The energy density of the stabiliser. Unit: GeV^8. Value: $\kappa\rho_\mathrm{s} = -H^2/16$. Status: ASSUMED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

S_stab is the action of a medium of the bulk; only its energy-momentum tensor $\mathcal{T}_\mathrm{stab}$ enters the field equations, and IN-7 states it in the frame of the author's metric. With the author's weights and Λ, the bulk of a universe whose ordinary space expands slowly (the late universe, $a_4' \ll H$) is not a vacuum solution: its frame curvature is that of the static block (FE-32), which requires exactly this source, a negative energy density $\rho_\mathrm{s} = -H^2/(16\kappa)$ with the isotropic pressure $p_\mathrm{s} = 5\rho_\mathrm{s}$ (DERIVED, `um3_geometry.py`, G4). The stabiliser couples only to gravity. It is absent in the primordial era, whose bulk is the author's vacuum (FE-31). Status: ASSUMED (its existence; its value is DERIVED from the requirement of a static bulk); its microscopic origin is OPEN.

2.5 The fold

$$\begin{aligned} &\int \mathrm{d}^8X\, \sqrt{|g|}\, \delta_\mathrm{D}(\sigma)\, \frac{1}{V_*}\, \mathbb{F} = \int \mathrm{d}^4x\, \sqrt{-\hat{h}}\, \mathcal{Z}\, \mathbb{F}\big|_{\sigma=0}, \qquad \mathrm{d}^4x = \mathrm{d}x_1\, \mathrm{d}x_2\, \mathrm{d}x_3\, \mathrm{d}x_4, \\ &\hat{h}_{\mu\nu} = \mathrm{diag}\big(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1\big), \qquad \sqrt{-\hat{h}} = \mathrm{e}^{3a_4}, \qquad \mathcal{Z} = \frac{V_\mathrm{t}\big(\mathrm{e}^{-a_4} b_4\big)^3}{V_*}, \qquad V_\mathrm{t} = \int \mathrm{d}x_5\, \mathrm{d}x_6\, \mathrm{d}x_7 \end{aligned} \tag{IN-8}$$

Symbols in (IN-8).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.

- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- δ_D: The Dirac delta function: $\int \mathrm{d}\sigma\,\delta_\mathrm{D}(\sigma)F(\sigma) = F(0)$. Unit: 1/length (for $\delta_\mathrm{D}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\mathbb{F}$: Any function of the fields at the fold (used to state a rule of integration). Status: STANDARD.
- x: The coordinates of the fold: x_i ($i = 1, 2, 3$) of ordinary space, x_4 the time, x_r ($r = 5, 6, 7$) the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- V_t: The coordinate volume of the (compact) extra times, $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$. Unit: length^3. Status: ASSUMED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

The fold is the seven-dimensional surface $\sigma = 0$ ($z = \pi/2$) with the coordinates $x_1, \ldots, x_7$. No field depends on the extra times (ASSUMED, FE-38), so the integral over them gives their coordinate volume V_t (the extra times are compact) times their scale factors $(\mathrm{e}^{-a_4}b_4)^3$ at the fold. V_* is the physical volume of the extra times on the fold in the late universe, where $\mathrm{e}^{-a_4}b_4$ is constant (FE-33); with this normalisation $\mathcal{Z} = 1$ in the whole late universe, the fields of the fold are four-dimensional fields in the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold, and their couplings are the measured four-dimensional ones. $\mathcal{Z}(x_4)$ is the factor by which the extra times rescale every coupling of the fold when $\mathrm{e}^{-a_4}b_4$ changes (FE-44). $\mathbb{F}$ stands for any function of the fields on the fold. Status: ASSUMED (the place of the fields), DERIVED (the reduction).

$$\mathcal{L}_{\mathrm{IG}} = \frac{\hat{R}}{2\kappa_4} \tag{IN-9}$$

Symbols in (IN-9).

- $\mathcal{L}_{\mathrm{IG}}$: The Lagrangian of the induced gravity of the fold. Unit: GeV^4. Status: ASSUMED.
- $\hat{R}$: The scalar curvature of the metric $\hat{h}_{\mu\nu}$ of the fold. Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- κ_4: The coupling of the induced gravity of the fold, $\kappa_4 = 8\pi G_{\mathrm{ind}}$; four-dimensional gravity on the fold has $8\pi G_{\mathrm{N}} = \kappa_4/(1 + 5\epsilon_{\mathrm{r}}/6)$ (the Friedmann equation of the fold to first order in ϵ_{r}). Unit: GeV^{-2}. Value: $\kappa_4 \simeq 8\pi G_{\mathrm{N}} = 8\pi/M_{\mathrm{Pl}}^2$. Status: ASSUMED.

The induced gravity of the fold: an Einstein-Hilbert term of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold ($\hat{R}$ its scalar curvature), with the coupling $\kappa_4 = 8\pi G_{\mathrm{ind}}$. Such a term is generated by the quantum fluctuations of every field that lives on the fold (induced gravity); UM-3 assumes that it dominates four-dimensional gravity, because the gravitons of the bulk move with the wrong speed (DERIVED, `um3_bulk.py`, B5: $c_{\mathrm{g}}^2 = 15/2$ at the static block). With it, four-dimensional gravity on the fold is general relativity with Newton's constant G_{N} (FE-59), up to a fraction ϵ_{r} of bulk gravity. Status: ASSUMED.

$$\mathcal{T}_{\mathrm{ff}}{}^{a}{}_{b} = \mathrm{diag}\big(p_{\mathrm{ff}}, p_{\mathrm{ff}}, p_{\mathrm{ff}}, -\rho_{\mathrm{ff}}, q_{\mathrm{ff}}, q_{\mathrm{ff}}, q_{\mathrm{ff}}\big) \tag{IN-10}$$

Symbols in (IN-10).

- $\mathcal{T}_{\mathrm{ff}}{}^{a}{}_{b}$: The energy-momentum tensor of the fold fluid (frame components of the fold, four-dimensional normalisation). Unit: GeV^4. Status: ASSUMED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- p_{ff}: The pressure of the fold fluid along ordinary space. Unit: GeV^4. Value: $-61HV_*/(8\kappa)$ at the static block. Status: ASSUMED.
- ρ_{ff}: The energy density of the fold fluid. Unit: GeV^4. Value: $31HV_*/(4\kappa)$ at the static block. Status: ASSUMED.
- q_{ff}: The pressure of the fold fluid along the extra times. Unit: GeV^4. Value: $-61HV_*/(8\kappa)$ at the static block. Status: ASSUMED.

The fold fluid: a medium at rest on the fold (the frame indices a, b run over the seven directions $1, \ldots, 7$ of the fold), with the energy density ρ_{ff}, the pressure p_{ff} along ordinary space and the pressure q_{ff} along the extra times (four-dimensional normalisation, IN-8). The kink of the author's metric at the fold requires a thin source (FE-4); the fold fluid is the part of it that the Standard Model and the portals do not supply: at the static block $\rho_{\mathrm{ff}} = 31HV_*/(4\kappa)$, $p_{\mathrm{ff}} = q_{\mathrm{ff}} = -61HV_*/(8\kappa)$ (FE-32), at the author's vacuum $\rho_{\mathrm{ff}} = 6HV_*/\kappa$, $p_{\mathrm{ff}} = q_{\mathrm{ff}} = -10HV_*/\kappa$ less the contribution of the induced gravity (FE-31). It plays the role of the brane tension of brane-world models. $\mathcal{L}_{\mathrm{ff}}$ stands for its Lagrangian; only $\mathcal{T}_{\mathrm{ff}}$ enters the field equations. Status: ASSUMED (its existence; its values DERIVED from the junction conditions); its microscopic origin is OPEN.

2.6 The Higgs doublet with the ad hoc vacuum

$$\mathcal{L}_{\mathrm{H}} = -\hat{h}^{\mu\nu}\big(D_\mu\Phi\big)^\dagger D_\nu\Phi - V(\Phi), \qquad V(\Phi) = \mu^2\,\Phi^\dagger\Phi + \lambda\big(\Phi^\dagger\Phi\big)^2, \qquad \lambda > 0, \quad \mu^2 < 0 \tag{IN-11}$$

Symbols in (IN-11).

- $\mathcal{L}_{\mathrm{H}}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.

- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $V(\Phi)$: The Higgs potential, $V = \mu^2\Phi^\dagger\Phi + \lambda(\Phi^\dagger\Phi)^2$. Unit: GeV^4. Status: AUTHOR.
- μ^2: The mass parameter of the Higgs potential, $\mu^2 < 0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2 = -\lambda v^2 = -m_h^2/2 = -(88.6\text{ GeV})^2$. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.

The Higgs doublet Φ (two complex components, hypercharge 1/2) with the author's potential. In the signature $(+,+,+,-)$ of the fold the kinetic term carries a minus sign, so that the energy density is positive. Status: AUTHOR (the potential), STANDARD (the rest).

$$\Phi = \frac{1}{\sqrt{2}}\begin{pmatrix}\mathcal{A}(x_4)\\ v+h\end{pmatrix}, \qquad \mathcal{A}(x_4) = v_{\mathcal{A}}\,\mathrm{e}^{-2a_{40}x_4}, \qquad a_{40}>0, \quad x_4 \geq 0, \qquad v^2 = -\frac{\mu^2}{\lambda} \tag{IN-12}$$

Symbols in (IN-12).

- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_{\mathcal{A}}\mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_\mathrm{F})^{-1/2}$ with the Fermi constant $G_\mathrm{F} = 1.1663788\times10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- h: The Higgs field: the fluctuation of the lower component of the Higgs doublet about v. Unit: GeV. Value: $m_h = 125.25$ GeV. Status: STANDARD.
- $v_{\mathcal{A}}$: The amplitude of almightyAdHoc at $x_4 = 0$; the author's almightyAdHoc = Exp$[-2\,a_{40}\,x_4]$ has $v_{\mathcal{A}} = 1$ in the units of his notebook. Unit: GeV. Value: 1 in the units of the author's notebook; UM-3 keeps it free (the examples take $v_{\mathcal{A}} = v$ and $10v$). Status: AUTHOR (the form), ASSUMED (a free amplitude).
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_{40}: The decay constant of almightyAdHoc, $a_{40} > 0$: $\mathcal{A}$ falls by e in the time $1/(2a_{40})$. Unit: GeV (1/length). Status: AUTHOR.
- μ^2: The mass parameter of the Higgs potential, $\mu^2 < 0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2 = -\lambda v^2 = -m_h^2/2 = -(88.6\text{ GeV})^2$. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.

The author's ad hoc vacuum: the upper component of the Higgs doublet is the prescribed function almightyAdHoc $\mathcal{A}(x_4) = \exp(-2a_{40}x_4)$ (the author writes it with the amplitude $v_{\mathcal{A}} = 1$ in the unit of the field; UM-3 keeps $v_{\mathcal{A}}$ to show where it enters), the lower component is the vacuum expectation value v plus the Higgs field h. The Goldstone fields are absorbed by the gauge choice (unitary gauge). $\mathcal{A}$ decays with the rate $2a_{40}$ from the beginning of the time $x_4 = 0$: it acts only in the primordial era. The ad hoc vacuum is not an exact solution of the Higgs equation; the residuals are FE-26. Status: AUTHOR.

2.7 The gauge fields

$$D_\mu = \partial_\mu + \hat{\omega}_\mu - \mathrm{i}g_3\, G^A_\mu\, t^A - \mathrm{i}g_2\, W^I_\mu\, \frac{\tau^I}{2} - \mathrm{i}g_1\, Y_f\, B_\mu,$$
$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4}\, G^A_{\mu\nu} G^{A\mu\nu} - \frac{1}{4}\, W^I_{\mu\nu} W^{I\mu\nu} - \frac{1}{4}\, B_{\mu\nu} B^{\mu\nu},$$
$$G^A_{\mu\nu} = \partial_\mu G^A_\nu - \partial_\nu G^A_\mu + g_3\, f^{ABC} G^B_\mu G^C_\nu, \qquad W^I_{\mu\nu} = \partial_\mu W^I_\nu - \partial_\nu W^I_\mu + g_2\, \epsilon^{IJK} W^J_\mu W^K_\nu, \qquad B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu \tag{IN-13}$$

Symbols in (IN-13).

- $D_M,\ D_\mu$: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- $\hat{\omega}_\mu$: The spin connection of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold acting on four-dimensional spinors. Unit: 1/length. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- g_3: The $SU(3)_C$ gauge coupling (the strong coupling g_s). Unit: dimensionless. Value: $g_3(m_Z) = 1.2177$ ($\alpha_s = 0.1180$). Status: STANDARD.
- $G^A_\mu,\ G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- B: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- C: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- t^A: The generators of $SU(3)$ on colour triplets, $t^A = \lambda^A_{\text{GM}}/2$ (the Gell-Mann matrices halved), $\mathrm{tr}(t^A t^B) = \delta^{AB}/2$. Unit: dimensionless. Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling (usually g). Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- $W^I_\mu,\ W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- J: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- K: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- τ^I: The Pauli matrices acting on weak doublets (τ^1, τ^2, τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- g_1: The $U(1)_Y$ gauge coupling (usually g'; not normalised for grand unification). Unit: dimensionless. Value: $g_1(m_Z) = 0.3574$. Status: STANDARD.
- Y_f: The hypercharge of a field, the charge of its coupling $g_1 Y_f B_\mu$ to the hypercharge field in the covariant derivative (1/6, 2/3, −1/3, −1/2, −1, 0 for $Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}$, 1/2 for Φ). Unit: dimensionless. Status: STANDARD.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- $B_\mu,\ B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- $\mathcal{L}_{\text{gauge}}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.
- f^{ABC}: The structure constants of $SU(3)$, $[t^A, t^B] = \mathrm{i} f^{ABC} t^C$. Unit: dimensionless. Status: STANDARD.
- ϵ^{IJK}: The Levi-Civita symbol of three indices (the structure constants of $SU(2)$), $\epsilon^{123} = 1$. Unit: dimensionless. Status: STANDARD.

The gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold, with the canonical normalisation (the couplings in

the covariant derivative): the gluons G^A_μ, the weak-isospin fields $W^1_\mu, W^2_\mu, W^3_\mu$ and the hypercharge field B_μ. The covariant derivative of a field acts with the part that fits the field: t^A on colour triplets, $\tau^I/2$ on weak doublets, the hypercharge Y_f of the field, and the four-dimensional spin connection $\hat{\omega}_\mu$ on spinors. The hypercharges (STANDARD) are $Y = 1/6$ for Q, $2/3$ for u_R, $-1/3$ for d_R, $-1/2$ for ℓ, -1 for e_R, 0 for ν_R and $1/2$ for Φ. The gauge fields have no components along the extra times (ASSUMED: such components would be scalars of wrong sign). Status: STANDARD.

2.8 The quarks and leptons

$$\mathcal{L}_\mathrm{f} = \sum_\psi \mathrm{i}\,\bar{\psi}\,\hat{\gamma}^\mu D_\mu \psi, \qquad \psi \in \{Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}\},$$
$$\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}, \qquad \bar{\psi} = \psi^\dagger\beta, \qquad \beta = \hat{\gamma}^{(4)}, \qquad \hat{\gamma}_5 = \mathrm{i}\,\hat{\gamma}^{(4)}\hat{\gamma}^{(1)}\hat{\gamma}^{(2)}\hat{\gamma}^{(3)},$$
$$\hat{P}_\mathrm{L} = \frac{\mathbf{1}-\hat{\gamma}_5}{2}, \quad \hat{P}_\mathrm{R} = \frac{\mathbf{1}+\hat{\gamma}_5}{2}, \qquad \hat{P}_\mathrm{L} Q = Q, \quad \hat{P}_\mathrm{L}\ell = \ell, \quad \hat{P}_\mathrm{R} u_\mathrm{R} = u_\mathrm{R}, \quad \hat{P}_\mathrm{R} d_\mathrm{R} = d_\mathrm{R}, \quad \hat{P}_\mathrm{R} e_\mathrm{R} = e_\mathrm{R}, \quad \hat{P}_\mathrm{R}\nu_\mathrm{R} = \nu_\mathrm{R} \tag{IN-14}$$

Symbols in (IN-14).

- $\mathcal{L}_\mathrm{f}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- ψ: The label of a Standard-Model fermion field in a sum: ψ runs over $Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}$. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ψ, $\bar{\psi}$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar{\psi} = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu\hat{\gamma}^{(a)}$, $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- u_R: The right-handed up-type quarks u, c, t (singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- d_R: The right-handed down-type quarks d, s, b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- e_R: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathbf{1}$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- β: The matrix of the four-dimensional Dirac adjoint, $\beta = \hat{\gamma}^{(4)}$ (Hermitian, $\beta^2 = \mathbf{1}$). Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}_5$: The four-dimensional chirality matrix, $\hat{\gamma}_5 = \mathrm{i}\hat{\gamma}^{(4)}\hat{\gamma}^{(1)}\hat{\gamma}^{(2)}\hat{\gamma}^{(3)}$, $\hat{\gamma}_5^2 = \mathbf{1}$. Unit: dimensionless. Status: STANDARD.
- $\hat{P}_\mathrm{L}$: The four-dimensional left-handed projector, $\hat{P}_\mathrm{L} = (\mathbf{1} - \hat{\gamma}_5)/2$. Unit: dimensionless. Status: STANDARD.
- $\hat{P}_\mathrm{R}$: The four-dimensional right-handed projector, $\hat{P}_\mathrm{R} = (\mathbf{1} + \hat{\gamma}_5)/2$. Unit: dimensionless. Status: STANDARD.

Every Standard-Model fermion is a four-dimensional Weyl field on the fold, written as a four-component spinor

with a fixed chirality: the quark doublets Q and the lepton doublets ℓ are left-handed, the singlets u_{R}, d_{R}, e_{R}, ν_{R} right-handed. Each symbol stands for the three generations (a vector in generation space) and, for quarks, the three colours. $\hat{\gamma}^{\mu} = \hat{\mathbf{e}}_a{}^{\mu}\hat{\gamma}^{(a)}$ are complex 4×4 Dirac matrices of the fold (with the frame matrices $\hat{\gamma}^{(a)}$, $a = 1,\ldots,4$, $(\hat{\gamma}^{(4)})^2 = \mathbf{1}$, $(\hat{\gamma}^{(1)})^2 = (\hat{\gamma}^{(2)})^2 = (\hat{\gamma}^{(3)})^2 = -\mathbf{1}$); $\beta = \hat{\gamma}^{(4)}$ is Hermitian and $(\hat{\gamma}^{\mu})^{\dagger} = \beta\hat{\gamma}^{\mu}\beta$. With these conventions the free Dirac equation is $(-\mathrm{i}\hat{\gamma}^{\mu}\partial_{\mu} + m)\psi = 0$ and $\bar{\psi}\beta\psi$-type energy densities are positive. Status: STANDARD.

$$\mathcal{L}_{\mathrm{Y}} = -\big(\bar{Q}\,\mathcal{Y}^{u}\,\tilde{\Phi}\,u_{\mathrm{R}} + \bar{Q}\,\mathcal{Y}^{d}\,\Phi\,d_{\mathrm{R}} + \bar{\ell}\,\mathcal{Y}^{e}\,\Phi\,e_{\mathrm{R}} + \bar{\ell}\,\mathcal{Y}^{\nu}\,\tilde{\Phi}\,\nu_{\mathrm{R}}\big) + \mathrm{h.c.}, \qquad \tilde{\Phi} = \mathrm{i}\tau^2\Phi^{*} \tag{IN-15}$$

Symbols in (IN-15).

- $\mathcal{L}_{\mathrm{Y}}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_{\mathrm{L}}, d_{\mathrm{L}})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^{u}$, $\mathcal{Y}^{d}$, $\mathcal{Y}^{e}$, $\mathcal{Y}^{\nu}$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi} = \mathrm{i}\tau^2\Phi^{*}$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- u_{R}: The right-handed up-type quarks u, c, t (singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- d_{R}: The right-handed down-type quarks d, s, b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_{\mathrm{L}}, e_{\mathrm{L}})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- e_{R}: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- ν_{R}: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- h.c.: The Hermitian conjugate of the preceding terms. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak doublets (τ^1, τ^2, τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- *: Complex conjugation of a number, or of each entry of a matrix or column. Status: STANDARD.

The Yukawa couplings, with the complex 3×3 matrices $\mathcal{Y}^{u}$, $\mathcal{Y}^{d}$, $\mathcal{Y}^{e}$, $\mathcal{Y}^{\nu}$ in generation space; $\tilde{\Phi}$ is the conjugate doublet (hypercharge $-1/2$). In the vacuum they give the quarks, the charged leptons and the neutrinos their Dirac masses (FE-46); with three right-handed neutrinos the neutrino masses are Dirac masses and the lepton mixing is that of the PMNS matrix. No Majorana mass of ν_{R} is included (lepton number is conserved). Status: STANDARD (with ν_{R}: accepted by the author, 2026-10-24 17:32).

2.9 The portals of dirac16complex

$$\begin{aligned}\mathcal{L}_{\mathrm{P}} = &-c_{\Phi}\,\big(\Phi^{\dagger}\Phi\big)\,\bar{\Psi}P_{\mathrm{L}}\Psi - \sum_X \Big[\frac{1}{4}\big(c_X^{G}\,G^{A}_{\mu\nu}G^{A\mu\nu} + c_X^{W}\,W^{I}_{\mu\nu}W^{I\mu\nu} + c_X^{B}\,B_{\mu\nu}B^{\mu\nu}\big)\,\bar{\Psi}P_X\Psi \\ &+ \frac{d_X}{2}\,B_{\mu\nu}\,\mathcal{D}_X^{\mu\nu} + \sum_f c_X^{f}\,J_f^{\mu}\,j_{X\mu}\Big], \qquad\qquad (\text{IN-16})\\ \mathcal{D}_X^{\mu\nu} = \mathrm{i}\,\bar{\Psi}\,\Sigma^{\mu\nu}P_X\Psi, &\qquad j_X^{\mu} = \mathrm{i}\big(\bar{\Psi}P_X D^{\mu}\Psi - (D^{\mu}\bar{\Psi})P_X\Psi\big), \qquad J_f^{\mu} = \bar{\psi}_f\,\hat{\gamma}^{\mu}\psi_f\end{aligned}$$

Symbols in (IN-16).

- $\mathcal{L}_{\mathrm{P}}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.
- c_{Φ}: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^{\dagger}\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_{L}: The author's left-handed chiral projector, $P_{\mathrm{L}} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.

- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- G_μ^A, $G_{\mu\nu}^A$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- W_μ^I, $W_{\mu\nu}^I$: The gauge field of $SU(2)_L$ on the fold, $W_\mu^1, W_\mu^2, W_\mu^3$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- $\mathcal{D}_X^{\mu\nu}$: The dipole (antisymmetric tensor) density of the half X of dirac16complex at the fold, $\mathcal{D}_X^{\mu\nu} = \mathrm{i}\bar{\Psi}\Sigma^{\mu\nu}P_X\Psi$ (real, DERIVED). Unit: GeV^7. Status: DERIVED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- J_f^μ: The fermion-number current of the Standard-Model multiplets of kind f, $J_f^\mu = \bar{\psi}_f\hat{\gamma}^\mu\psi_f$ summed over the three generations and the gauge components; with other labels the currents that are the sources of the gauge fields. Unit: GeV^3. Status: STANDARD.
- j_X^μ: The convective current of the half X of dirac16complex, $j_X^\mu = \mathrm{i}(\bar{\Psi}P_X D^\mu\Psi - (D^\mu\bar{\Psi})P_X\Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu\mathbf{e}_b{}^\nu\Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- ψ, $\bar{\psi}$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar{\psi} = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu\hat{\gamma}^{(a)}$, $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.

The new interactions of UM-3 (INVENTED), the order of the author that the left-handed and the right-handed halves of dirac16complex each couple to every Standard-Model field with its own coupling, and that only the left-handed half couples to the Higgs doublet. Ψ is the value of dirac16complex at the fold. Every term is a product of a gauge-invariant Standard-Model operator and a Lorentz-tensor bilinear of one half X of dirac16complex, so every term is invariant under $SU(3)_C \times SU(2)_L \times U(1)_Y$ and under the Lorentz transformations of the fold; none changes the flavour of a quark or a lepton (consistent with QCD and with the electroweak theory, DERIVED by construction):

- the Higgs portal: the scalar density $\bar{\Psi}P_\mathrm{L}\Psi$ of the left-handed half times $\Phi^\dagger\Phi$, with the coupling c_Φ; the right-handed half has no Higgs coupling;
- the gauge-kinetic portals: the scalar density of each half times the gauge-invariant kinetic densities of the

gluons, of the W fields and of the B field, with the six couplings c_X^G, c_X^W, c_X^B: they make the gauge couplings of the fold depend on the density of dirac16complex at the fold (FE-16 to FE-18) and couple the bound states of dirac16complex to gluons, photons and weak bosons;
- the dipole portal: the antisymmetric tensor $\mathcal{D}_X^{\mu\nu}$ of each half times the hypercharge field strength (the only Standard-Model field strength that is gauge invariant by itself), with the couplings d_X: an electric and magnetic dipole moment of dirac16complex with respect to the photon and the Z;
- the current portals: the convective current j_X^μ of each half (the only real vector current that joins a half with itself: $\bar\Psi\gamma^\mu P_X\Psi$ joins Ψ_L with Ψ_R and is not real, DERIVED, `um3_fields.py`, F1) times the fermion-number current J_f^μ of each kind f of Standard-Model multiplet, with the twelve couplings c_X^f; J_f^μ sums over the three generations and over the gauge components, so the portal is flavour blind.

The 21 couplings are c_Φ, c_L^G, c_R^G, c_L^W, c_R^W, c_L^B, c_R^B, d_L, d_R and c_L^f, c_R^f for $f = Q, u, d, \ell, e, \nu$. Every term is bilinear in Ψ: the action is invariant under a change of the phase of Ψ by a constant (the charge of dirac16complex is conserved, and its lightest state is stable), and baryon and lepton numbers are conserved. Status: INVENTED.

3 Gravity

3.1 The Einstein-Lovelock equations in the bulk

$$\sum_{p=1}^{3} \alpha_{(p)}\, {\mathcal{E}_{(p)}}^M{}_N + \Lambda\, \delta^M_N = \kappa\big({\mathcal{T}_\Psi}^M{}_N + {\mathcal{T}_\mathrm{stab}}^M{}_N\big) \qquad (\sigma \neq 0) \tag{FE-1}$$

Symbols in (FE-1).

- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- ${\mathcal{E}_{(p)}}^M{}_N$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$ (the author writes the pure number $\Lambda_\mathrm{N} = 10$ in $0 = -\Lambda_\mathrm{N} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}} = 1$ for $X = \mathrm{L}$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex, $\mathcal{T}_{MN} = -(2/\sqrt{|g|})\,\delta S_\Psi/\delta g^{MN}$. Unit: GeV^8. Status: STANDARD.
- ${\mathcal{T}_\mathrm{stab}}^a{}_b$: The energy-momentum tensor of the stabiliser of the bulk (frame components). Unit: GeV^8. Status: ASSUMED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

The variation of S_tot with respect to the metric, away from the fold. The left-hand side is the Lovelock tensor of the author's theory; the right-hand side is κ times the energy-momentum tensor of the fields of the bulk: dirac16complex (FE-6) and the stabiliser (IN-7). With the energy-momentum tensor defined by $\mathcal{T}_{MN} = -(2/\sqrt{|g|})\,\delta S_\mathrm{matter}/\delta g^{MN}$, the time-time component $\mathcal{T}_{x_4x_4} = -{\mathcal{T}^{x_4}}_{x_4}$ is the energy density seen by an observer at rest. Status: AUTHOR (the theory), STANDARD (the variation).

$$
\begin{aligned}
\mathcal{E}_{(p)}{}^{M}{}_{N} &= -\frac{1}{2^{p+1}}\,\delta^{M M_1 N_1 \cdots M_p N_p}_{N P_1 Q_1 \cdots P_p Q_p}\, R^{P_1 Q_1}{}_{M_1 N_1} \cdots R^{P_p Q_p}{}_{M_p N_p}, \\
\mathcal{E}_{(1)}{}^{M}{}_{N} &= R^{M}{}_{N} - \frac{1}{2}\,\delta^{M}_{N}\,R, \\
\mathcal{E}_{(2)}{}^{M}{}_{N} &= 2\big(R\,R^{M}{}_{N} - 2R^{M}{}_{P}R^{P}{}_{N} - 2R^{MP}{}_{NQ}R^{Q}{}_{P} + R^{MPQS}R_{NPQS}\big) - \frac{1}{2}\,\delta^{M}_{N}\,\mathcal{L}_{(2)}
\end{aligned}
\tag{FE-2}
$$

Symbols in (FE-2).

- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^{M}_{N} = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}} = 1$ for $X = \mathrm{L}$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1 \cdots M_n}_{N_1 \cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor (convention $[\nabla_P, \nabla_Q]V^M = R^{M}{}_{NPQ}V^N$), with two indices the Ricci tensor $R_{MN} = R^{P}{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\text{length}^2$. Status: STANDARD.
- S: A coordinate index of eight dimensions, running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\text{length}^{2p}$. Status: STANDARD.

The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric. For $p = 1$ it is the Einstein tensor, for $p = 2$ the Gauss-Bonnet tensor (written out in the third line); the cubic tensor $\mathcal{E}_{(3)}$ is the first line with $p = 3$. UM-3 evaluates the first line exactly by enumerating the non-zero frame components of the curvature (`um3_lovelock.py`); the second and third lines are independent checks of it (DERIVED, `um3_geometry.py`, G3: the Einstein tensor from the coordinate Ricci tensor and the explicit Gauss-Bonnet tensor at random points agree with the first line). Status: STANDARD.

$$
\mathcal{E}_{(p)}{}^{M}{}_{M} = -\frac{8 - 2p}{2}\,\mathcal{L}_{(p)}, \qquad \nabla_M\,\mathcal{E}_{(p)}{}^{M}{}_{N} = 0
\tag{FE-3}
$$

Symbols in (FE-3).

- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

- $\mathcal{L}_{(p)}$: The Lovelock density of order p (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols ${\Gamma^P}_{MN}$). Status: STANDARD.

Two identities of every Lovelock tensor in eight dimensions: its trace is a multiple of the Lovelock density, and its covariant divergence vanishes (the contracted Bianchi identity). The second makes the energy-momentum tensor of the sources conserved (FE-11). Status: STANDARD (DERIVED for the author's metric with general $a_4(x_4)$, $b_4(x_4)$: `um3_geometry.py`, G3).

3.2 The junction conditions at the fold

$$\begin{aligned} \kappa\, {\mathcal{S}^a}_b &= \lim_{\varepsilon\to 0}\int_{-\varepsilon}^{\varepsilon} \mathrm{d}\sigma \sum_{p=1}^{3} \alpha_{(p)}\, {\mathcal{E}_{(p)}}^a{}_b = {\mathfrak{F}^a}_b\big|_{\sigma=0^+} - {\mathfrak{F}^a}_b\big|_{\sigma=0^-}, \\ \frac{\partial \mathfrak{F}^a{}_b}{\partial(\partial_\sigma \mathfrak{f})} &= \frac{\partial}{\partial(\partial_\sigma^2 \mathfrak{f})} \sum_{p=1}^{3} \alpha_{(p)}\, {\mathcal{E}_{(p)}}^a{}_b, \qquad \mathfrak{f} \in \{\mathfrak{a}, \mathfrak{b}, \mathfrak{n}\} \end{aligned} \tag{FE-4}$$

Symbols in (FE-4).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- ${\mathcal{S}^a}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- ε: The half-thickness of a layer around the fold, taken to zero. Unit: length. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- ${\mathcal{E}_{(p)}}^M{}_N$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- ${\mathfrak{F}^a}_b$: The junction potential: a function of the first σ-derivatives of the metric functions whose derivatives are the coefficients of the second σ-derivatives in $\sum_p \alpha_{(p)} {\mathcal{E}_{(p)}}^a{}_b$ (FE-4). Unit: GeV. Status: DERIVED.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.
- $\mathfrak{f}$: Any one of the three metric functions $\mathfrak{a}, \mathfrak{b}, \mathfrak{n}$ of the general warped metric (FE-34). Unit: dimensionless. Status: DERIVED.
- $\mathfrak{a}$, $\mathfrak{a}_1$: The logarithm of the scale factor of ordinary space in the general warped metric FE-34, a function of σ and x_4; for the author's metric $\mathfrak{a} = a_4 - H|\sigma|$; $\mathfrak{a}_n$ its term of order n in the power series FE-35 ($\mathfrak{a}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- $\mathfrak{b}$, $\mathfrak{b}_1$: The logarithm of the scale factor of the extra times in the general warped metric FE-34; for the author's metric $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$; $\mathfrak{b}_n$ its term of order n ($\mathfrak{b}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- $\mathfrak{n}$, $\mathfrak{n}_1$: The logarithm of the lapse in the general warped metric FE-34 ($g_{x_4 x_4} = -\mathrm{e}^{2\mathfrak{n}}$); for the author's metric $\mathfrak{n} = 0$; $\mathfrak{n}_n$ its term of order n ($\mathfrak{n}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.

The field equations at the fold. The author's metric (FE-27) has a kink at $\sigma = 0$; in general (FE-34) the metric functions $\mathfrak{a}, \mathfrak{b}, \mathfrak{n}$ have first σ-derivatives that jump at the fold, so their second derivatives contain $\delta_{\mathrm{D}}(\sigma)$. Because every Lovelock tensor is linear in the second σ-derivatives (quasi-linearity, DERIVED for the general metric: `um3_fold.py`, J3), FE-1 extended across the fold reads $\sum_p \alpha_{(p)}\mathcal{E}_{(p)} + \Lambda = \kappa(\mathcal{T}_{\mathrm{bulk}} + \mathcal{S}\,\delta_{\mathrm{D}}(\sigma))$: the fold carries the thin source $\mathcal{S}^a{}_b$ (energy per unit area of the fold; a, b along the fold), equal to the integral of the Lovelock tensor across the fold. The coefficients of the second derivatives form an exact differential of the first derivatives (DERIVED, J4), with the potential $\mathfrak{F}^a{}_b$; the integral is therefore the difference of $\mathfrak{F}$ on the two sides. The (σ,σ) and (σ, x_4) components contain no second σ-derivative: they carry no thin part and hold on each side separately. Status: DERIVED.

$$\mathcal{S}^a{}_b = \frac{1}{V_*}\Big(\mathcal{T}_{\mathrm{H}}{}^a{}_b + \mathcal{T}_{\mathrm{gauge}}{}^a{}_b + \mathcal{T}_{\mathrm{f}}{}^a{}_b + \mathcal{T}_{\mathrm{P}}{}^a{}_b + \mathcal{T}_{\mathrm{ff}}{}^a{}_b + \mathcal{T}_{\mathrm{IG}}{}^a{}_b\Big), \qquad \mathcal{T}_{\mathrm{IG}}{}^\mu{}_\nu = -\frac{1}{\kappa_4}\,\hat{G}^\mu{}_\nu, \qquad \mathcal{T}^r{}_r = \mathcal{L}_{\mathrm{fold}} \tag{FE-5}$$

Symbols in (FE-5).

- $\mathcal{S}^a{}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\mathcal{T}_{\mathrm{H}}$: The energy-momentum tensor of the Higgs doublet (four-dimensional normalisation). Unit: GeV^4. Status: STANDARD.
- $\mathcal{T}_{\mathrm{gauge}}$: The energy-momentum tensor of the gauge fields. Unit: GeV^4. Status: STANDARD.
- $\mathcal{T}_{\mathrm{f}}$: The energy-momentum tensor of the quarks and leptons. Unit: GeV^4. Status: STANDARD.
- $\mathcal{T}_{\mathrm{P}}$: The energy-momentum tensor of the portals of dirac16complex. Unit: GeV^4. Status: DERIVED.
- $\mathcal{T}_{\mathrm{ff}}{}^a{}_b$: The energy-momentum tensor of the fold fluid (frame components of the fold, four-dimensional normalisation). Unit: GeV^4. Status: ASSUMED.
- $\mathcal{T}_{\mathrm{IG}}$: The energy-momentum tensor of the induced gravity of the fold, $\mathcal{T}_{\mathrm{IG}}{}^\mu{}_\nu = -\hat{G}^\mu{}_\nu/\kappa_4$. Unit: GeV^4. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- κ_4: The coupling of the induced gravity of the fold, $\kappa_4 = 8\pi G_{\mathrm{ind}}$; four-dimensional gravity on the fold has $8\pi G_{\mathrm{N}} = \kappa_4/(1 + 5\epsilon_{\mathrm{r}}/6)$ (the Friedmann equation of the fold to first order in ϵ_{r}). Unit: GeV^{-2}. Value: $\kappa_4 \simeq 8\pi G_{\mathrm{N}} = 8\pi/M_{\mathrm{Pl}}^2$. Status: ASSUMED.
- $\hat{G}^\mu{}_\nu$: The Einstein tensor of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold. Unit: 1/length^2. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- $\mathcal{T}$: The energy-momentum tensor of any one term of the fold (in $\mathcal{T}^r{}_r = \mathcal{L}_{\mathrm{fold}}$). Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\mathrm{fold}}$: The Lagrangian of any one term of the fold (four-dimensional normalisation). Unit: GeV^4. Status: STANDARD.

The thin source of the fold is the energy-momentum tensor of everything that lives on the fold, divided by V_* (IN-8): the Higgs doublet (FE-7), the gauge fields (FE-8), the quarks and leptons (FE-9), the portals (FE-10), the fold fluid (IN-10) and the induced gravity, whose energy-momentum tensor is $-1/\kappa_4$ times the Einstein tensor $\hat{G}^\mu{}_\nu$ of the metric of the fold. Because the fields of the fold do not depend on the extra times and have no components along them, the extra times enter only through $\sqrt{|g|}$: for every term of the fold the component along an extra time is the Lagrangian of the term, $\mathcal{T}^r{}_r = \mathcal{L}_{\mathrm{fold}}$ (DERIVED: $\delta\sqrt{|g|} = -\frac{1}{2}\sqrt{|g|}g_{rr}\delta g^{rr}$). Combining FE-4 and FE-5 gives the Einstein equations of the fold (FE-59). Status: DERIVED.

3.3 The energy-momentum tensors

$$\mathcal{T}_{\Psi\,MN} = -\frac{1}{4}\Big(\bar{\Psi}\gamma_M D_N\Psi + \bar{\Psi}\gamma_N D_M\Psi - (D_M\bar{\Psi})\gamma_N\Psi - (D_N\bar{\Psi})\gamma_M\Psi\Big) + g_{MN}\,\mathcal{L}_\Psi \tag{FE-6}$$

Symbols in (FE-6).

- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex, $\mathcal{T}_{MN} = -(2/\sqrt{|g|})\,\delta S_\Psi/\delta g^{MN}$. Unit: GeV^8. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- $\mathcal{L}_\Psi$: The Lagrangian of dirac16complex, the square bracket of IN-4. Unit: GeV^8. Status: RECORD.

The symmetric energy-momentum tensor of dirac16complex, from the variation of the vielbein in IN-4 ($\mathcal{L}_\Psi$ is the Lagrangian in the square bracket of IN-4; the variation of the spin connection adds only the divergence of the spin current, which drops out of the symmetric part). On solutions of FE-12, $\mathcal{L}_\Psi = \frac{\lambda_\Psi}{2}(\bar\Psi\Psi)^2$. For a stationary state $\Psi \propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ of the free field the energy density is $\mathcal{T}_{\Psi\,x_4x_4} = -\omega\,\mathrm{i}\bar\Psi\gamma^{x_4}\Psi$: its sign is the product of the sign of the frequency and of the Krein density (FE-43). Status: STANDARD (DERIVED for the energy density).

$$\mathcal{T}_{\mathrm{H}\mu\nu} = (D_\mu\Phi)^\dagger D_\nu\Phi + (D_\nu\Phi)^\dagger D_\mu\Phi + \hat h_{\mu\nu}\,\mathcal{L}_\mathrm{H} \tag{FE-7}$$

Symbols in (FE-7).

- $\mathcal{T}_\mathrm{H}$: The energy-momentum tensor of the Higgs doublet (four-dimensional normalisation). Unit: GeV^4. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\hat h_{\mu\nu}$: The four-dimensional metric of the fold, $\hat h_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat h^{\mu\nu}$ its inverse, $\hat h$ its determinant, $\sqrt{-\hat h} = \mathrm{e}^{3a_4}$; $\hat h^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{L}_\mathrm{H}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.

The energy-momentum tensor of the Higgs doublet (four-dimensional normalisation): its time-time component is $|D_4\Phi|^2 + \hat h^{ij}(D_i\Phi)^\dagger D_j\Phi + V \geq V$. Status: STANDARD.

$$\mathcal{T}_{\mathrm{gauge}\,\mu\nu} = G^A_{\mu\rho}G^A{}_\nu{}^\rho + W^I_{\mu\rho}W^I{}_\nu{}^\rho + B_{\mu\rho}B_\nu{}^\rho + \hat h_{\mu\nu}\,\mathcal{L}_\mathrm{gauge} \tag{FE-8}$$

Symbols in (FE-8).

- $\mathcal{T}_\mathrm{gauge}$: The energy-momentum tensor of the gauge fields. Unit: GeV^4. Status: STANDARD.

- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- ρ: A four-dimensional coordinate index (as μ). Status: STANDARD.
- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{L}_{\mathrm{gauge}}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.

The energy-momentum tensor of the gauge fields; for each field its energy density is $(\vec{E}^2 + \vec{B}^2)/2$ in terms of the electric and magnetic components. Status: STANDARD.

$$\mathcal{T}_{\mathrm{f}\mu\nu} = -\frac{\mathrm{i}}{4}\sum_{\psi}\Big(\bar{\psi}\hat{\gamma}_\mu D_\nu\psi + \bar{\psi}\hat{\gamma}_\nu D_\mu\psi - (D_\mu\bar{\psi})\hat{\gamma}_\nu\psi - (D_\nu\bar{\psi})\hat{\gamma}_\mu\psi\Big) + \hat{h}_{\mu\nu}\big(\mathcal{L}_{\mathrm{f}} + \mathcal{L}_{\mathrm{Y}}\big) \tag{FE-9}$$

Symbols in (FE-9).

- $\mathcal{T}_{\mathrm{f}}$: The energy-momentum tensor of the quarks and leptons. Unit: GeV^4. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ψ: The label of a Standard-Model fermion field in a sum: ψ runs over $Q, u_{\mathrm{R}}, d_{\mathrm{R}}, \ell, e_{\mathrm{R}}, \nu_{\mathrm{R}}$. Status: STANDARD.
- ψ, $\bar{\psi}$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar{\psi} = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu\hat{\gamma}^{(a)}$, $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{L}_{\mathrm{f}}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\mathrm{Y}}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.

The energy-momentum tensor of the quarks and leptons (with the kinetic term in its symmetric form). For a free particle of energy E at rest in the frame of the fold its time-time component is $E\,\psi^\dagger\psi > 0$. Status: STANDARD.

$$\begin{aligned}\mathcal{T}_{\mathrm{P}\mu\nu} = \sum_X \Big[&\big(c^G_X\, G^A_{\mu\rho}G^A{}_\nu{}^\rho + c^W_X\, W^I_{\mu\rho}W^I{}_\nu{}^\rho + c^B_X\, B_{\mu\rho}B_\nu{}^\rho\big)\,\bar{\Psi}P_X\Psi \\ &+ \frac{d_X}{2}\big(B_{\mu\rho}\,\mathcal{D}_X{}_\nu{}^\rho + B_{\nu\rho}\,\mathcal{D}_X{}_\mu{}^\rho\big) + \sum_f \frac{c^f_X}{2}\big(J_{f\mu}\,j_{X\nu} + J_{f\nu}\,j_{X\mu}\big)\Big] + \hat{h}_{\mu\nu}\,\mathcal{L}_{\mathrm{P}}\end{aligned} \tag{FE-10}$$

Symbols in (FE-10).

- $\mathcal{T}_{\mathrm{P}}$: The energy-momentum tensor of the portals of dirac16complex. Unit: GeV^4. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- G_μ^A, $G_{\mu\nu}^A$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- ρ: A four-dimensional coordinate index (as μ). Status: STANDARD.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- W_μ^I, $W_{\mu\nu}^I$: The gauge field of $SU(2)_L$ on the fold, $W_\mu^1, W_\mu^2, W_\mu^3$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_{\mathrm{L}}$ for $X = \mathrm{L}$, P_{R} for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- $\mathcal{D}_X^{\mu\nu}$: The dipole (antisymmetric tensor) density of the half X of dirac16complex at the fold, $\mathcal{D}_X^{\mu\nu} = \mathrm{i}\bar{\Psi}\Sigma^{\mu\nu}P_X\Psi$ (real, DERIVED). Unit: GeV^7. Status: DERIVED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- J_f^μ: The fermion-number current of the Standard-Model multiplets of kind f, $J_f^\mu = \bar{\psi}_f \hat{\gamma}^\mu \psi_f$ summed over the three generations and the gauge components; with other labels the currents that are the sources of the gauge fields. Unit: GeV^3. Status: STANDARD.
- j_X^μ: The convective current of the half X of dirac16complex, $j_X^\mu = \mathrm{i}(\bar{\Psi}P_X D^\mu \Psi - (D^\mu \bar{\Psi})P_X\Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}_{\mu\nu}^{(1)}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{L}_{\mathrm{P}}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

The energy-momentum tensor of the portals (from the variation of IN-16 with respect to the metric of the fold, which enters through the raised indices, through $\hat{\gamma}^\mu$ in J_f^μ and through $\sqrt{-\hat{h}}$); the Higgs portal enters only through $\hat{h}_{\mu\nu}\mathcal{L}_{\mathrm{P}}$. It is a source of the fold. Status: DERIVED.

3.4 The conservation laws

$$\begin{aligned} &\nabla_M \big(\mathcal{T}_\Psi{}^M{}_N + \mathcal{T}_{\mathrm{stab}}{}^M{}_N\big) = 0 \qquad (\sigma \neq 0), \\ &\hat{\nabla}_a \, \mathcal{S}^a{}_b = -\Big(\mathcal{T}_{\mathrm{bulk}}{}^{(0)}{}_b\big|_{\sigma=0^+} - \mathcal{T}_{\mathrm{bulk}}{}^{(0)}{}_b\big|_{\sigma=0^-}\Big) \end{aligned} \tag{FE-11}$$

Symbols in (FE-11).

- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols $\Gamma^P{}_{MN}$). Status: STANDARD.
- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex, $\mathcal{T}_{MN} = -(2/\sqrt{|g|})\,\delta S_\Psi / \delta g^{MN}$. Unit: GeV^8. Status: STANDARD.

- $\mathcal{T}_{\text{stab}}{}^{a}{}_{b}$: The energy-momentum tensor of the stabiliser of the bulk (frame components). Unit: GeV^8. Status: ASSUMED.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \operatorname{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\hat{\nabla}$: The covariant derivative of the fold: with the Levi-Civita connection of the metric of the fold ($\hat{h}_{\mu\nu}$ for four-dimensional tensors, the induced metric of the seven-dimensional fold for $\hat{\nabla}_a \mathcal{S}^a{}_b$). Status: STANDARD.
- $\mathcal{S}^a{}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- $\mathcal{T}_{\text{bulk}}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\text{stab}}$; $\mathcal{T}_{\text{bulk}}{}^{(0)}{}_{b}$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.

The energy and momentum of the bulk are conserved away from the fold (FE-3 and FE-1). At the fold the thin source exchanges energy and momentum with the bulk: its covariant divergence along the fold equals minus the jump of the flux $\mathcal{T}_{\text{bulk}}{}^{(0)}{}_{b}$ of the bulk sources along σ (the frame component $(0), b$). For the author's metric with general $a_4(x_4)$, $b_4(x_4)$ this is an identity of the junction formula FE-30 and the bulk equation FE-29 (DERIVED, `um3_fold.py`, J6). The fluxes of dirac16complex into the bulk carry its share of this exchange. Status: DERIVED.

4 The matter fields

4.1 dirac16complex

$$\gamma^M D_M \Psi - m_\Psi \Psi - \lambda_\Psi \big(\bar{\Psi}\Psi\big)\Psi = 0 \qquad (\sigma \neq 0) \tag{FE-12}$$

Symbols in (FE-12).

- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\text{GeV}^{7/2}$. Status: AUTHOR.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- λ_Ψ: The coupling of the contact self-interaction of dirac16complex. Unit: GeV^{-6}. Status: RECORD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \operatorname{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

The field equation of dirac16complex in the bulk: the variation of IN-4 with respect to $\bar{\Psi}$ (the second half of the symmetric kinetic term, integrated by parts with $D_M\gamma^M = 0$, doubles the first). With the author's real matrices the operator $\gamma^M D_M - m_\Psi$ is real: the field equation does not contain i, and a real solution exists for every complex one. Status: RECORD.

$$\Psi\big|_{\sigma=0^+} = \exp\Big(\frac{1}{V_*}\,\gamma^{(0)}\,\mathbb{M}\Big)\,\Psi\big|_{\sigma=0^-},$$
$$\mathbb{M} = c_\Phi\big(\Phi^\dagger\Phi\big)P_{\mathrm{L}} + \sum_X\Big[\frac{1}{4}\big(c_X^G\,G^A_{\mu\nu}G^{A\mu\nu} + c_X^W\,W^I_{\mu\nu}W^{I\mu\nu} + c_X^B\,B_{\mu\nu}B^{\mu\nu}\big)P_X + \frac{d_X}{2}\,B_{\mu\nu}\,\mathrm{i}\Sigma^{\mu\nu}P_X \qquad \text{(FE-13)}$$
$$+ \sum_f c_X^f\big(2\mathrm{i}\,J_f^\mu P_X D_\mu + \mathrm{i}(\hat\nabla_\mu J_f^\mu)P_X\big)\Big]$$

Symbols in (FE-13).

- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- $\mathbb{M}$: The fold operator of dirac16complex: the 16×16 matrix (with a derivative along the fold) built from the Standard-Model fields at the fold by varying the portals with respect to $\bar\Psi$ (FE-13). Unit: GeV^{-3} (so that $\mathbb{M}/V_*$ is dimensionless). Status: DERIVED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- P_{L}: The author's left-handed chiral projector, $P_{\mathrm{L}} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_{\mathrm{L}}$ for $X = \mathrm{L}$, P_{R} for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu \mathbf{e}_b{}^\nu \Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.

- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- J_f^μ: The fermion-number current of the Standard-Model multiplets of kind f, $J_f^\mu = \bar\psi_f \hat\gamma^\mu \psi_f$ summed over the three generations and the gauge components; with other labels the currents that are the sources of the gauge fields. Unit: GeV^3. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\hat\nabla$: The covariant derivative of the fold: with the Levi-Civita connection of the metric of the fold ($\hat h_{\mu\nu}$ for four-dimensional tensors, the induced metric of the seven-dimensional fold for $\hat\nabla_a \mathcal{S}^a{}_b$). Status: STANDARD.

The equation of dirac16complex at the fold: the variation of the portals IN-16 with respect to $\bar\Psi$ adds $-\delta_\mathrm{D}(\sigma)\,\mathbb{M}\Psi/V_*$ to FE-12. $\mathbb{M}$ is the fold operator, a 16×16 matrix (with a derivative along the fold from the current portals) built from the Standard-Model fields at the fold. Integrating FE-12 with this term across a fold of vanishing thickness gives the matching condition in the first line: the values of dirac16complex on the two sides of the fold differ by the exponential of $\gamma^{(0)}\mathbb{M}/V_*$ (the exact solution of $\partial_\sigma\Psi = \gamma^{(0)}\mathbb{M}\,\delta_\mathrm{D}(\sigma)\Psi/V_*$ in a layer in which the fold term dominates; $(\gamma^{(0)})^2 = \mathbf{1}$). The Higgs term acts on the left-handed half only. Status: DERIVED (from INVENTED terms).

$$\gamma^M\Omega_M = -3H\,\mathrm{sgn}(\sigma)\,\gamma^{(0)} + \frac{3}{2}\,\frac{b_4'}{b_4}\,\gamma^{(4)} \neq 0 \qquad \text{(FE-14)}$$

Symbols in (FE-14).

- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Ω_M: The spin connection acting on dirac16complex, $\Omega_M = \frac{1}{2}\varpi_{Mab}\Sigma^{ab}$. Unit: 1/length. Status: RECORD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.

The coupling of dirac16complex to gravity through the canonical spin connection, for the author's metric: the contraction $\gamma^M\Omega_M$ that enters the Dirac operator. The first term comes from the warp of the six directions $x_1, x_2, x_3, x_5, x_6, x_7$ (each contributes $\frac{1}{2}\partial_\sigma$ of the logarithm of its scale factor), the second from the time dependence of their scale factors (a_4 cancels between ordinary space and the extra times). The first term never vanishes ($H > 0$, and it only changes sign at the fold); the second is non-zero whenever b_4 depends on x_4, as the author requires. Hence the coupling of dirac16complex to gravity through the spin connection is never identically zero (DERIVED, `um3_geometry.py`, G5, with the author's matrices; the vielbein postulate $D_M\gamma^N = 0$ holds, G5). For the general warped metric FE-34, $\gamma^M\Omega_M = \frac{1}{2}\partial_\sigma(3\mathfrak{a} + 3\mathfrak{b} + \mathfrak{n})\gamma^{(0)} + \frac{1}{2}\mathrm{e}^{-\mathfrak{n}}\partial_{x_4}(3\mathfrak{a} + 3\mathfrak{b})\gamma^{(4)}$. Status: DERIVED.

4.2 The Higgs doublet

$$\frac{1}{\sqrt{-\hat{h}}\,\mathcal{Z}}\,D_\mu\big(\sqrt{-\hat{h}}\,\mathcal{Z}\,\hat{h}^{\mu\nu}D_\nu\Phi\big)-\mu^2\Phi-2\lambda\big(\Phi^\dagger\Phi\big)\Phi-c_\Phi\,\bar{\Psi}P_{\rm L}\Psi\,\Phi=\Theta,$$
$$\Theta=\begin{pmatrix}\bar{d}_{\rm R}(\mathcal{Y}^d)^\dagger u_{\rm L}+\bar{e}_{\rm R}(\mathcal{Y}^e)^\dagger\nu_{\rm L}-\bar{d}_{\rm L}\mathcal{Y}^u u_{\rm R}-\bar{e}_{\rm L}\mathcal{Y}^\nu\nu_{\rm R}\\ \bar{d}_{\rm R}(\mathcal{Y}^d)^\dagger d_{\rm L}+\bar{e}_{\rm R}(\mathcal{Y}^e)^\dagger e_{\rm L}+\bar{u}_{\rm L}\mathcal{Y}^u u_{\rm R}+\bar{\nu}_{\rm L}\mathcal{Y}^\nu\nu_{\rm R}\end{pmatrix}\qquad\text{(FE-15)}$$

Symbols in (FE-15).

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu}=\mathrm{diag}(\mathrm{e}^{2a_4},\mathrm{e}^{2a_4},\mathrm{e}^{2a_4},-1)$ in the coordinates $x_1,\dots,x_4$ (the induced metric at $\sigma=0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}}=\mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4)=V_{\rm t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z}=1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M=\partial_M+\Omega_M$ (with $D_M\bar{\Psi}=\partial_M\bar{\Psi}-\bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu=\hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A=\partial_\mu X^A+g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1,x_2,x_3,x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- μ^2: The mass parameter of the Higgs potential, $\mu^2<0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2=-\lambda v^2=-m_h^2/2=-(88.6\ \mathrm{GeV})^2$. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda>0$. Unit: dimensionless. Value: $\lambda=m_h^2/(2v^2)=0.129$. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi}=\Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma=0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- $P_{\rm L}$: The author's left-handed chiral projector, $P_{\rm L}=(\mathbf{1}-\Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- Θ: The Yukawa source of the Higgs equation, $\Theta=-\partial\mathcal{L}_{\rm Y}/\partial\Phi^\dagger$ (a doublet of fermion bilinears). Unit: GeV^3. Status: STANDARD.
- $d_{\rm R}$: The right-handed down-type quarks d,s,b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $u_{\rm L}$: The upper (up-type, left-handed) components of the quark doublets. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $e_{\rm R}$: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\nu_{\rm L}$: The upper (neutral, left-handed) components of the lepton doublets: the left-handed neutrinos. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $d_{\rm L}$: The lower (down-type, left-handed) components of the quark doublets. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $u_{\rm R}$: The right-handed up-type quarks u,c,t (singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\nu_{\rm R}$: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- $e_{\rm L}$: The lower (charged, left-handed) components of the lepton doublets. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.

The field equation of the Higgs doublet on the fold: the variation of IN-11, IN-15 and IN-16 with respect to $\Phi^\dagger$. The factor $\mathcal{Z}$ (IN-8) is 1 in the late universe. The Higgs portal adds the term $-c_\Phi\bar{\Psi}P_{\rm L}\Psi\,\Phi$: the scalar density of the left-handed half of dirac16complex at the fold shifts the mass parameter of the Higgs field. Θ is the Yukawa source, $-\partial\mathcal{L}_{\rm Y}/\partial\Phi^\dagger$, written with the components $Q=(u_{\rm L},d_{\rm L})$ and $\ell=(\nu_{\rm L},e_{\rm L})$ of the doublets (generation indices contracted with the Yukawa matrices). Status: STANDARD (AUTHOR potential; INVENTED portal term).

4.3 The gauge fields

$$\frac{1}{\sqrt{-\hat{h}}\,\mathcal{Z}}\Big(\partial_\mu\big[\sqrt{-\hat{h}}\,\mathcal{Z}\,\mathcal{K}_G\,G^{A\mu\nu}\big]+g_3\,f^{ABC}G^B_\mu\,\sqrt{-\hat{h}}\,\mathcal{Z}\,\mathcal{K}_G\,G^{C\mu\nu}\Big)=-g_3\,J_{\rm c}^{A\nu},\qquad \mathcal{K}_G=1+\sum_X c_X^G\,\bar\Psi P_X\Psi,$$
$$J_{\rm c}^{A\nu}=\bar Q\,\hat\gamma^\nu t^A Q+\bar u_{\rm R}\,\hat\gamma^\nu t^A u_{\rm R}+\bar d_{\rm R}\,\hat\gamma^\nu t^A d_{\rm R}$$

(FE-16)

Symbols in (FE-16).

- $\hat h_{\mu\nu}$: The four-dimensional metric of the fold, $\hat h_{\mu\nu}=\mathrm{diag}(\mathrm{e}^{2a_4},\mathrm{e}^{2a_4},\mathrm{e}^{2a_4},-1)$ in the coordinates $x_1,\dots,x_4$ (the induced metric at $\sigma=0$ without the extra times); $\hat h^{\mu\nu}$ its inverse, $\hat h$ its determinant, $\sqrt{-\hat h}=\mathrm{e}^{3a_4}$; $\hat h^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4)=V_{\rm t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z}=1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1,x_2,x_3,x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $\mathcal{K}_G$, $\mathcal{K}_W$, $\mathcal{K}_B$: The portal factors of the gauge kinetic terms, $\mathcal{K}_G=1+\sum_X c_X^G\bar\Psi P_X\Psi$ (likewise for W and B): the gauge couplings at the fold are divided by $\sqrt{\mathcal{K}}$. Unit: dimensionless. Value: 1 where dirac16complex vanishes. Status: DERIVED.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A=1,\dots,8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1,\dots,8$; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- g_3: The $SU(3)_C$ gauge coupling (the strong coupling g_s). Unit: dimensionless. Value: $g_3(m_Z)=1.2177$ ($\alpha_s=0.1180$). Status: STANDARD.
- f^{ABC}: The structure constants of $SU(3)$, $[t^A,t^B]=\mathrm{i}f^{ABC}t^C$. Unit: dimensionless. Status: STANDARD.
- B: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- C: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- $J_{\rm c}^{A\nu}$: The colour current of the quarks. Unit: GeV^3. Status: STANDARD.
- X: The label of a half of dirac16complex: $X\in\{\mathrm{L},\mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi=\Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma=0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X=P_{\rm L}$ for $X=\mathrm{L}$, $P_{\rm R}$ for $X=\mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- Q, $\bar Q$: The left-handed quark doublets of the three generations, $Q=(u_{\rm L},d_{\rm L})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat\gamma^\mu=\hat{\mathbf{e}}_a{}^\mu\hat\gamma^{(a)}$, $\{\hat\gamma^\mu,\hat\gamma^\nu\}=-2\hat h^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu=\hat h_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.
- t^A: The generators of $SU(3)$ on colour triplets, $t^A=\lambda^A_{\rm GM}/2$ (the Gell-Mann matrices halved), $\mathrm{tr}(t^At^B)=\delta^{AB}/2$. Unit: dimensionless. Status: STANDARD.
- $u_{\rm R}$: The right-handed up-type quarks u,c,t (singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $d_{\rm R}$: The right-handed down-type quarks d,s,b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.

The Yang-Mills equation of the gluons (with the convention of the signature $(+,+,+,-)$: $\nabla_\mu F^{\mu\nu}=-J^\nu$ for a Lagrangian $-\frac14F^2+A_\mu J^\mu$). The gluon-kinetic portals multiply the gluon kinetic term by $\mathcal{K}_G$: the strong coupling at the fold is $g_3/\sqrt{\mathcal{K}_G}$ where dirac16complex is dense. The colour current $J_{\rm c}^{A\nu}$ of the quarks is the source. Status: STANDARD (INVENTED portal factor).

$$\frac{1}{\sqrt{-\hat{h}}\,\mathcal{Z}}\Big(\partial_\mu\big[\sqrt{-\hat{h}}\,\mathcal{Z}\,\mathcal{K}_W\,W^{I\mu\nu}\big]+g_2\,\epsilon^{IJK}W^J_\mu\sqrt{-\hat{h}}\,\mathcal{Z}\,\mathcal{K}_W\,W^{K\mu\nu}\Big)=-g_2\big(J^{I\nu}_{\rm w}+J^{I\nu}_{\Phi}\big),$$
$$\mathcal{K}_W=1+\sum_X c^W_X\,\bar\Psi P_X\Psi,\qquad J^{I\nu}_{\rm w}=\bar Q\,\hat\gamma^\nu\frac{\tau^I}{2}Q+\bar\ell\,\hat\gamma^\nu\frac{\tau^I}{2}\ell,\qquad J^{I\nu}_{\Phi}=\mathrm{i}\Big[(D^\nu\Phi)^\dagger\frac{\tau^I}{2}\Phi-\Phi^\dagger\frac{\tau^I}{2}D^\nu\Phi\Big] \tag{FE-17}$$

Symbols in (FE-17).

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu}=\mathrm{diag}(\mathrm{e}^{2a_4},\mathrm{e}^{2a_4},\mathrm{e}^{2a_4},-1)$ in the coordinates $x_1,\dots,x_4$ (the induced metric at $\sigma=0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}}=\mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4)=V_{\rm t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z}=1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1,x_2,x_3,x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $\mathcal{K}_G$, $\mathcal{K}_W$, $\mathcal{K}_B$: The portal factors of the gauge kinetic terms, $\mathcal{K}_G=1+\sum_X c^G_X\bar\Psi P_X\Psi$ (likewise for W and B): the gauge couplings at the fold are divided by $\sqrt{\mathcal{K}}$. Unit: dimensionless. Value: 1 where dirac16complex vanishes. Status: DERIVED.
- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, W^1_μ,W^2_μ,W^3_μ, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1,2,3$; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling (usually g). Unit: dimensionless. Value: $g_2(m_Z)=0.6517$. Status: STANDARD.
- ϵ^{IJK}: The Levi-Civita symbol of three indices (the structure constants of $SU(2)$), $\epsilon^{123}=1$. Unit: dimensionless. Status: STANDARD.
- J: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- K: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- $J^{I\nu}_{\rm w}$: The weak-isospin current of the left-handed fermion doublets. Unit: GeV^3. Status: STANDARD.
- $J^{I\nu}_{\Phi}$: The weak-isospin current of the Higgs doublet (it contains the mass terms of the W fields); $J^{2\,x_4}_{\Phi}$ its isospin charge density with $I=2$. Unit: GeV^3. Status: STANDARD.
- X: The label of a half of dirac16complex: $X\in\{\mathrm{L},\mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c^W_X: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi=\Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma=0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X=P_{\rm L}$ for $X=\mathrm{L}$, $P_{\rm R}$ for $X=\mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- Q, $\bar Q$: The left-handed quark doublets of the three generations, $Q=(u_{\rm L},d_{\rm L})$ (colour triplets, hypercharge $1/6$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat\gamma^\mu=\hat{\mathbf{e}}_a{}^\mu\hat\gamma^{(a)}$, $\{\hat\gamma^\mu,\hat\gamma^\nu\}=-2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu=\hat{h}_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak doublets (τ^1,τ^2,τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- ℓ, $\bar\ell$: The left-handed lepton doublets of the three generations, $\ell=(\nu_{\rm L},e_{\rm L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M=\partial_M+\Omega_M$ (with $D_M\bar\Psi=\partial_M\bar\Psi-\bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu=\hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A=\partial_\mu X^A+g_3f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge $1/2$), a field of the fold. Unit: GeV. Status: AUTHOR.

- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.

The Yang-Mills equation of the $SU(2)_L$ fields. The sources are the weak-isospin current of the left-handed doublets and the isospin current of the Higgs doublet (which contains the mass terms of the W fields: in the vacuum $J^{I\nu}_\Phi \ni -g_2(v^2+\mathcal{A}^2)W^{I\nu}/4$, so that FE-17 becomes $\nabla_\mu W^{I\mu\nu} = m_{\mathrm{W}}^2 W^{I\nu} + \ldots$). Status: STANDARD (INVENTED portal factor).

$$\frac{1}{\sqrt{-\hat{h}}\,\mathcal{Z}}\,\partial_\mu\Big[\sqrt{-\hat{h}}\,\mathcal{Z}\Big(\mathcal{K}_B\,B^{\mu\nu}+\sum_X d_X\,\mathcal{D}_X^{\mu\nu}\Big)\Big] = -g_1\,J_Y^\nu, \qquad \mathcal{K}_B = 1+\sum_X c_X^B\,\bar\Psi P_X\Psi,$$
$$J_Y^\nu = \sum_\psi Y_\psi\,\bar\psi\,\hat\gamma^\nu\psi + \frac{\mathrm{i}}{2}\big[(D^\nu\Phi)^\dagger\Phi - \Phi^\dagger D^\nu\Phi\big] \tag{FE-18}$$

Symbols in (FE-18).

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $\mathcal{K}_G$, $\mathcal{K}_W$, $\mathcal{K}_B$: The portal factors of the gauge kinetic terms, $\mathcal{K}_G = 1 + \sum_X c_X^G \bar\Psi P_X \Psi$ (likewise for W and B): the gauge couplings at the fold are divided by $\sqrt{\mathcal{K}}$. Unit: dimensionless. Value: 1 where dirac16complex vanishes. Status: DERIVED.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- $\mathcal{D}_X^{\mu\nu}$: The dipole (antisymmetric tensor) density of the half X of dirac16complex at the fold, $\mathcal{D}_X^{\mu\nu} = \mathrm{i}\bar\Psi\Sigma^{\mu\nu}P_X\Psi$ (real, DERIVED). Unit: GeV^7. Status: DERIVED.
- g_1: The $U(1)_Y$ gauge coupling (usually g'; not normalised for grand unification). Unit: dimensionless. Value: $g_1(m_Z) = 0.3574$. Status: STANDARD.
- J_Y^ν: The hypercharge current of the fermions and of the Higgs doublet. Unit: GeV^3. Status: STANDARD.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_{\mathrm{L}}$ for $X = \mathrm{L}$, P_{R} for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- ψ: The label of a Standard-Model fermion field in a sum: ψ runs over $Q, u_{\mathrm{R}}, d_{\mathrm{R}}, \ell, e_{\mathrm{R}}, \nu_{\mathrm{R}}$. Status: STANDARD.
- Y_f: The hypercharge of a field, the charge of its coupling $g_1 Y_f B_\mu$ to the hypercharge field in the covariant derivative (1/6, 2/3, −1/3, −1/2, −1, 0 for $Q, u_{\mathrm{R}}, d_{\mathrm{R}}, \ell, e_{\mathrm{R}}, \nu_{\mathrm{R}}$, 1/2 for Φ). Unit: dimensionless. Status: STANDARD.
- ψ, $\bar\psi$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar\psi = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat\gamma^\mu = \hat{\mathbf{e}}_a{}^\mu\hat\gamma^{(a)}$, $\{\hat\gamma^\mu, \hat\gamma^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu = \hat{h}_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the

fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2 \epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.

- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.

The Maxwell equation of the hypercharge field. The dipole portal adds the dipole density of dirac16complex to the field strength, as a polarisation adds to the electric field in a medium: dirac16complex at the fold carries electric and magnetic dipole moments. The source is the hypercharge current of the fermions and of the Higgs doublet. Status: STANDARD (INVENTED portal terms).

4.4 The quarks and leptons

$$\mathrm{i}\hat{\gamma}^\mu \big(D_\mu + \mathrm{i}\,\mathcal{V}^Q_\mu\big) Q = \mathcal{Y}^u \tilde{\Phi}\, u_\mathrm{R} + \mathcal{Y}^d \Phi\, d_\mathrm{R}, \qquad \mathcal{V}^f_\mu = \sum_X c^f_X\, j_{X\mu} \tag{FE-19}$$

Symbols in (FE-19).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu \hat{\gamma}^{(a)}$, $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M \bar{\Psi} = \partial_M \bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2 \epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^f_\mu$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_\mu = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi} = \mathrm{i}\tau^2 \Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- u_R: The right-handed up-type quarks u, c, t (singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- d_R: The right-handed down-type quarks d, s, b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c^f_X: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c^ℓ_X and c^ν_X are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- j^μ_X: The convective current of the half X of dirac16complex, $j^\mu_X = \mathrm{i}(\bar{\Psi} P_X D^\mu \Psi - (D^\mu \bar{\Psi}) P_X \Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.

The Dirac equation of the quark doublets: the variation of IN-14, IN-15 and IN-16 with respect to $\bar{Q}$. The current portals act on every Standard-Model fermion of kind f as a vector potential $\mathcal{V}^f_\mu$ built from the convective currents of the two halves of dirac16complex at the fold; it is the same for the three generations (flavour blind). Status: STANDARD (INVENTED potential).

$$\mathrm{i}\hat{\gamma}^{\mu}\big(D_{\mu}+\mathrm{i}\,\mathcal{V}^{u}_{\mu}\big)u_{\mathrm{R}}=(\mathcal{Y}^{u})^{\dagger}\tilde{\Phi}^{\dagger}Q \tag{FE-20}$$

Symbols in (FE-20).

- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^{\mu}$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^{\mu}=\hat{\mathbf{e}}_a{}^{\mu}\hat{\gamma}^{(a)}$, $\{\hat{\gamma}^{\mu},\hat{\gamma}^{\nu}\}=-2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_{\mu}=\hat{h}_{\mu\nu}\hat{\gamma}^{\nu}$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1,x_2,x_3,x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_{μ}: The covariant derivative: on dirac16complex $D_M=\partial_M+\Omega_M$ (with $D_M\bar{\Psi}=\partial_M\bar{\Psi}-\bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_{μ} (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^{\mu}=\hat{h}^{\mu\nu}D_{\nu}$; on an adjoint field $(D_{\mu}X)^{A}=\partial_{\mu}X^{A}+g_3f^{ABC}G^{B}_{\mu}X^{C}$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^{f}_{\mu}$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^{f}_{\mu}=\sum_X c^{f}_{X}j_{X\mu}$ ($\mathcal{V}^{Q},\mathcal{V}^{u},\ldots,\mathcal{V}^{\nu}$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- u_{R}: The right-handed up-type quarks u,c,t (singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\mathcal{Y}^{u}$, $\mathcal{Y}^{d}$, $\mathcal{Y}^{e}$, $\mathcal{Y}^{\nu}$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi}=\mathrm{i}\tau^2\Phi^{*}$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q=(u_{\mathrm{L}},d_{\mathrm{L}})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

$$\mathrm{i}\hat{\gamma}^{\mu}\big(D_{\mu}+\mathrm{i}\,\mathcal{V}^{d}_{\mu}\big)d_{\mathrm{R}}=(\mathcal{Y}^{d})^{\dagger}\Phi^{\dagger}Q \tag{FE-21}$$

Symbols in (FE-21).

- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^{\mu}$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^{\mu}=\hat{\mathbf{e}}_a{}^{\mu}\hat{\gamma}^{(a)}$, $\{\hat{\gamma}^{\mu},\hat{\gamma}^{\nu}\}=-2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_{\mu}=\hat{h}_{\mu\nu}\hat{\gamma}^{\nu}$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1,x_2,x_3,x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_{μ}: The covariant derivative: on dirac16complex $D_M=\partial_M+\Omega_M$ (with $D_M\bar{\Psi}=\partial_M\bar{\Psi}-\bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_{μ} (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^{\mu}=\hat{h}^{\mu\nu}D_{\nu}$; on an adjoint field $(D_{\mu}X)^{A}=\partial_{\mu}X^{A}+g_3f^{ABC}G^{B}_{\mu}X^{C}$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^{f}_{\mu}$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^{f}_{\mu}=\sum_X c^{f}_{X}j_{X\mu}$ ($\mathcal{V}^{Q},\mathcal{V}^{u},\ldots,\mathcal{V}^{\nu}$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- d_{R}: The right-handed down-type quarks d,s,b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\mathcal{Y}^{u}$, $\mathcal{Y}^{d}$, $\mathcal{Y}^{e}$, $\mathcal{Y}^{\nu}$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q=(u_{\mathrm{L}},d_{\mathrm{L}})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

$$\mathrm{i}\hat{\gamma}^{\mu}\big(D_{\mu}+\mathrm{i}\,\mathcal{V}^{\ell}_{\mu}\big)\ell=\mathcal{Y}^{e}\Phi\,e_{\mathrm{R}}+\mathcal{Y}^{\nu}\tilde{\Phi}\,\nu_{\mathrm{R}} \tag{FE-22}$$

Symbols in (FE-22).

- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.

- $\hat{\gamma}^{\mu}$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^{\mu} = \hat{\mathbf{e}}_a{}^{\mu}\hat{\gamma}^{(a)}$, $\{\hat{\gamma}^{\mu}, \hat{\gamma}^{\nu}\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_{\mu} = \hat{h}_{\mu\nu}\hat{\gamma}^{\nu}$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_{μ}: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_{μ} (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^{\mu} = \hat{h}^{\mu\nu}D_{\nu}$; on an adjoint field $(D_{\mu}X)^A = \partial_{\mu}X^A + g_3 f^{ABC}G^B_{\mu}X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^f_{\mu}$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_{\mu} = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^{\nu}$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_{\mathrm{L}}, e_{\mathrm{L}})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^{\nu}$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge $1/2$), a field of the fold. Unit: GeV. Status: AUTHOR.
- e_{R}: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi} = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- ν_{R}: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.

$$\mathrm{i}\hat{\gamma}^{\mu}\big(D_{\mu} + \mathrm{i}\,\mathcal{V}^e_{\mu}\big)e_{\mathrm{R}} = (\mathcal{Y}^e)^{\dagger}\Phi^{\dagger}\ell \tag{FE-23}$$

Symbols in (FE-23).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^{\mu}$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^{\mu} = \hat{\mathbf{e}}_a{}^{\mu}\hat{\gamma}^{(a)}$, $\{\hat{\gamma}^{\mu}, \hat{\gamma}^{\nu}\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_{\mu} = \hat{h}_{\mu\nu}\hat{\gamma}^{\nu}$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_{μ}: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_{μ} (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^{\mu} = \hat{h}^{\mu\nu}D_{\nu}$; on an adjoint field $(D_{\mu}X)^A = \partial_{\mu}X^A + g_3 f^{ABC}G^B_{\mu}X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^f_{\mu}$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_{\mu} = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^{\nu}$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- e_{R}: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^{\nu}$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge $1/2$), a field of the fold. Unit: GeV. Status: AUTHOR.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_{\mathrm{L}}, e_{\mathrm{L}})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

$$\mathrm{i}\hat{\gamma}^{\mu}\big(D_{\mu} + \mathrm{i}\,\mathcal{V}^{\nu}_{\mu}\big)\nu_{\mathrm{R}} = (\mathcal{Y}^{\nu})^{\dagger}\tilde{\Phi}^{\dagger}\ell \tag{FE-24}$$

Symbols in (FE-24).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^{\mu}$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^{\mu} = \hat{\mathbf{e}}_a{}^{\mu}\hat{\gamma}^{(a)}$, $\{\hat{\gamma}^{\mu}, \hat{\gamma}^{\nu}\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_{\mu} = \hat{h}_{\mu\nu}\hat{\gamma}^{\nu}$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.

- $D_M,\ D_\mu$: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^f_\mu$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_\mu = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- $\mathcal{Y}^u,\ \mathcal{Y}^d,\ \mathcal{Y}^e,\ \mathcal{Y}^\nu$: The Yukawa matrices ($3\times 3$ complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\tilde\Phi$: The conjugate Higgs doublet, $\tilde\Phi = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- $\ell,\ \bar\ell$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

The Dirac equations of the up-quark singlets, the down-quark singlets, the lepton doublets, the charged-lepton singlets and the right-handed neutrinos (FE-20 to FE-24), each the variation of IN-14, IN-15 and IN-16 with respect to the conjugate field. Each fermion sees the gauge fields (in D_μ), the Higgs doublet (through the Yukawa matrices) and dirac16complex (through $\mathcal{V}^f_\mu$). The right-handed neutrinos see no gauge field; they couple to the Standard Model only through the neutrino Yukawa matrix, which gives the neutrinos their Dirac masses. Status: STANDARD (INVENTED potentials).

$$\begin{aligned}\hat\nabla_\mu J^\mu_\ell &= -\hat\nabla_\mu\big(J^\mu_e + J^\mu_\nu\big) = -\mathrm{i}\big(\bar\ell\,\mathcal{Y}^e\Phi\,e_\mathrm{R} + \bar\ell\,\mathcal{Y}^\nu\tilde\Phi\,\nu_\mathrm{R}\big) + \mathrm{h.c.},\\ \hat\nabla_\mu J^\mu_Q &= -\hat\nabla_\mu\big(J^\mu_u + J^\mu_d\big) = -\mathrm{i}\big(\bar Q\,\mathcal{Y}^u\tilde\Phi\,u_\mathrm{R} + \bar Q\,\mathcal{Y}^d\Phi\,d_\mathrm{R}\big) + \mathrm{h.c.}\end{aligned} \qquad \text{(FE-25)}$$

Symbols in (FE-25).

- $\hat\nabla$: The covariant derivative of the fold: with the Levi-Civita connection of the metric of the fold ($\hat h_{\mu\nu}$ for four-dimensional tensors, the induced metric of the seven-dimensional fold for $\hat\nabla_a\mathcal{S}^a{}_b$). Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- J^μ_f: The fermion-number current of the Standard-Model multiplets of kind f, $J^\mu_f = \bar\psi_f\hat\gamma^\mu\psi_f$ summed over the three generations and the gauge components; with other labels the currents that are the sources of the gauge fields. Unit: GeV^3. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\ell,\ \bar\ell$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u,\ \mathcal{Y}^d,\ \mathcal{Y}^e,\ \mathcal{Y}^\nu$: The Yukawa matrices ($3\times 3$ complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge $1/2$), a field of the fold. Unit: GeV. Status: AUTHOR.
- e_R: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\tilde\Phi$: The conjugate Higgs doublet, $\tilde\Phi = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- h.c.: The Hermitian conjugate of the preceding terms. Status: STANDARD.
- $Q,\ \bar Q$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge $1/6$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- u_R: The right-handed up-type quarks u, c, t (singlets of $SU(2)_L$, hypercharge $2/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- d_R: The right-handed down-type quarks d, s, b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.

The divergences of the fermion-number currents of the six kinds of multiplets (DERIVED from FE-19 to FE-24): the Yukawa couplings turn doublets into singlets, so the current of each kind is not conserved by itself, while the lepton number $J_\ell + J_e + J_\nu$ and the baryon number $(J_Q + J_u + J_d)/3$ are conserved. The divergences enter the fold operator $\mathbb{M}$ (FE-13). Status: DERIVED.

4.5 The ad hoc Higgs vacuum

$$\begin{aligned}\mathcal{R}_{\mathcal{A}} &= \mathcal{A}'' + 3\frac{b_4'}{b_4}\mathcal{A}' + \lambda\mathcal{A}^3 + c_\Phi\,\bar\Psi P_{\rm L}\Psi\,\mathcal{A} = \Big(4a_{40}^2 - 6a_{40}\frac{b_4'}{b_4}\Big)\mathcal{A} + \lambda\mathcal{A}^3 + c_\Phi\,\bar\Psi P_{\rm L}\Psi\,\mathcal{A},\\ \mathcal{R}_v &= \lambda\mathcal{A}^2 v + c_\Phi\,\bar\Psi P_{\rm L}\Psi\,v, \qquad J_\Phi^{2\,x_4} = a_{40}\,v\,\mathcal{A}, \qquad J_\Phi^{1\,x_4} = J_\Phi^{3\,x_4} = 0\end{aligned} \tag{FE-26}$$

Symbols in (FE-26).

- $\mathcal{R}_{\mathcal{A}}$, $\mathcal{R}_v$: The residuals of the ad hoc Higgs vacuum in the upper ($\mathcal{R}_{\mathcal{A}}$) and lower ($\mathcal{R}_v$) components of the Higgs equation: what the ad hoc vacuum leaves unbalanced. Unit: GeV^3. Status: DERIVED.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_{\mathcal{A}}\mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: GeV$^{7/2}$. Status: AUTHOR.
- $P_{\rm L}$: The author's left-handed chiral projector, $P_{\rm L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- a_{40}: The decay constant of almightyAdHoc, $a_{40} > 0$: $\mathcal{A}$ falls by e in the time $1/(2a_{40})$. Unit: GeV (1/length). Status: AUTHOR.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_{\rm F})^{-1/2}$ with the Fermi constant $G_{\rm F} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- $J_\Phi^{I\nu}$: The weak-isospin current of the Higgs doublet (it contains the mass terms of the W fields); $J_\Phi^{2\,x_4}$ its isospin charge density with $I = 2$. Unit: GeV^3. Status: STANDARD.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.

The author's ad hoc vacuum IN-12 inserted into the Higgs equation FE-15 (homogeneous fields, no gauge fields, the fold factor $\mathcal{Z} \propto b_4^3\mathrm{e}^{-3a_4}$ so that the time derivative of $\sqrt{-\hat h}\mathcal{Z}$ gives $3b_4'/b_4$). The upper and lower components of FE-15 are not satisfied: $\mathcal{R}_{\mathcal{A}}$ and $\mathcal{R}_v$ are what remains (the residuals; FE-15 holds exactly if the source Θ is replaced by $\Theta - (\mathcal{R}_{\mathcal{A}}, \mathcal{R}_v)/\sqrt{2}$). The ad hoc vacuum is an exact solution only in the limit $\mathcal{A} \to 0$. It also carries a weak-isospin charge density $J_\Phi^{2\,x_4} = a_{40}v\mathcal{A}$ (the component along x_4 of the isospin current of FE-17 with $I = 2$), which the W^2 equation FE-17 must balance with a background field $W_{x_4}^2$ (DERIVED, `um3_fields.py`, F5: the masses of the W and Z, the direction of the photon, the charge density, the residual of the upper component). Status: DERIVED.

5 The background

The background is the author's metric with the two functions $a_4(x_4)$ and $b_4(x_4)$ of the time, on which every other field moves; the matter fields correct it by the next terms of a power series (FE-35). This section writes out its geometry and the exact equations that its two functions obey, in the bulk and at the fold.

5.1 The author's metric and the proper chart

$$\begin{aligned}g_{MN} &= \mathrm{diag}\Big(\cot^2 z,\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ -1,\\ &\qquad -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6}\Big), \qquad z = 6Hx_0,\\ \sigma &= \mathrm{sgn}\Big(\frac{\pi}{2} - z\Big)\frac{\ln\sin z}{6H}, \qquad \mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0,\\ g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N &= \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\Big[\mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{e}^{-2a_4}b_4^2\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big] - \mathrm{d}x_4^2\end{aligned} \tag{FE-27}$$

Symbols in (FE-27).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

The author's metric of the primordial gravitational field (his order of 2026-10-24: the metric of `Dirac_Einstein-Lovelock_4+4_Nash.nb` with $b_4 \to b_4^2$; $(\sin^2 z)^{1/6}$ is the positive branch of `Sin[6 H x0]^(1/3)`, the author's rule of 2026-10-16), and the same metric in the proper chart σ of the hidden direction. In the author's chart the metric is smooth but degenerate at the fold $z = \pi/2$ ($\cot z = 0$); the proper distance from the fold is $|\sigma|$, and σ runs from $-\infty$ (the tip $z \to 0$) through 0 (the fold) to $+\infty$ (the tip $z \to \pi$): one period of x_0 is the whole line, and the two halves of a period are mirror images under $z \to \pi - z$ ($\sigma \to -\sigma$). Ordinary space and the extra times are warped by $\mathrm{e}^{-H|\sigma|}$ (with a kink at the fold), the time x_4 is not warped. a_4 and b_4 are functions of x_4; b_4 is never identically 1. Status: AUTHOR (the metric), DERIVED (the chart, `um3_fold.py`, J1).

$$
\begin{aligned}
&\mathbf{e}^{(0)} = \mathrm{d}\sigma, \quad \mathbf{e}^{(i)} = \mathrm{e}^{a_4 - H|\sigma|}\mathrm{d}x_i, \quad \mathbf{e}^{(4)} = \mathrm{d}x_4, \quad \mathbf{e}^{(r)} = b_4\,\mathrm{e}^{-a_4 - H|\sigma|}\mathrm{d}x_r,\\
&\sqrt{|g|} = b_4^3\,\mathrm{e}^{-6H|\sigma|}, \qquad \mathsf{A} = \frac{a_4'}{H}, \qquad \mathsf{U} = \frac{b_4'}{Hb_4}, \qquad \mathsf{A}_1 = \frac{\mathsf{A}'}{H}, \qquad \mathsf{U}_1 = \frac{\mathsf{U}'}{H},\\
&R^{(0)(i)}{}_{(0)(i)} = R^{(0)(r)}{}_{(0)(r)} = -H^2, \qquad R^{(0)(4)}{}_{(0)(4)} = 0, \qquad R^{(i)(j)}{}_{(i)(j)} = (\mathsf{A}^2 - 1)H^2,\\
&R^{(i)(4)}{}_{(i)(4)} = (\mathsf{A}^2 + \mathsf{A}_1)H^2, \qquad R^{(i)(r)}{}_{(i)(r)} = -(1 + \mathsf{A}^2 - \mathsf{A}\mathsf{U})H^2,\\
&R^{(4)(r)}{}_{(4)(r)} = \big((\mathsf{U} - \mathsf{A})^2 + \mathsf{U}_1 - \mathsf{A}_1\big)H^2, \qquad R^{(r)(s)}{}_{(r)(s)} = \big((\mathsf{U} - \mathsf{A})^2 - 1\big)H^2,\\
&R^{(0)(i)}{}_{(4)(i)} = \mathrm{sgn}(\sigma)\,\mathsf{A}H^2, \qquad R^{(0)(r)}{}_{(4)(r)} = \mathrm{sgn}(\sigma)\,(\mathsf{U} - \mathsf{A})H^2
\end{aligned}
\tag{FE-28}
$$

Symbols in (FE-28).

- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x: The coordinates of the fold: x_i ($i = 1, 2, 3$) of ordinary space, x_4 the time, x_r ($r = 5, 6, 7$) the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+, +, +, +, -, -, -, -$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor (convention $[\nabla_P, \nabla_Q]V^M = R^M{}_{NPQ}V^N$), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: 1/length^2. Status: STANDARD.
- j: A second generation or mass-state index (as i). Status: STANDARD.
- s: A second extra-time direction of the fold (as r, different from it). Status: STANDARD.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.

The vielbein and the volume element (i, j any two different directions of ordinary space, r, s any two different

extra times; x_i and x_r the coordinates of these directions), the four dimensionless rates of the background (A the expansion rate of ordinary space in units of H, U the rate of change of b_4), and every non-zero frame component of the Riemann tensor, up to the symmetries $R^{ab}{}_{cd} = -R^{ba}{}_{cd} = -R^{ab}{}_{dc} = R_{cd}{}^{ab}$ (the last with the frame metric). No component depends on σ except through the sign of the two mixed ones, which change sign at the fold: the bulk on each side of the fold is homogeneous along σ (DERIVED, `um3_geometry.py`, G2, exact; `um3_fold.py`, J2). The frame curvature of the hidden direction is $-H^2$: each half of a period is a space of constant negative curvature along σ, like a slice of anti-de Sitter space, with the time not warped. Status: DERIVED.

5.2 The bulk equations of the background

$$
\begin{aligned}
\frac{\kappa\,\mathcal{T}_{\text{bulk}}{}^{(0)}{}_{(0)}}{H^2} &= -\frac{1}{16}\Big(\mathsf{A}^6 - 3\mathsf{A}^5\mathsf{U} + 12\mathsf{A}^4\mathsf{U}^2 + 3\mathsf{A}^4\mathsf{U}_1 + 3\mathsf{A}^4 \\
&\quad + 12\mathsf{A}^3\mathsf{A}_1\mathsf{U} - 19\mathsf{A}^3\mathsf{U}^3 - 12\mathsf{A}^3\mathsf{U}\mathsf{U}_1 - 6\mathsf{A}^3\mathsf{U} - 18\mathsf{A}^2\mathsf{A}_1\mathsf{U}^2 + 9\mathsf{A}^2\mathsf{U}^4 \\
&\quad + 9\mathsf{A}^2\mathsf{U}^2\mathsf{U}_1 - 18\mathsf{A}^2\mathsf{U}^2 - 6\mathsf{A}^2\mathsf{U}_1 - 9\mathsf{A}^2 + 6\mathsf{A}\mathsf{A}_1\mathsf{U}^3 - 12\mathsf{A}\mathsf{A}_1\mathsf{U} \\
&\quad + 21\mathsf{A}\mathsf{U}^3 + 12\mathsf{A}\mathsf{U}\mathsf{U}_1 + 9\mathsf{A}\mathsf{U} + 6\mathsf{A}_1\mathsf{U}^2 + 3\mathsf{U}^4 + 3\mathsf{U}^2\mathsf{U}_1 \\
&\quad + 18\mathsf{U}^2 + 3\mathsf{U}_1 + 5\Big), \\
\frac{\kappa\,\mathcal{T}_{\text{bulk}}{}^{(i)}{}_{(i)}}{H^2} &= -\frac{1}{16}\Big(\mathsf{A}^6 - 6\mathsf{A}^5\mathsf{U} - 5\mathsf{A}^4\mathsf{A}_1 + 12\mathsf{A}^4\mathsf{U}^2 + 3\mathsf{A}^4\mathsf{U}_1 \\
&\quad + 3\mathsf{A}^4 + 12\mathsf{A}^3\mathsf{A}_1\mathsf{U} - 10\mathsf{A}^3\mathsf{U}^3 - 6\mathsf{A}^3\mathsf{U}\mathsf{U}_1 + 12\mathsf{A}^3\mathsf{U} - 9\mathsf{A}^2\mathsf{A}_1\mathsf{U}^2 \\
&\quad + 6\mathsf{A}^2\mathsf{A}_1 + 3\mathsf{A}^2\mathsf{U}^4 + 3\mathsf{A}^2\mathsf{U}^2\mathsf{U}_1 - 36\mathsf{A}^2\mathsf{U}^2 - 6\mathsf{A}^2\mathsf{U}_1 - 9\mathsf{A}^2 \\
&\quad + 2\mathsf{A}\mathsf{A}_1\mathsf{U}^3 - 12\mathsf{A}\mathsf{A}_1\mathsf{U} + 18\mathsf{A}\mathsf{U}^3 + 6\mathsf{A}\mathsf{U}\mathsf{U}_1 - 6\mathsf{A}\mathsf{U} + 3\mathsf{A}_1\mathsf{U}^2 \\
&\quad - \mathsf{A}_1 + 3\mathsf{U}^4 + 3\mathsf{U}^2\mathsf{U}_1 + 24\mathsf{U}^2 + 3\mathsf{U}_1 + 5\Big), \\
\frac{\kappa\,\mathcal{T}_{\text{bulk}}{}^{(4)}{}_{(4)}}{H^2} &= \frac{1}{16}\Big(5\mathsf{A}^6 - 15\mathsf{A}^5\mathsf{U} + 15\mathsf{A}^4\mathsf{U}^2 - 9\mathsf{A}^4 - 5\mathsf{A}^3\mathsf{U}^3 \\
&\quad + 18\mathsf{A}^3\mathsf{U} + 3\mathsf{A}^2 - 9\mathsf{A}\mathsf{U}^3 - 3\mathsf{A}\mathsf{U} - 3\mathsf{U}^2 + 1\Big), \\
\frac{\kappa\,\mathcal{T}_{\text{bulk}}{}^{(r)}{}_{(r)}}{H^2} &= -\frac{1}{16}\Big(\mathsf{A}^6 + 5\mathsf{A}^4\mathsf{A}_1 - 3\mathsf{A}^4\mathsf{U}^2 - 2\mathsf{A}^4\mathsf{U}_1 + 3\mathsf{A}^4 \\
&\quad - 8\mathsf{A}^3\mathsf{A}_1\mathsf{U} + 2\mathsf{A}^3\mathsf{U}^3 + 2\mathsf{A}^3\mathsf{U}\mathsf{U}_1 - 24\mathsf{A}^3\mathsf{U} + 3\mathsf{A}^2\mathsf{A}_1\mathsf{U}^2 - 6\mathsf{A}^2\mathsf{A}_1 \\
&\quad + 18\mathsf{A}^2\mathsf{U}^2 - 9\mathsf{A}^2 + 6\mathsf{A}\mathsf{U}^3 + 6\mathsf{A}\mathsf{U}\mathsf{U}_1 + 24\mathsf{A}\mathsf{U} + 3\mathsf{A}_1\mathsf{U}^2 \\
&\quad + \mathsf{A}_1 + 9\mathsf{U}^2 + 2\mathsf{U}_1 + 5\Big), \\
\frac{\kappa\,\mathcal{T}_{\text{bulk}}{}^{(0)}{}_{(4)}}{H^2} &= \operatorname{sgn}(\sigma)\,\frac{3}{16}\,\mathsf{U}\,\mathcal{G}, \qquad \mathcal{G} = 5\mathsf{A}^2(\mathsf{A}-\mathsf{U})^2 - 6\mathsf{A}^2 + 6\mathsf{A}\mathsf{U} + \mathsf{U}^2 + 1
\end{aligned}
\tag{FE-29}
$$

Symbols in (FE-29).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\text{bulk}}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\text{stab}}$; $\mathcal{T}_{\text{bulk}}{}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- sgn: The sign function: $\operatorname{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\mathcal{G}$: The factor of the (σ, x_4) bulk equation of the author's metric, $\mathcal{G} = 5\mathsf{A}^2(\mathsf{A}-\mathsf{U})^2 - 6\mathsf{A}^2 + 6\mathsf{A}\mathsf{U} + \mathsf{U}^2 + 1$. Unit: dimensionless. Status: DERIVED.

The bulk equations FE-1 for the author's metric, divided by H^2: the left-hand side of FE-1 with the author's weights and $\Lambda = -10H^2$, evaluated exactly (DERIVED, `um3_geometry.py`, G3 and G4, with the identities FE-3 checked; `um3_fold.py`, J2, by an independent general computation). They state what the bulk sources must be for given $a_4(x_4)$, $b_4(x_4)$: ${\mathcal{T}^{(4)}}_{(4)} = -\rho$ gives the energy density, ${\mathcal{T}^{(0)}}_{(0)}$, ${\mathcal{T}^{(i)}}_{(i)}$, ${\mathcal{T}^{(r)}}_{(r)}$ the pressures along σ, ordinary space and the extra times, and ${\mathcal{T}^{(0)}}_{(4)}$ the energy flux along σ, which has opposite signs on the two sides of the fold (it flows toward or away from the fold on both sides). Because the frame components do not depend on σ, the author's metric can be exact only with sources whose frame components do not depend on σ; sources that depend on σ require the next terms of the power series of the metric (FE-35). The time-time equation contains no A_1 and no U_1: it is the energy constraint of the background. The flux equation factorises: with a source without flux along σ, either $\mathsf{U} = 0$ (b_4 constant) or $\mathcal{G}(\mathsf{A}, \mathsf{U}) = 0$, which has real roots U only for $0.3249 \le |\mathsf{A}| \le 1.3764$ ($5\mathsf{A}^4 - 10\mathsf{A}^2 + 1 \le 0$). Status: DERIVED.

5.3 The fold equations of the background

$$
\begin{aligned}
\frac{\kappa\,{\mathcal{S}^{(i)}}_{(i)}}{H} &= \frac{1}{24}\Big(9\mathsf{A}^4 - 24\mathsf{A}^3\mathsf{U} - 12\mathsf{A}^2\mathsf{A}_1 + 27\mathsf{A}^2\mathsf{U}^2 + 12\mathsf{A}^2\mathsf{U}_1 - 66\mathsf{A}^2 + 24\mathsf{A}\mathsf{A}_1\mathsf{U} - 6\mathsf{A}\mathsf{U}^3 - 12\mathsf{A}\mathsf{U}\mathsf{U}_1 + 120\mathsf{A}\mathsf{U} \\
&\qquad - 6\mathsf{A}_1\mathsf{U}^2 + 20\mathsf{A}_1 - 6\mathsf{U}^4 - 6\mathsf{U}^2\mathsf{U}_1 - 111\mathsf{U}^2 - 60\mathsf{U}_1 - 183\Big), \\
\frac{\kappa\,{\mathcal{S}^{(4)}}_{(4)}}{H} &= -\frac{1}{4}\Big(3\mathsf{A}^4 - 6\mathsf{A}^3\mathsf{U} - 10\mathsf{A}^2 + 3\mathsf{A}\mathsf{U}^3 + 10\mathsf{A}\mathsf{U} + 10\mathsf{U}^2 + 31\Big), \\
\frac{\kappa\,{\mathcal{S}^{(r)}}_{(r)}}{H} &= \frac{1}{24}\Big(9\mathsf{A}^4 - 12\mathsf{A}^3\mathsf{U} + 12\mathsf{A}^2\mathsf{A}_1 + 9\mathsf{A}^2\mathsf{U}^2 - 66\mathsf{A}^2 - 12\mathsf{A}\mathsf{U}^3 - 12\mathsf{A}\mathsf{U}\mathsf{U}_1 + 12\mathsf{A}\mathsf{U} - 6\mathsf{A}_1\mathsf{U}^2 - 20\mathsf{A}_1 \\
&\qquad - 57\mathsf{U}^2 - 40\mathsf{U}_1 - 183\Big)
\end{aligned}
\tag{FE-30}
$$

Symbols in (FE-30).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- ${\mathcal{S}^a}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.

The junction conditions FE-4 for the author's metric: the thin source that the fold must carry for given $a_4(x_4)$, $b_4(x_4)$ (the off-diagonal and (σ, σ) components have no thin part). They follow from the potential $\mathfrak{F}$ evaluated with $\partial_\sigma \mathfrak{a} = \partial_\sigma \mathfrak{b} = -H\,\mathrm{sgn}(\sigma)$ on the two sides (DERIVED, `um3_fold.py`, J4 and J5, and independently by smoothing the kink, $\partial_\sigma \mathfrak{a} = -H\tanh(\sigma/\varepsilon)$, and integrating the full left-hand side of FE-1 across the layer: the result converges linearly in ε to FE-30). With the bulk flux of FE-29 they satisfy the conservation law FE-11 identically (DERIVED, J6). Status: DERIVED.

5.4 The author's vacuum

$$
\begin{aligned}
&a_4 = Hx_4, \quad b_4 = \text{constant } (\mathsf{A}=1,\ \mathsf{U}=\mathsf{A}_1=\mathsf{U}_1=0): \qquad \sum_{p=1}^{3} \alpha_{(p)}\, {\mathcal{E}_{(p)}}^{M}{}_{N} + \Lambda\, \delta^{M}_{N} = 0 \quad (\sigma \neq 0),\\
&\kappa\, {\mathcal{S}^{(i)}}_{(i)} = \kappa\, {\mathcal{S}^{(r)}}_{(r)} = -10H, \qquad \kappa\, {\mathcal{S}^{(4)}}_{(4)} = -6H,\\
&\kappa\rho_{\text{bulk}} = -H^2\Big[\frac{3}{4}\big((\mathsf{A}-1-\mathsf{U})^2 + (\mathsf{A}-1)^2\big) + \dots\Big], \qquad \mathcal{G}(1,\mathsf{U}) = 2\mathsf{U}(3\mathsf{U}-2)
\end{aligned}
\tag{FE-31}
$$

Symbols in (FE-31).

- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4} b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- ${\mathcal{E}_{(p)}}^{M}{}_{N}$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \dots$): it runs over $x_0, x_1, \dots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \dots$), running over $x_0, \dots, x_7$; a repeated index is summed. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$ (the author writes the pure number $\Lambda_{\mathrm{N}} = 10$ in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}} = 1$ for $X = \mathrm{L}$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1 \cdots M_n}_{N_1 \cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or −1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- ${\mathcal{S}^{a}}_{b}$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- ρ_{bulk}: The energy density of the bulk sources, $\rho_{\text{bulk}} = -{\mathcal{T}_{\text{bulk}}}^{(4)}{}_{(4)}$. Unit: GeV^8. Status: DERIVED.
- $\mathcal{G}$: The factor of the (σ, x_4) bulk equation of the author's metric, $\mathcal{G} = 5\mathsf{A}^2(\mathsf{A}-\mathsf{U})^2 - 6\mathsf{A}^2 + 6\mathsf{A}\mathsf{U} + \mathsf{U}^2 + 1$. Unit: dimensionless. Status: DERIVED.

The author's solution (`Dirac_Einstein-Lovelock_4+4_Nash.nb`, In[134]: $a_4' = H$, the weights $(1, 1/32, 1/1152)$ and $\Lambda_N = 10$): ordinary space inflates with the rate H forever and the extra times deflate ($e^{-a_4} b_4 \propto e^{-Hx_4}$), "eternal inflation/deflation". With the new metric (b_4^2) it is still an exact vacuum solution of the bulk equations for every constant b_4 (DERIVED, `um3_geometry.py`, G4). Three facts about it (DERIVED):

- The fold must carry a thin source: $\kappa\mathcal{S} = \mathrm{diag}(-10H, -6H, -10H)$ for (ordinary space, time, extra times), an energy density $6H/\kappa$ with the pressure $-10H/\kappa$ (`um3_fold.py`, J5).
- The vacuum is degenerate: near it the energy density of any bulk source is $\kappa\rho_{\mathrm{bulk}} = -H^2[\frac{3}{4}((\mathsf{A}-1-\mathsf{U})^2 + (\mathsf{A}-1)^2) + \dots]$ (the dots are of third order in $\mathsf{A}-1$ and U): the linear terms vanish, so a small positive energy density of the bulk cannot be carried by a small change of a_4 and b_4 (`um3_geometry.py`, G4). The transverse traceless gravitons of the bulk have no time-kinetic term at this point (`um3_bulk.py`, B5).
- b_4 is a function of x_4, never identically 1 (the author's order): a constant $b_4 \neq 1$ is equivalent to $b_4 = 1$ by a rescaling of x_5, x_6, x_7 unless their coordinate period is fixed, and $\mathsf{U} \neq 0$ is not a vacuum solution near $\mathsf{A} = 1$: it needs a bulk source with a flux along σ, $\kappa\mathcal{T}^{(0)}{}_{(4)} = \mathrm{sgn}(\sigma)\frac{3}{16}\mathsf{U}\,\mathcal{G}(1,\mathsf{U})H^2 = \mathrm{sgn}(\sigma)\frac{3}{8}\mathsf{U}^2(3\mathsf{U}-2)H^2$, and a non-positive energy density. At $\mathsf{A} = 1$ the flux vanishes for $\mathsf{U} = 2/3$ ($b_4 \propto e^{2Hx_4/3}$), which needs the constant source $\kappa\mathcal{T} = (-14/27, -2/9, 2/27, -22/27)H^2$ for (σ, ordinary space, time, extra times), with $\kappa\rho_{\mathrm{bulk}} = -2H^2/27$ (`um3_geometry.py`, G4).

Status: AUTHOR (the solution), DERIVED (the three facts).

5.5 The static block and the frozen extra times

$$\begin{aligned} &\mathsf{A} = \mathsf{U} = \mathsf{A}_1 = \mathsf{U}_1 = 0: \qquad \kappa\,\mathcal{T}_{\mathrm{bulk}}{}^{a}{}_{b} = H^2\,\mathrm{diag}\Big(-\frac{5}{16}, -\frac{5}{16}, -\frac{5}{16}, -\frac{5}{16}, \frac{1}{16}, -\frac{5}{16}, -\frac{5}{16}, -\frac{5}{16}\Big), \\ &\kappa\,\mathcal{S}^{(i)}{}_{(i)} = \kappa\,\mathcal{S}^{(r)}{}_{(r)} = -\frac{61}{8}H, \qquad \kappa\,\mathcal{S}^{(4)}{}_{(4)} = -\frac{31}{4}H \end{aligned} \tag{FE-32}$$

Symbols in (FE-32).

- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\mathrm{bulk}}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\mathrm{stab}}$; $\mathcal{T}_{\mathrm{bulk}}{}^{(0)}{}_{b}$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \dots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \dots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \dots, x_7$ for a tensor of eight dimensions, $x_1, \dots, x_7$ for a tensor of the fold, $x_1, \dots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- $\mathcal{S}^a{}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.

The static block: a_4 and b_4 constant. Its bulk is (each half of a period) a seven-dimensional space of constant negative curvature $-H^2$ of the directions $\sigma, x_1, x_2, x_3, x_5, x_6, x_7$ times the unwarped time (FE-28 with $\mathsf{A} = \mathsf{U} = 0$). It is not a vacuum solution: it requires the bulk source of the first line, which is the stabiliser IN-7 ($\kappa\rho = -H^2/16$, $\kappa p = -5H^2/16$), and the thin source of the second line at the fold, which is the fold fluid at rest ($\kappa\rho_{\mathrm{ff}}/V_* = 31H/4$,

$\kappa p_{\rm ff}/V_* = \kappa q_{\rm ff}/V_* = -61H/8$, $\rho + p = H/(8\kappa)$ per V_*, positive). The late universe, whose expansion rate is tiny in units of H, is the static block plus small corrections (FE-33, FE-59). Status: DERIVED.

$$
\begin{aligned}
&\mathsf{U} = \mathsf{A}, \quad \mathsf{U}_1 = \mathsf{A}_1 \ \left(\mathrm{e}^{-a_4} b_4 = \text{constant},\ \mathcal{Z} = 1\right): \\
&\frac{\kappa\rho_{\rm bulk}}{H^2} = \frac{3\mathsf{A}^2 - 1}{16}, \qquad \frac{\kappa\,{\mathcal{T}_{\rm bulk}}^{(0)}{}_{(0)}}{H^2} = -\frac{3\mathsf{A}^4 + 3\mathsf{A}^2\mathsf{A}_1 + 18\mathsf{A}^2 + 3\mathsf{A}_1 + 5}{16}, \\
&\frac{\kappa\,{\mathcal{T}_{\rm bulk}}^{(i)}{}_{(i)}}{H^2} = -\frac{9\mathsf{A}^2 + 2\mathsf{A}_1 + 5}{16}, \qquad \frac{\kappa\,{\mathcal{T}_{\rm bulk}}^{(r)}{}_{(r)}}{H^2} = -\frac{3\mathsf{A}^4 + 3\mathsf{A}^2\mathsf{A}_1 + 24\mathsf{A}^2 + 3\mathsf{A}_1 + 5}{16}, \\
&\frac{\kappa\,{\mathcal{T}_{\rm bulk}}^{(0)}{}_{(4)}}{H^2} = \mathrm{sgn}(\sigma)\,\frac{3\mathsf{A}(1 + \mathsf{A}^2)}{16}, \qquad \frac{\kappa\,{\mathcal{S}}^{(i)}{}_{(i)}}{H} = -\frac{57\mathsf{A}^2 + 40\mathsf{A}_1 + 183}{24}, \\
&\frac{\kappa\,{\mathcal{S}}^{(4)}{}_{(4)}}{H} = -\frac{10\mathsf{A}^2 + 31}{4}, \qquad \frac{\kappa\,{\mathcal{S}}^{(r)}{}_{(r)}}{H} = -\frac{2\mathsf{A}^4 + 2\mathsf{A}^2\mathsf{A}_1 + 37\mathsf{A}^2 + 20\mathsf{A}_1 + 61}{8}
\end{aligned}
\tag{FE-33}
$$

Symbols in (FE-33).

- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4} b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_{\rm t}(\mathrm{e}^{-a_4} b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\rho_{\rm bulk}$: The energy density of the bulk sources, $\rho_{\rm bulk} = -{\mathcal{T}_{\rm bulk}}^{(4)}{}_{(4)}$. Unit: GeV^8. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- $\mathcal{T}_{\rm bulk}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\rm stab}$; ${\mathcal{T}_{\rm bulk}}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- ${\mathcal{S}^a}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.

The frozen extra times: the physical size $\mathrm{e}^{-a_4} b_4$ of the extra times is constant while ordinary space expands. Then the fold factor $\mathcal{Z}$ is constant (IN-8) and every coupling of the Standard Model is constant in time, as observation requires (the fine-structure constant has not changed by more than parts in 10^6 since the CMB); this is the condition of the late universe. It gives $b_4 = c_b\,\mathrm{e}^{a_4}$ with a constant c_b: b_4 grows with the scale factor of ordinary space, so it is a non-trivial function of x_4, never identically 1, as the author requires. The equations of FE-29 and FE-30 on this branch are written out (DERIVED, `um3_geometry.py`, G4, `um3_fold.py`, J7). For $\mathsf{A} \to 0$ they reduce to the static block FE-32. The extra energy of the fold above the static value is $(5/2)a_4'^2/(\kappa H)$ per V_*: on the separable metric the fold energy fixes the expansion rate. A fold fluid on top of the static values cannot be separately conserved on

the separable metric, for any equation of state: its continuity equation fails by $(3/8)a_4'^3/(\kappa H)$, the part of the jump of the bulk energy flux at the fold that the static source does not balance (DERIVED, J7), so ordinary matter on the fold requires the next terms of the power series (FE-35, FE-59). Status: DERIVED.

5.6 The general warped metric and the power series of the metric

$$
\begin{aligned}
&g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N = \mathrm{d}\sigma^2 + \mathrm{e}^{2\mathfrak{a}}\big(\mathrm{d}x_1^2+\mathrm{d}x_2^2+\mathrm{d}x_3^2\big) - \mathrm{e}^{2\mathfrak{n}}\,\mathrm{d}x_4^2 - \mathrm{e}^{2\mathfrak{b}}\big(\mathrm{d}x_5^2+\mathrm{d}x_6^2+\mathrm{d}x_7^2\big),\\
&R^{(0)(i)}{}_{(0)(i)} = -\partial_\sigma^2\mathfrak{a} - (\partial_\sigma\mathfrak{a})^2, \qquad R^{(0)(4)}{}_{(0)(4)} = -\partial_\sigma^2\mathfrak{n} - (\partial_\sigma\mathfrak{n})^2, \qquad R^{(i)(j)}{}_{(i)(j)} = -(\partial_\sigma\mathfrak{a})^2 + \mathrm{e}^{-2\mathfrak{n}}(\partial_{x_4}\mathfrak{a})^2,\\
&R^{(i)(r)}{}_{(i)(r)} = -\partial_\sigma\mathfrak{a}\,\partial_\sigma\mathfrak{b} + \mathrm{e}^{-2\mathfrak{n}}\,\partial_{x_4}\mathfrak{a}\,\partial_{x_4}\mathfrak{b}, \qquad R^{(i)(4)}{}_{(i)(4)} = -\partial_\sigma\mathfrak{a}\,\partial_\sigma\mathfrak{n} + \mathrm{e}^{-2\mathfrak{n}}\big(\partial_{x_4}^2\mathfrak{a} + (\partial_{x_4}\mathfrak{a})^2 - \partial_{x_4}\mathfrak{a}\,\partial_{x_4}\mathfrak{n}\big),\\
&R^{(0)(i)}{}_{(4)(i)} = -\mathrm{e}^{-\mathfrak{n}}\big(\partial_\sigma\partial_{x_4}\mathfrak{a} + \partial_\sigma\mathfrak{a}\,\partial_{x_4}\mathfrak{a} - \partial_{x_4}\mathfrak{a}\,\partial_\sigma\mathfrak{n}\big)
\end{aligned}
\tag{FE-34}
$$

Symbols in (FE-34).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\dots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \dots$): it runs over $x_0, x_1, \dots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \dots$), running over $x_0, \dots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \dots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \dots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $\mathfrak{a}$, $\mathfrak{a}_1$: The logarithm of the scale factor of ordinary space in the general warped metric FE-34, a function of σ and x_4; for the author's metric $\mathfrak{a} = a_4 - H|\sigma|$; $\mathfrak{a}_n$ its term of order n in the power series FE-35 ($\mathfrak{a}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $\mathfrak{n}$, $\mathfrak{n}_1$: The logarithm of the lapse in the general warped metric FE-34 ($g_{x_4x_4} = -\mathrm{e}^{2\mathfrak{n}}$); for the author's metric $\mathfrak{n} = 0$; $\mathfrak{n}_n$ its term of order n ($\mathfrak{n}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\mathfrak{b}$, $\mathfrak{b}_1$: The logarithm of the scale factor of the extra times in the general warped metric FE-34; for the author's metric $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$; $\mathfrak{b}_n$ its term of order n ($\mathfrak{b}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor (convention $[\nabla_P, \nabla_Q]V^M = R^M{}_{NPQ}V^N$), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- j: A second generation or mass-state index (as i). Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.

- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.

The general metric with the symmetries of the author's metric (homogeneous and isotropic in ordinary space and in the extra times, independent of the extra times) but with three arbitrary functions $\mathfrak{a}$, $\mathfrak{b}$, $\mathfrak{n}$ of σ and x_4, and its frame curvature: the components with the extra times follow from those with ordinary space by replacing $\mathfrak{a}$ by $\mathfrak{b}$ and (i) by (r) (${R^{(0)(r)}}_{(0)(r)}$, ${R^{(r)(s)}}_{(r)(s)}$, ${R^{(r)(4)}}_{(r)(4)}$, ${R^{(0)(r)}}_{(4)(r)}$); all others vanish (DERIVED, `um3_fold.py`, J2). Its Lovelock tensor is FE-2 with these components; it has only diagonal components and the (σ, x_4) pair. The author's metric is $\mathfrak{a} = a_4 - H|\sigma|$, $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$, $\mathfrak{n} = 0$, and the general equations reduce to FE-29 on both sides of the fold (DERIVED, J2). Status: DERIVED.

$$\begin{aligned} &\mathfrak{a} = a_4 - H|\sigma| + \sum_{n\geq 1} \epsilon_{\mathrm{r}}^n\, \mathfrak{a}_n(\sigma, x_4), \qquad \mathfrak{b} = \ln b_4 - a_4 - H|\sigma| + \sum_{n\geq 1} \epsilon_{\mathrm{r}}^n\, \mathfrak{b}_n(\sigma, x_4), \\ &\mathfrak{n} = \sum_{n\geq 1} \epsilon_{\mathrm{r}}^n\, \mathfrak{n}_n(\sigma, x_4), \qquad \hat{h}_{\mu\nu} \to \hat{h}_{\mu\nu} + \epsilon_{\mathrm{c}}\, \hat{h}^{(1)}_{\mu\nu} + \dots \end{aligned} \tag{FE-35}$$

Symbols in (FE-35).

- $\mathfrak{a}$, $\mathfrak{a}_1$: The logarithm of the scale factor of ordinary space in the general warped metric FE-34, a function of σ and x_4; for the author's metric $\mathfrak{a} = a_4 - H|\sigma|$; $\mathfrak{a}_n$ its term of order n in the power series FE-35 ($\mathfrak{a}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- n: The order $n = 1, 2, \dots$ of a term of the power series of the metric (also its exponent of ϵ_{r}); $\mathfrak{a}_n$, $\mathfrak{b}_n$, $\mathfrak{n}_n$ are the terms of order n. Status: STANDARD.
- ϵ_{r}: The bookkeeping parameter of the response of the bulk to the matter of the fold: the ratio of the bulk to the fold gravity, $\epsilon_{\mathrm{r}} = \kappa_4 V_*/(\kappa H)$. Unit: dimensionless. Value: $\lesssim 10^{-15}$ (an estimate: if a fraction ϵ_{r} of the gravitational waves of the fold travelled with the bulk gravitons, whose speed squared is 15/2 times that of light, their speed would differ from that of light by about $3\epsilon_{\mathrm{r}}$, and the gravitational waves and the light of the neutron-star merger GW170817 arrived together to about 10^{-15}). Status: DERIVED (the definition), ESTIMATE (the bound).
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\mathfrak{b}$, $\mathfrak{b}_1$: The logarithm of the scale factor of the extra times in the general warped metric FE-34; for the author's metric $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$; $\mathfrak{b}_n$ its term of order n ($\mathfrak{b}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4} b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- $\mathfrak{n}$, $\mathfrak{n}_1$: The logarithm of the lapse in the general warped metric FE-34 ($g_{x_4 x_4} = -\mathrm{e}^{2\mathfrak{n}}$); for the author's metric $\mathfrak{n} = 0$; $\mathfrak{n}_n$ its term of order n ($\mathfrak{n}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \dots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- ϵ_{c}: The bookkeeping parameter of the cosmological perturbations of the metric of the fold (the size of the CMB anisotropies). Unit: dimensionless. Value: $\sim 10^{-5}$. Status: STANDARD.

The metric as a power series of terms of decreasing magnitude (the author's request): the order-0 term is the author's metric; the terms of order n in ϵ_{r} are the response of the bulk to the matter of the fold and of dirac16complex, which

depends on σ; $\epsilon_c \sim 10^{-5}$ (the size of the CMB anisotropies) counts the cosmological perturbations of the metric of the fold. The bookkeeping parameter ϵ_r is the ratio of the bulk to the fold gravity, $\epsilon_r = \kappa_4 V_*/(\kappa H)$ (FE-59). Each order obeys the linearisation of FE-1 and FE-4 around the previous orders, with the sources of that order. Around the static block the static first-order terms obey linear equations with constant coefficients; if the bulk medium does not respond, their only solution is the difference mode $\mathfrak{a}_1' = -\mathfrak{b}_1' \propto \mathrm{e}^{6H|\sigma|}$, which grows away from the fold, while a responding medium admits the radion mode, whose jumps at the fold are in the ratio $[\mathfrak{a}_1'] : [\mathfrak{b}_1'] : [\mathfrak{n}_1'] = 1 : -1 : 5$ and change only the extra-time component of FE-4 (DERIVED, `um3_series.py`, SR1 to SR3; FE-59). Status: DERIVED (the scheme and the first-order modes).

6 dirac16complex: modes and Kohn-Sham equations

dirac16complex lives in the bulk; its four-dimensional particles are its modes along the hidden direction. This section writes the equation of those modes on the static block (the late universe; the primordial era replaces a_4, b_4 by their values there), the conditions at the fold, the states bound to the fold, and the Kohn-Sham equations that make the description self-consistent with the self-interaction, with gravity and with the mean fields of the Standard Model.

6.1 The equation of the hidden direction

$$\Psi = \mathrm{e}^{3H|\sigma|}\,\chi(\sigma)\,\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)}, \qquad \gamma^{(0)}\chi' - \mathrm{i}\omega\,\gamma^{(4)}\chi + \mathrm{i}\,\mathrm{e}^{H|\sigma|}\,\vec{k}\cdot\vec{\gamma}\,\chi - m_\Psi\chi = 0 \qquad (\sigma \neq 0) \tag{FE-36}$$

Symbols in (FE-36).

- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\vec{k}$: The (comoving) momentum along ordinary space, $\vec{k}^2$ its square. Unit: GeV. Status: STANDARD.
- $\vec{x}$: The position in ordinary space, $\vec{x} = (x_1, x_2, x_3)$. Unit: length. Status: AUTHOR.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\vec{\gamma}$: The three frame Dirac matrices of dirac16complex along ordinary space, $\vec{\gamma} = (\gamma^{(1)}, \gamma^{(2)}, \gamma^{(3)})$. Unit: dimensionless. Status: AUTHOR.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.

The free field equation FE-12 on the static block (a_4 and b_4 constant, ordinary space normalised so that $\mathrm{e}^{a_4} = 1$ at the fold) for a mode of definite momentum $\vec{k}$ along ordinary space and frequency ω, independent of the extra times. The factor $\mathrm{e}^{3H|\sigma|}$ removes the spin connection exactly ($\gamma^{(0)}\partial_\sigma \mathrm{e}^{3H|\sigma|} = 3H\,\mathrm{sgn}(\sigma)\gamma^{(0)}\mathrm{e}^{3H|\sigma|}$ cancels the first term of FE-14), so that the profile $\chi(\sigma)$ obeys a Dirac equation of one variable whose only dependence on σ is the warp $\mathrm{e}^{H|\sigma|}$ of the momentum. The four-dimensional norm density is $\sqrt{|g|}\,\Psi^\dagger\Psi = \chi^\dagger\chi$ (times b_4^3). Status: DERIVED.

$$\chi' = \mathbb{K}\,\chi, \qquad \mathbb{K} = \gamma^{(0)}\big(m_\Psi + \mathrm{i}\omega\gamma^{(4)}\big), \qquad \mathbb{K}^2 = \big(m_\Psi^2 - \omega^2\big)\mathbf{1}, \qquad \chi \propto \mathrm{e}^{\pm q\sigma}, \qquad q = \sqrt{m_\Psi^2 - \omega^2} \tag{FE-37}$$

Symbols in (FE-37).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\mathbb{K}$: The transfer matrix of the profile equation at zero momentum, $\chi' = \mathbb{K}\chi$. Unit: GeV. Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- $\mathbf{1}$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- q: The decay rate of a profile along σ at frequency ω, $q = \sqrt{m_\Psi^2 - \omega^2}$ (imaginary in the continuum). Unit: GeV. Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

At zero momentum along ordinary space ($\vec{k} = 0$) the profile equation has constant coefficients on each side of the fold: χ is a combination of $\mathrm{e}^{+q\sigma}$ and $\mathrm{e}^{-q\sigma}$ (eight solutions each). For $|\omega| \geq m_\Psi$, q is imaginary and the modes are waves along σ: the bulk continuum, which starts at the bulk mass m_Ψ. For $|\omega| < m_\Psi$ the solutions grow or decay exponentially, and only the fold terms (FE-39) can join decaying solutions on the two sides into a normalisable state. With chirality-dependent masses $m_\mathrm{L}P_\mathrm{L} + m_\mathrm{R}P_\mathrm{R}$ in place of m_Ψ the threshold would be $\sqrt{m_\mathrm{L}m_\mathrm{R}}$ (tachyonic for $m_\mathrm{L}m_\mathrm{R} < 0$); UM-3 keeps the record's single bulk mass and puts the difference of the halves into the fold terms (DERIVED, `um3_spectra.py`, S1, and `um3_bulk.py`, B4, with the author's matrices). Status: DERIVED.

$$\omega^2 = m_\Psi^2 + p_\sigma^2 + \mathrm{e}^{2H|\sigma|-2a_4}\,\vec{k}^2 - \mathrm{e}^{2H|\sigma|+2a_4}\,b_4^{-2}\,\vec{\kappa}^2 \tag{FE-38}$$

Symbols in (FE-38).

- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- p_σ: The momentum along the hidden direction (local wave number of a profile along σ). Unit: GeV. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $\vec{k}$: The (comoving) momentum along ordinary space, $\vec{k}^2$ its square. Unit: GeV. Status: STANDARD.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- $\vec{\kappa}$: A momentum along the extra times (three components); the fields of UM-3 have $\vec{\kappa} = 0$. Unit: GeV. Status: DERIVED.

The local dispersion relation of dirac16complex (WKB: for profiles that vary slowly compared with the warp), with the momentum p_σ along the hidden direction and a momentum $\vec{\kappa}$ along the extra times. A momentum along a

time-like direction enters with the negative sign: modes that depend on the extra times have $\omega^2 < 0$ for large $|\vec{\kappa}|$, they grow exponentially in x_4 (tachyons). The same holds for every field (the sign comes from $g^{rr} < 0$). UM-3 therefore keeps only the fields that do not depend on the extra times (ASSUMED); making the extra times small does not help, because it only raises the lowest non-zero $|\vec{\kappa}|$. Status: DERIVED (the dispersion, from the identity $(\mathcal{D}+m_\Psi)(\mathcal{D}-m_\Psi)=\mathcal{D}^2-m_\Psi^2$ of the Dirac operator $\mathcal{D}$), ASSUMED (the restriction); the stability of the theory with dependence on the extra times is OPEN.

6.2 The fold and the states bound to it

$$\chi(0^+) = \exp\Big(\gamma^{(0)}\big(\mu_{\rm L}P_{\rm L}+\mu_{\rm R}P_{\rm R}\big)\Big)\,\chi(0^-),$$
$$\mu_X = \frac{1}{V_*}\Big[c_\Phi\,\langle\Phi^\dagger\Phi\rangle\,\delta_{X{\rm L}} + \frac{1}{4}\big(c_X^G\langle G^2\rangle + c_X^W\langle W^2\rangle + c_X^B\langle B^2\rangle\big) + 2\omega\sum_f c_X^f\,n_f\Big], \qquad \langle\Phi^\dagger\Phi\rangle = \frac{v^2+\mathcal{A}^2}{2} \tag{FE-39}$$

Symbols in (FE-39).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- μ_X, $\mu_{\rm L}$, $\mu_{\rm R}$: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_{\rm L}P_{\rm L}+\mu_{\rm R}P_{\rm R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- $P_{\rm L}$: The author's left-handed chiral projector, $P_{\rm L} = (\mathbf{1}-\Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- $P_{\rm R}$: The author's right-handed chiral projector, $P_{\rm R} = (\mathbf{1}+\Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- X: The label of a half of dirac16complex: $X \in \{{\rm L},{\rm R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\rm t}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta_N^M = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X{\rm L}} = 1$ for $X = {\rm L}$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n\times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- $\langle G^2\rangle$: The mean square of the gluon field strength, $\langle G^A_{\mu\nu}G^{A\mu\nu}\rangle$ (the gluon condensate in the QCD vacuum, a thermal value in the plasma). Unit: GeV^4. Status: STANDARD.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1,\ldots,8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- $\langle W^2\rangle$: The mean square of the $SU(2)$ field strength, $\langle W^I_{\mu\nu}W^{I\mu\nu}\rangle$. Unit: GeV^4. Status: STANDARD.
- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- $\langle B^2\rangle$: The mean square of the hypercharge field strength, $\langle B_{\mu\nu}B^{\mu\nu}\rangle$. Unit: GeV^4. Status: STANDARD.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.

- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- n_f: The number density of the Standard-Model fermions of kind f (particles minus antiparticles, summed over generations and gauge components), the time component of the mean current $\langle J_f^\mu \rangle$ in the rest frame of the medium. Unit: GeV^3. Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_\mathrm{F})^{-1/2}$ with the Fermi constant $G_\mathrm{F} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_\mathcal{A} \mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.

The matching condition FE-13 for a mode, with the Standard-Model fields replaced by their mean values in a homogeneous, isotropic medium at rest on the fold (no mean field strength, no mean spatial current): the fold acts on dirac16complex as a chirality-dependent mass concentrated at $\sigma = 0$, with the dimensionless strengths μ_L and μ_R. The Higgs portal acts on the left-handed half only ($\delta_{X\mathrm{L}} = 1$ for $X = \mathrm{L}$, 0 for $X = \mathrm{R}$); the gauge-kinetic portals act through the mean squares $\langle G^2 \rangle = \langle G^A_{\mu\nu} G^{A\mu\nu} \rangle$ (the gluon condensate of the QCD vacuum, or of the plasma), $\langle W^2 \rangle$, $\langle B^2 \rangle$; the current portals act through the number densities n_f of the Standard-Model fermions of kind f (they give an energy-dependent strength). In the vacuum of the late universe only the Higgs term and the gluon condensate contribute. $(\gamma^{(0)}(\mu_\mathrm{L} P_\mathrm{L} + \mu_\mathrm{R} P_\mathrm{R}))^2 = \mu_\mathrm{L}\mu_\mathrm{R}\mathbf{1}$, so the exponential is $\cosh\varsigma + (\sinh\varsigma/\varsigma)\gamma^{(0)}(\mu_\mathrm{L} P_\mathrm{L} + \mu_\mathrm{R} P_\mathrm{R})$ with $\varsigma^2 = \mu_\mathrm{L}\mu_\mathrm{R}$; for $\mu_\mathrm{R} = 0$ it is exactly $1 + \mu_\mathrm{L}\gamma^{(0)} P_\mathrm{L}$ (DERIVED, `um3_spectra.py`, S2). Status: DERIVED.

$$q_\mathrm{b} = -m_\Psi \left(\mu_\mathrm{L} + \mu_\mathrm{R}\right) \frac{\tanh\varsigma}{2\varsigma}, \qquad \varsigma = \sqrt{\mu_\mathrm{L}\mu_\mathrm{R}}, \qquad \omega_\mathrm{b} = \sqrt{m_\Psi^2 - q_\mathrm{b}^2}, \qquad |\chi_\mathrm{b}(\sigma)| \propto \mathrm{e}^{-q_\mathrm{b}|\sigma|} \tag{FE-40}$$

Symbols in (FE-40).

- q_b: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- μ_X, μ_L, μ_R: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_\mathrm{L} P_\mathrm{L} + \mu_\mathrm{R} P_\mathrm{R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- ς: The auxiliary variable of the bound-state condition, $\varsigma = \sqrt{\mu_\mathrm{L}\mu_\mathrm{R}}$ ($\varsigma_- = \sqrt{-\mu_\mathrm{L}\mu_\mathrm{R}}$ when $\mu_\mathrm{L}\mu_\mathrm{R} < 0$). Unit: dimensionless. Status: DERIVED.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- χ_b: The profile of a state of dirac16complex bound to the fold. Unit: GeV^2. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

The states of dirac16complex bound to the fold at zero momentum: the decaying solutions of FE-37 on the two sides, joined by FE-39. A bound state exists if and only if $0 < q_\mathrm{b} < m_\Psi$, that is if the fold is attractive, $\mu_\mathrm{L} + \mu_\mathrm{R} < 0$ (for $\mu_\mathrm{L}\mu_\mathrm{R} < 0$, $\tanh(\varsigma)/\varsigma$ means $\tan(\varsigma_-)/\varsigma_-$ with $\varsigma_- = \sqrt{-\mu_\mathrm{L}\mu_\mathrm{R}}$). Its energy ω_b is the four-dimensional rest mass of a particle that lives near the fold, with the profile $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$ of width $1/q_\mathrm{b}$ along σ: the "x_0 mass" of dirac16complex. The bound states are eight-fold degenerate at $+\omega_\mathrm{b}$ and at $-\omega_\mathrm{b}$ (DERIVED, `um3_spectra.py`, S3: the roots of the 8×8 matching determinant with the author's 16×16 matrices agree with this formula to 10^{-7} in seven cases, with and without bound states). For the Higgs portal alone ($\mu_\mathrm{R} = 0$): $q_\mathrm{b} = -m_\Psi \mu_\mathrm{L}/2$, $\omega_\mathrm{b} = m_\Psi\sqrt{1 - \mu_\mathrm{L}^2/4}$. Status: DERIVED.

6.3 The Kohn-Sham equations

$$\begin{aligned}&\big[\gamma^M D_M - m_\Psi - \lambda_\Psi\, n_\mathrm{s} + \lambda_\Psi\, \mathcal{N}\big]\varphi_o = \delta_\mathrm{D}(\sigma)\,\frac{1}{V_*}\,\langle\mathbb{M}\rangle\,\varphi_o,\\ &\mathcal{N} = \sum_o \mathsf{n}_o\,\varphi_o\bar\varphi_o, \qquad n_\mathrm{s} = \mathrm{tr}\,\mathcal{N} = \sum_o \mathsf{n}_o\,\bar\varphi_o\varphi_o, \qquad \mathsf{n}_o = \frac{1}{\mathrm{e}^{(E_o-\mu_\Psi)/\mathsf{T}_\Psi}+1}\end{aligned} \tag{FE-41}$$

Symbols in (FE-41).

- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- λ_Ψ: The coupling of the contact self-interaction of dirac16complex. Unit: GeV^{-6}. Status: RECORD.
- n_s: The scalar density of dirac16complex in the Kohn-Sham state, $n_\mathrm{s} = \langle\bar\Psi\Psi\rangle = \mathrm{tr}\,\mathcal{N}$. Unit: GeV^7. Status: DERIVED.
- $\mathcal{N}$: The one-body density matrix of dirac16complex, $\mathcal{N} = \sum_o \mathsf{n}_o\varphi_o\bar\varphi_o$ (a 16×16 matrix at each point). Unit: GeV^7. Status: DERIVED.
- φ_o, $\bar\varphi_o$: A Kohn-Sham orbital of dirac16complex (a single-particle solution of FE-41) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- o: The label of a Kohn-Sham orbital (a single-particle state of dirac16complex); a sum over o runs over the orbitals. Status: STANDARD.
- δ_D: The Dirac delta function: $\int \mathrm{d}\sigma\,\delta_\mathrm{D}(\sigma)F(\sigma) = F(0)$. Unit: 1/length (for $\delta_\mathrm{D}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\mathbb{M}$: The fold operator of dirac16complex: the 16×16 matrix (with a derivative along the fold) built from the Standard-Model fields at the fold by varying the portals with respect to $\bar\Psi$ (FE-13). Unit: GeV^{-3} (so that $\mathbb{M}/V_*$ is dimensionless). Status: DERIVED.
- n_o: The occupation of the Kohn-Sham orbital o (between 0 and 1). Unit: dimensionless. Status: RECORD.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- μ_Ψ: The chemical potential of dirac16complex. Unit: GeV. Status: DERIVED.
- T_Ψ: The temperature of dirac16complex. Unit: GeV. Status: DERIVED.

The Kohn-Sham equations of dirac16complex coupled to itself, to gravity and to every Standard-Model field: the quantised field is described by single-particle orbitals φ_o with the occupations n_o (Fermi-Dirac with the temperature T_Ψ and the chemical potential μ_Ψ of dirac16complex; $\mathsf{n}_o = 0$ or 1 in a degenerate state). Each orbital obeys the field equation with the mean fields: gravity through D_M and γ^M (the background metric and its spin connection), the self-interaction through the Hartree term $\lambda_\Psi n_\mathrm{s}$ (the scalar density) and the exact exchange term $-\lambda_\Psi\mathcal{N}$ (the one-body density matrix, a 16×16 matrix at each point), and every Standard-Model field through the fold operator with the mean fields of the Standard Model, $\langle\mathbb{M}\rangle$ (FE-13, FE-39). The mean-field decomposition of $-\frac{\lambda_\Psi}{2}(\bar\Psi\Psi)^2$ is $-\lambda_\Psi n_\mathrm{s}\bar\Psi\Psi + \lambda_\Psi\bar\Psi\mathcal{N}\Psi$ (the second term with the sign of fermion exchange); correlation beyond exchange is neglected and the negative-energy sea is not included (no-sea approximation). Status: RECORD (the Kohn-Sham treatment),

DERIVED (the exchange term), ASSUMED (no correlation, no sea).

$$\langle \mathcal{T}_{\Psi\,MN} \rangle = -\frac{1}{4}\sum_o \mathsf{n}_o \Big(\bar\varphi_o \gamma_M D_N \varphi_o + \bar\varphi_o \gamma_N D_M \varphi_o - (D_M \bar\varphi_o)\gamma_N \varphi_o - (D_N \bar\varphi_o)\gamma_M \varphi_o \Big) + \frac{\lambda_\Psi}{2}\, g_{MN} \big(n_{\mathrm{s}}^2 - \operatorname{tr} \mathcal{N}^2 \big) \quad \text{(FE-42)}$$

Symbols in (FE-42).

- $\langle \cdots \rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex, $\mathcal{T}_{MN} = -(2/\sqrt{|g|})\,\delta S_\Psi / \delta g^{MN}$. Unit: GeV^8. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- o: The label of a Kohn-Sham orbital (a single-particle state of dirac16complex); a sum over o runs over the orbitals. Status: STANDARD.
- n_o: The occupation of the Kohn-Sham orbital o (between 0 and 1). Unit: dimensionless. Status: RECORD.
- φ_o, $\bar\varphi_o$: A Kohn-Sham orbital of dirac16complex (a single-particle solution of FE-41) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- λ_Ψ: The coupling of the contact self-interaction of dirac16complex. Unit: GeV^{-6}. Status: RECORD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- n_{s}: The scalar density of dirac16complex in the Kohn-Sham state, $n_{\mathrm{s}} = \langle\bar\Psi\Psi\rangle = \operatorname{tr}\mathcal{N}$. Unit: GeV^7. Status: DERIVED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- $\mathcal{N}$: The one-body density matrix of dirac16complex, $\mathcal{N} = \sum_o \mathsf{n}_o \varphi_o \bar\varphi_o$ (a 16×16 matrix at each point). Unit: GeV^7. Status: DERIVED.

The energy-momentum tensor of the Kohn-Sham state, the source of dirac16complex in FE-1 and, through the portals, in FE-5: the sum over the occupied orbitals of FE-6, and the mean value of the self-interaction (on solutions of FE-41, $\langle\mathcal{L}_\Psi\rangle = \frac{\lambda_\Psi}{2}(n_{\mathrm{s}}^2 - \operatorname{tr}\mathcal{N}^2)$). It closes the loop: the orbitals determine the densities, the densities the mean fields and the metric, the metric and the mean fields the orbitals. Status: DERIVED.

$$Q_\Psi = \int \mathrm{d}\sigma\, \mathrm{d}^3x\, \sqrt{|g|}\; \Psi^\dagger \mathfrak{B} \Psi, \qquad \mathfrak{B} = \mathrm{i}\,\sigma_{16}\,\gamma^{(4)}, \qquad \mathfrak{B}^\dagger = \mathfrak{B}, \qquad \mathfrak{B}^2 = \mathbf{1}, \qquad E_o = -\omega_o\,\mathcal{Q}_o \qquad \text{(FE-43)}$$

Symbols in (FE-43).

- Q_Ψ: The conserved charge of dirac16complex (the Noether charge of a change of the phase of Ψ by a constant), a Krein form (indefinite). Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \operatorname{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- x: The coordinates of the fold: x_i $(i = 1,2,3)$ of ordinary space, x_4 the time, x_r $(r = 5,6,7)$ the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\dots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi=\Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma=0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathfrak{B}$: The Krein operator, $\mathfrak{B}=\mathrm{i}\sigma_{16}\gamma^{(4)}$: a Hermitian involution with the signature (8, 8); $\Psi^\dagger\mathfrak{B}\Psi$ is the charge density. Unit: dimensionless. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- σ_{16}: The matrix of the Dirac adjoint, $\sigma_{16}=T_0T_1T_2T_3$, real and symmetric; $\sigma_{16}T_a$ is real and antisymmetric. Unit: dimensionless. Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M=\mathbf{e}_a{}^MT^a$, $\gamma_M=g_{MN}\gamma^N$; $\gamma^{(a)}=T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- o: The label of a Kohn-Sham orbital (a single-particle state of dirac16complex); a sum over o runs over the orbitals. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- $\mathcal{Q}_o$: The charge Q_Ψ of the Kohn-Sham orbital o (positive or negative). Unit: dimensionless. Status: DERIVED.

The conserved charge of dirac16complex (the Noether charge of a constant change of the phase of Ψ, whose density is $\mathrm{i}\bar\Psi\gamma^{x_4}\Psi$) is a Krein form: $\mathfrak{B}$ is a Hermitian involution with eight eigenvalues $+1$ and eight eigenvalues -1, so the charge, and with it the norm of the quantised field, is indefinite (the consequence of the signature (4,4); DERIVED, `um3_fields.py`, F3). $\mathfrak{B}$ commutes with $\gamma^{(0)},\dots,\gamma^{(4)}$ and anticommutes with $\gamma^{(5)},\gamma^{(6)},\gamma^{(7)}$ and with Γ_8: every term that treats the two halves differently (here: the fold terms with $\mu_\mathrm{L}\neq\mu_\mathrm{R}$) mixes the two Krein sectors. The energy of a stationary orbital of frequency ω_o and charge $\mathcal{Q}_o$ is $E_o=-\omega_o\mathcal{Q}_o$ (from FE-6), positive when the frequency and the charge have opposite signs. Of the eight bound states at $+\omega_\mathrm{b}$, four have positive and four negative charge (DERIVED, `um3_spectra.py`, S4; no null states): four carry positive and four negative energy. The stability of the quantised theory with both kinds of states is OPEN. Status: DERIVED.

7 The Standard Model on the fold

7.1 The four-dimensional couplings

$$g_{1,\mathrm{eff}}=\frac{g_1}{\sqrt{\mathcal{Z}}},\quad g_{2,\mathrm{eff}}=\frac{g_2}{\sqrt{\mathcal{Z}}},\quad g_{3,\mathrm{eff}}=\frac{g_3}{\sqrt{\mathcal{Z}}},\qquad \lambda_\mathrm{eff}=\frac{\lambda}{\mathcal{Z}},$$
$$\mathcal{Y}_\mathrm{eff}=\frac{\mathcal{Y}}{\sqrt{\mathcal{Z}}},\qquad G_{\mathrm{N,eff}}=\frac{G_\mathrm{N}}{\mathcal{Z}},\qquad \mathcal{Z}=\frac{V_\mathrm{t}\left(\mathrm{e}^{-a_4}b_4\right)^3}{V_*} \tag{FE-44}$$

Symbols in (FE-44).

- $g_{k,\mathrm{eff}}$: The effective (canonically normalised) gauge couplings of the fold when the fold factor is $\mathcal{Z}$, $g_{k,\mathrm{eff}}=g_k/\sqrt{\mathcal{Z}}$ ($k=1,2,3$). Unit: dimensionless. Status: DERIVED.
- g_1: The $U(1)_Y$ gauge coupling (usually g'; not normalised for grand unification). Unit: dimensionless. Value: $g_1(m_Z)=0.3574$. Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling (usually g). Unit: dimensionless. Value: $g_2(m_Z)=0.6517$. Status: STANDARD.
- g_3: The $SU(3)_C$ gauge coupling (the strong coupling g_s). Unit: dimensionless. Value: $g_3(m_Z)=1.2177$ ($\alpha_s=0.1180$). Status: STANDARD.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4)=V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z}=1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- λ_eff: The effective quartic Higgs coupling, $\lambda/\mathcal{Z}$. Unit: dimensionless. Status: DERIVED.

- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- $\mathcal{Y}_{\text{eff}}$: The effective Yukawa matrices, $\mathcal{Y}/\sqrt{\mathcal{Z}}$ (for each of $\mathcal{Y}^u, \mathcal{Y}^d, \mathcal{Y}^e, \mathcal{Y}^\nu$). Unit: dimensionless. Status: DERIVED.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $G_{\text{N,eff}}$: The effective Newton constant of the fold, $G_{\text{N}}/\mathcal{Z}$. Unit: GeV^{-2}. Status: DERIVED.
- G_{N}: Newton's constant of four-dimensional gravity on the fold, $G_{\text{N}} = 1/M_{\text{Pl}}^2$. Unit: GeV^{-2}. Value: $6.708{\times}10^{-39}$ GeV^{-2}. Status: STANDARD.
- V_{t}: The coordinate volume of the (compact) extra times, $V_{\text{t}} = \int \mathrm{d}x_5\, \mathrm{d}x_6\, \mathrm{d}x_7$. Unit: length^3. Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4} b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\text{t}}(\mathrm{e}^{-a_4} b_4)^3$ at a time when $\mathrm{e}^{-a_4} b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.

When the extra times have the size $\mathrm{e}^{-a_4} b_4$, every term of the fold carries the factor $\mathcal{Z}$ (IN-8); rescaling every field of the fold by $\sqrt{\mathcal{Z}}$ restores the canonical normalisation and leaves the effective couplings of FE-44, while the masses (products of a coupling and a vacuum value) are unchanged. In the late universe $\mathcal{Z} = 1$ (frozen extra times, FE-33); in the primordial era at the author's vacuum $\mathrm{e}^{-a_4} b_4 \propto \mathrm{e}^{-Hx_4}$, so $\mathcal{Z} \propto \mathrm{e}^{-3Hx_4}$ and all couplings grow as $\mathrm{e}^{3Hx_4/2}$ until the extra times freeze. Status: DERIVED.

7.2 The electroweak vacuum with the ad hoc component

$$m_{\text{W}}^2 = \frac{g_2^2\,(v^2+\mathcal{A}^2)}{4}, \qquad m_{\text{Z}}^2 = \frac{(g_1^2+g_2^2)(v^2+\mathcal{A}^2)}{4}, \qquad \frac{G_{\text{F}}}{\sqrt{2}} = \frac{g_2^2}{8m_{\text{W}}^2} = \frac{1}{2(v^2+\mathcal{A}^2)}, \qquad m_h^2 = 2\lambda v^2,$$
$$A_\mu = \frac{g_2\,B_\mu - g_1\,n^I W_\mu^I}{\sqrt{g_1^2+g_2^2}}, \qquad n^I = \frac{\left(2\mathcal{A}v,\ 0,\ \mathcal{A}^2 - v^2\right)}{\mathcal{A}^2+v^2} \tag{FE-45}$$

Symbols in (FE-45).

- m_{W}: The mass of the W bosons. Unit: GeV. Value: 80.37 GeV ($\mathcal{A} = 0$). Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling (usually g). Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_{\text{F}})^{-1/2}$ with the Fermi constant $G_{\text{F}} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_{\mathcal{A}}\mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- m_{Z}: The mass of the Z boson. Unit: GeV. Value: 91.19 GeV ($\mathcal{A} = 0$). Status: STANDARD.
- g_1: The $U(1)_Y$ gauge coupling (usually g'; not normalised for grand unification). Unit: dimensionless. Value: $g_1(m_Z) = 0.3574$. Status: STANDARD.
- G_{F}: The Fermi constant, $G_{\text{F}} = 1/(\sqrt{2}\,v^2)$ in the Standard-Model vacuum (its measured value); in FE-45 the Fermi constant of the ad hoc vacuum, which is $G_{\text{F,loc}}$ of FE-49. Unit: GeV^{-2}. Value: 1.1664×10^{-5} GeV^{-2}. Status: STANDARD.
- m_h: The mass of the Higgs boson. Unit: GeV. Value: 125.25 GeV (Particle Data Group 2022). Status: STANDARD.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- A_μ: The photon field: the massless combination of W^1, W^2, W^3 and B (canonically normalised). Unit: GeV. Status: STANDARD.

- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- n^I: The unit isovector of the ad hoc vacuum, $n = (2\mathcal{A}v, 0, \mathcal{A}^2 - v^2)/(\mathcal{A}^2 + v^2)$, which fixes which combination of W^1, W^2, W^3 joins B in the photon. Unit: dimensionless. Value: $(0, 0, -1)$ for $\mathcal{A} = 0$. Status: DERIVED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.

The masses of the gauge bosons and the photon in the author's ad hoc vacuum (the eigenvalues and eigenvectors of the mass matrix of W^1, W^2, W^3, B from $|D_\mu\Phi|^2$ with $\Phi = (\mathcal{A}, v)/\sqrt{2}$; DERIVED, `um3_fields.py`, F5): the vacuum value that breaks the electroweak symmetry is $\sqrt{v^2 + \mathcal{A}^2}$, and the unbroken direction, the photon A_μ, is rotated from the usual combination by the unit isovector n^I (for $\mathcal{A} = 0$: $n = (0, 0, -1)$ and $A_\mu = (g_2 B_\mu + g_1 W^3_\mu)/\sqrt{g_1^2 + g_2^2}$). The Fermi constant measured in muon decay is that of the total vacuum value. As $\mathcal{A}$ decays (rate $2a_{40}$), the electroweak vacuum becomes the Standard-Model vacuum. The Higgs mass is that of the Standard-Model vacuum ($\mathcal{A} \to 0$). Status: DERIVED.

7.3 The masses and the mixing of quarks and leptons

$$
\begin{aligned}
&\frac{v}{\sqrt{2}}\,\mathcal{Y}^u = U_{u\mathrm{L}}\,\mathrm{diag}(m_u, m_c, m_t)\,U^\dagger_{u\mathrm{R}}, \qquad \frac{v}{\sqrt{2}}\,\mathcal{Y}^d = U_{d\mathrm{L}}\,\mathrm{diag}(m_d, m_s, m_b)\,U^\dagger_{d\mathrm{R}},\\
&\frac{v}{\sqrt{2}}\,\mathcal{Y}^e = U_{e\mathrm{L}}\,\mathrm{diag}(m_e, m_\mu, m_\tau)\,U^\dagger_{e\mathrm{R}}, \qquad \frac{v}{\sqrt{2}}\,\mathcal{Y}^\nu = U_{\nu\mathrm{L}}\,\mathrm{diag}(m_1, m_2, m_3)\,U^\dagger_{\nu\mathrm{R}},\\
&V_{\mathrm{CKM}} = U^\dagger_{u\mathrm{L}} U_{d\mathrm{L}}, \qquad U_{\mathrm{PMNS}} = U^\dagger_{e\mathrm{L}} U_{\nu\mathrm{L}}, \qquad \nu_\alpha = \sum_{i=1}^{3} \big(U_{\mathrm{PMNS}}\big)_{\alpha i}\,\nu_i,\\
&\mathcal{L}_{\mathrm{CC}} = \frac{g_2}{\sqrt{2}}\,W^+_\mu \Big(\bar{\hat{u}}_{\mathrm{L}}\,\hat{\gamma}^\mu V_{\mathrm{CKM}}\,\hat{d}_{\mathrm{L}} + \sum_\alpha \bar{\nu}_\alpha\,\hat{\gamma}^\mu \hat{e}_{\mathrm{L}\alpha}\Big) + \mathrm{h.c.}, \qquad W^\pm_\mu = \frac{W^1_\mu \mp \mathrm{i} W^2_\mu}{\sqrt{2}}
\end{aligned}
\tag{FE-46}
$$

Symbols in (FE-46).

- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_{\mathrm{F}})^{-1/2}$ with the Fermi constant $G_{\mathrm{F}} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $U_{f\mathrm{L}}$, $U_{f\mathrm{R}}$: The unitary 3×3 matrices that diagonalise the mass matrices of the up quarks, the down quarks, the charged leptons and the neutrinos (left-handed and right-handed rotations). Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- $m_u, \ldots, m_3$: The masses of the quarks (m_u, m_c, m_t; m_d, m_s, m_b), of the charged leptons (m_e, m_μ, m_τ) and of the three neutrino mass states (m_1, m_2, m_3, Dirac masses). Unit: GeV. Value: measured values; $\Delta m^2_{21} = 7.41 \times 10^{-5}$ eV^2, $|\Delta m^2_{31}| = 2.511 \times 10^{-3}$ eV^2 (the global fit NuFIT 5.2 of the oscillation data, November 2022). Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- V_{CKM}: The quark mixing matrix (Cabibbo-Kobayashi-Maskawa). Unit: dimensionless. Status: STANDARD.
- U_{PMNS}: The lepton mixing matrix (Pontecorvo-Maki-Nakagawa-Sakata) of the three Dirac neutrinos; $U_{\alpha i}$ are its entries. Unit: dimensionless. Value: $\theta_{12} = 33.40°$, $\theta_{23} = 49.14°$, $\theta_{13} = 8.54°$ (NuFIT 5.2, November 2022, normal ordering, with the Super-Kamiokande atmospheric data); the Dirac phase is set to zero (ASSUMED; the fit prefers 197°). Status: STANDARD, ASSUMED (the phase).
- ν_α, ν_i: A neutrino field: ν_α the flavour states ($\alpha = e, \mu, \tau$), ν_i the mass states ($i = 1, 2, 3$). Unit: GeV$^{3/2}$. Status: STANDARD.
- α: A lepton-flavour index, $\alpha \in \{e, \mu, \tau\}$. Status: STANDARD.

- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\mathcal{L}_{\mathrm{CC}}$: The charged-current interaction of the quarks and leptons with the W bosons. Unit: GeV^4. Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling (usually g). Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $\hat{u}_{\mathrm{L}}$, $\hat{d}_{\mathrm{L}}$, $\hat{e}_{\mathrm{L}\alpha}$: The left-handed quarks and charged leptons in the mass basis ($\hat{e}_{\mathrm{L}\alpha}$ the charged lepton of flavour α). Unit: GeV$^{3/2}$. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu \hat{\gamma}^{(a)}$, $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- h.c.: The Hermitian conjugate of the preceding terms. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.

The Dirac masses of the quarks, the charged leptons and the three neutrinos (in the vacuum with $\mathcal{A} = 0$; with $\mathcal{A} \neq 0$ the same with the rotated vacuum), the unitary matrices that diagonalise the Yukawa matrices, the quark mixing matrix V_{CKM} and the lepton mixing matrix U_{PMNS}, and the charged-current interaction in the mass basis (fields with a hat; the neutrino flavour states ν_α are the partners of the charged leptons of definite mass). This is the flavour structure of the Standard Model with three Dirac neutrinos; the portals of dirac16complex add nothing to it (they are flavour blind, IN-16). The values of the masses and mixing angles are the measured ones (STANDARD inputs). Status: STANDARD.

7.4 The four-dimensional couplings of the bound states of dirac16complex

$$\mathfrak{g}_\Phi = \frac{\partial\omega_{\mathrm{b}}}{\partial\mu_{\mathrm{L}}}\frac{c_\Phi}{V_*}, \qquad \mathfrak{g}_G = \sum_X \frac{\partial\omega_{\mathrm{b}}}{\partial\mu_X}\frac{c^G_X}{4V_*}, \qquad \mathfrak{g}_f = \sum_X \frac{\partial\omega_{\mathrm{b}}}{\partial\mu_X}\frac{2\omega_{\mathrm{b}}\,c^f_X}{V_*}, \qquad \frac{\partial\omega_{\mathrm{b}}}{\partial\mu_X} = -\frac{q_{\mathrm{b}}}{\omega_{\mathrm{b}}}\frac{\partial q_{\mathrm{b}}}{\partial\mu_X},$$
$$\Delta\omega_{\mathrm{b}} = \mathfrak{g}_\Phi\,\Delta\langle\Phi^\dagger\Phi\rangle + \mathfrak{g}_G\,\Delta\langle G^2\rangle + \sum_f \mathfrak{g}_f\,\Delta n_f, \qquad \mathfrak{g}_{h\chi} = \mathfrak{g}_\Phi\, v = -\frac{m^2_\Psi\,\mu^2_{\mathrm{L}}}{2\,\omega_{\mathrm{b}}\,v} \quad (\mu_{\mathrm{R}} = 0) \tag{FE-47}$$

Symbols in (FE-47).

- $\mathfrak{g}_\Phi$, $\mathfrak{g}_G$, $\mathfrak{g}_f$, $\mathfrak{g}_{h\chi}$: The four-dimensional couplings of the bound states χ of dirac16complex to the Standard Model (FE-47): to $\Phi^\dagger\Phi$, to the gluon kinetic density, to the number densities of the fermions of kind f, and the Yukawa coupling of the Higgs boson to χ. Unit: $\mathfrak{g}_\Phi$: GeV^{-1}; $\mathfrak{g}_G$: GeV^{-3}; $\mathfrak{g}_f$: GeV^{-2}; $\mathfrak{g}_{h\chi}$: dimensionless. Status: DERIVED.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- ω_{b}: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- μ_X, μ_{L}, μ_{R}: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_{\mathrm{L}}P_{\mathrm{L}} + \mu_{\mathrm{R}}P_{\mathrm{R}})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- c^G_X: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c^f_X: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c^ℓ_X and c^ν_X are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.

- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- q_b: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- Δ: A change, difference or small variation of the quantity that follows ($\Delta\rho$ a perturbation, Δm^2 a difference of squared masses, ΔN_eff an addition to N_eff). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\langle G^2\rangle$: The mean square of the gluon field strength, $\langle G^A_{\mu\nu} G^{A\mu\nu}\rangle$ (the gluon condensate in the QCD vacuum, a thermal value in the plasma). Unit: GeV^4. Status: STANDARD.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- n_f: The number density of the Standard-Model fermions of kind f (particles minus antiparticles, summed over generations and gauge components), the time component of the mean current $\langle J^\mu_f\rangle$ in the rest frame of the medium. Unit: GeV^3. Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_\mathrm{F})^{-1/2}$ with the Fermi constant $G_\mathrm{F} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.

The couplings of the bound states (the particles χ of mass ω_b) to the Standard Model, by the theorem of Feynman and Hellmann: a slowly varying Standard-Model field shifts the strengths μ_X of FE-39 and with them the energy of the bound state, so the coupling of the bound state to a Standard-Model operator is the derivative of ω_b with respect to its mean value (valid for momentum transfers small compared with q_b). $\mathfrak{g}_{h\chi}$ is the Yukawa coupling of the Higgs boson h to the bound state ($\Delta\langle\Phi^\dagger\Phi\rangle = vh$), here for the Higgs portal alone; it is fixed by the mass of the bound state and the bulk mass. The couplings to W and B follow like $\mathfrak{g}_G$; the dipole portal gives the bound states magnetic and electric dipole moments. These couplings govern the annihilation (FE-61), the scattering on nuclei and the production of the bound states. Status: DERIVED.

8 The three neutrino flavours

The neutrinos are the upper components ν_L of the three lepton doublets on the fold; with the three right-handed neutrinos they have Dirac masses and the lepton mixing of FE-46. Their flavour conversion follows from FE-22 and FE-24 for neutrinos that propagate through a medium of electrons, nucleons, other neutrinos and dirac16complex, by the standard reduction of the Dirac equations of ultrarelativistic neutrinos to equations for their flavour content (the mean-field quantum kinetic equations of Sigl and Raffelt, 1993). The author asks for correct flavour conversion because it shapes the fate of core-collapse supernovae: Gogilashvili and Tamborra ("Neutrino Flavor Conversion Shapes the Rate of Failed Core-collapse Supernovae", arXiv:2605.16504v3) simulate the collapse of 195 stars of 9 to 120 solar masses and find that flavour equipartition below a critical baryon density raises the fraction of failed explosions from 25.6% (without flavour conversion) to 50.8%, 77.4%, 93.3% and 96.4% for critical densities of 10^9, 10^{10}, 10^{11} and 10^{13} g/cm^3 (from 27.0% to between 47.6% and 88.3% when the stars are weighted with the Salpeter initial mass function). This section writes the equations of motion of the flavour content and shows exactly which terms UM-3 adds to those of the Standard Model.

8.1 The equations of motion of the flavour content

$$
\begin{aligned}
&\mathrm{i}\big(\partial_{x_4} + \hat{v}\cdot\vec{\nabla}\big)\,\varrho_{\vec{p}} = \big[\mathsf{H}_{\vec{p}},\ \varrho_{\vec{p}}\big] + \mathrm{i}\,\mathcal{C}_{\vec{p}}, \qquad \mathrm{i}\big(\partial_{x_4} + \hat{v}\cdot\vec{\nabla}\big)\,\bar{\varrho}_{\vec{p}} = \big[\bar{\mathsf{H}}_{\vec{p}},\ \bar{\varrho}_{\vec{p}}\big] + \mathrm{i}\,\bar{\mathcal{C}}_{\vec{p}},\\
&\mathsf{H}_{\vec{p}} = \frac{v_\mathrm{loc}^2}{v^2}\,\frac{U_\mathrm{PMNS}\,\mathsf{M}^2\,U_\mathrm{PMNS}^\dagger}{2E} + \mathsf{V}_\mathrm{m} + \mathsf{V}_{\nu\nu} + V_\Psi\mathbf{1}, \qquad \mathsf{M}^2 = \mathrm{diag}\big(m_1^2,\ m_2^2,\ m_3^2\big),\\
&\bar{\mathsf{H}}_{\vec{p}} = -\frac{v_\mathrm{loc}^2}{v^2}\,\frac{U_\mathrm{PMNS}\,\mathsf{M}^2\,U_\mathrm{PMNS}^\dagger}{2E} + \mathsf{V}_\mathrm{m} + \mathsf{V}_{\nu\nu} + V_\Psi\mathbf{1}
\end{aligned}
\tag{FE-48}
$$

Symbols in (FE-48).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\hat{v}$, $\hat{v}^\mu$: The direction of motion of a neutrino: the unit vector $\hat{v} = \vec{p}/|\vec{p}|$ ($\hat{v}_{\vec{p}}$, $\hat{v}_{\vec{q}}$ for the momenta $\vec{p}$, $\vec{q}$), and the four-vector $\hat{v}^\mu = (1, \hat{v})$ in the order $(x_4; x_1, x_2, x_3)$; $\mathrm{d}^2\hat{v}$ is the element of solid angle of the directions. Unit: dimensionless. Status: STANDARD.
- $\vec{\nabla}$: The gradient along ordinary space, $\vec{\nabla} = (\partial_{x_1}, \partial_{x_2}, \partial_{x_3})$ (in the local frame of the medium). Unit: 1/length. Status: STANDARD.
- $\varrho_{\vec{p}}$, $\bar{\varrho}_{\vec{p}}$: The 3×3 flavour density matrices of the neutrinos and of the antineutrinos of momentum $\vec{p}$ in the basis e, μ, τ: the diagonal entries are the occupation numbers of the three flavours, the off-diagonal entries the coherence between the flavours; $\bar{\varrho}$ is defined with the flavour indices in the reversed order, so that it obeys an equation of the same form. Unit: dimensionless. Status: STANDARD.
- $\vec{p}$: The momentum of a neutrino (three components along ordinary space, in the rest frame of the medium); $|\vec{p}| = E$ for a neutrino of negligible mass. Unit: GeV. Status: STANDARD.
- $\mathsf{H}_{\vec{p}}$, $\bar{\mathsf{H}}_{\vec{p}}$: The flavour Hamiltonians (3×3 Hermitian matrices) of the neutrinos and of the antineutrinos of momentum $\vec{p}$. Unit: GeV. Status: STANDARD.
- $\mathcal{C}_{\vec{p}}$, $\bar{\mathcal{C}}_{\vec{p}}$: The collision terms ($3 \times 3$ matrices) of the neutrinos and the antineutrinos of momentum $\vec{p}$: the rates of change of $\varrho_{\vec{p}}$ and $\bar{\varrho}_{\vec{p}}$ by emission, absorption, scattering and pair processes with the matter. Unit: GeV. Status: STANDARD.
- v_{loc}: The local value of the electroweak vacuum, $v_{\mathrm{loc}}^2 = 2\langle\Phi^\dagger\Phi\rangle$: it sets the local masses of the quarks and leptons (each mass times v_{loc}/v) and the local Fermi constant; $v_{\mathrm{loc}} = v$ in the Standard-Model vacuum. Unit: GeV. Value: v in the late universe outside dense dirac16complex. Status: DERIVED.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_{\mathrm{F}})^{-1/2}$ with the Fermi constant $G_{\mathrm{F}} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- U_{PMNS}: The lepton mixing matrix (Pontecorvo-Maki-Nakagawa-Sakata) of the three Dirac neutrinos; $U_{\alpha i}$ are its entries. Unit: dimensionless. Value: $\theta_{12} = 33.40°$, $\theta_{23} = 49.14°$, $\theta_{13} = 8.54°$ (NuFIT 5.2, November 2022, normal ordering, with the Super-Kamiokande atmospheric data); the Dirac phase is set to zero (ASSUMED; the fit prefers 197°). Status: STANDARD, ASSUMED (the phase).
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- M^2: The diagonal matrix of the squared masses of the three neutrino mass states, $\mathsf{M}^2 = \mathrm{diag}(m_1^2, m_2^2, m_3^2)$ (M appears only squared). Unit: GeV^2 (of M^2). Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- V_{m}: The matter potential of the neutrinos (a 3×3 matrix in flavour space) from their forward scattering on the electrons, positrons and neutrons of the medium. Unit: GeV. Status: STANDARD.
- $\mathsf{V}_{\nu\nu}$: The neutrino-neutrino potential (a 3×3 matrix in flavour space) from the forward scattering of a neutrino on the other neutrinos and antineutrinos of the medium; it depends on their flavour content and their directions. Unit: GeV. Status: STANDARD.
- V_Ψ: The potential energy that dirac16complex gives every neutrino through the current portal of the lepton doublets, $V_\Psi = \mathcal{V}^\ell_{x_4}$ (the same for the three flavours). Unit: GeV. Status: DERIVED.
- $\mathbf{1}$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- $m_u, \ldots, m_3$: The masses of the quarks (m_u, m_c, m_t; m_d, m_s, m_b), of the charged leptons (m_e, m_μ, m_τ) and of the three neutrino mass states (m_1, m_2, m_3, Dirac masses). Unit: GeV. Value: measured values; $\Delta m^2_{21} = 7.41 \times 10^{-5}$ eV^2, $|\Delta m^2_{31}| = 2.511 \times 10^{-3}$ eV^2 (the global fit NuFIT 5.2 of the oscillation data, November 2022). Status: STANDARD.

The equations of motion of the 3×3 flavour density matrices of the neutrinos ($\varrho_{\vec{p}}$) and of the antineutrinos ($\bar{\varrho}_{\vec{p}}$,

defined with the flavour indices in the reversed order) of momentum $\vec{p}$, energy $E = |\vec{p}|$ and direction $\hat{v}$; the diagonal entries are the occupation numbers of the three flavours, the off-diagonal entries the coherence between them. Free streaming stands on the left, the commutator with the flavour Hamiltonian and the collisions $\mathcal{C}$ (emission, absorption, scattering and pair processes) on the right. The Hamiltonian contains the vacuum term of the three Dirac masses (with the opposite sign for antineutrinos in this convention), multiplied by v_{loc}^2/v^2 because the masses are the Yukawa couplings times the local value of the Higgs field (FE-49); the matter potential; the neutrino-neutrino potential; and the potential V_Ψ of dirac16complex. Status: STANDARD (UM-3 adds V_Ψ and the factor v_{loc}^2/v^2).

$$
\begin{aligned}
&\mathsf{V}_{\mathrm{m}} = \sqrt{2}\, G_{\mathrm{F,loc}}\, \mathrm{diag}\big(n_e - n_{\bar{e}},\ 0,\ 0\big) - \frac{G_{\mathrm{F,loc}}}{\sqrt{2}}\, n_n \mathbf{1}, \qquad \mathsf{V}_{\nu\nu} = \sqrt{2}\, G_{\mathrm{F,loc}} \int \frac{\mathrm{d}^3\vec{q}}{(2\pi)^3} \big(1 - \hat{v}_{\vec{p}} \cdot \hat{v}_{\vec{q}}\big) \big(\varrho_{\vec{q}} - \bar{\varrho}_{\vec{q}}\big), \\
&V_\Psi = \mathcal{V}^\ell_{x_4} = \sum_X c^\ell_X\, j_{X\, x_4}, \qquad G_{\mathrm{F,loc}} = \frac{1}{\sqrt{2}\, v_{\mathrm{loc}}^2}, \\
&v_{\mathrm{loc}}^2 = 2\langle \Phi^\dagger \Phi \rangle, \qquad v_{\mathrm{loc}}^2 = v^2 + \mathcal{A}^2 \ \text{ (the ad hoc vacuum)}, \\
&v_{\mathrm{loc}}^2 = v^2 - \frac{c_\Phi}{\lambda} \langle \bar{\Psi} P_{\mathrm{L}} \Psi \rangle \ \text{ (the relaxed vacuum with dirac16complex)}
\end{aligned}
\tag{FE-49}
$$

Symbols in (FE-49).

- V_{m}: The matter potential of the neutrinos (a 3×3 matrix in flavour space) from their forward scattering on the electrons, positrons and neutrons of the medium. Unit: GeV. Status: STANDARD.
- $G_{\mathrm{F,loc}}$: The local Fermi constant, $G_{\mathrm{F,loc}} = 1/(\sqrt{2}\, v_{\mathrm{loc}}^2)$ ($= G_{\mathrm{F}}$ where $v_{\mathrm{loc}} = v$). Unit: GeV^{-2}. Status: DERIVED.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- n_e: The number density of electrons (bound and free; in a supernova all are free, in the recombining universe $x_e n_e$ is the density of the free electrons). Unit: GeV^3 (cm^{-3}). Status: STANDARD.
- $n_{\bar{e}}$: The number density of positrons. Unit: GeV^3. Status: STANDARD.
- n_n: The number density of neutrons. Unit: GeV^3. Status: STANDARD.
- $\mathbf{1}$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- $\mathsf{V}_{\nu\nu}$: The neutrino-neutrino potential (a 3×3 matrix in flavour space) from the forward scattering of a neutrino on the other neutrinos and antineutrinos of the medium; it depends on their flavour content and their directions. Unit: GeV. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $\vec{q}$: The momentum of another neutrino of the medium (the integration variable of the neutrino-neutrino potential); $\mathrm{d}^3\vec{q}$ its volume element. Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\hat{v}$, $\hat{v}^\mu$: The direction of motion of a neutrino: the unit vector $\hat{v} = \vec{p}/|\vec{p}|$ ($\hat{v}_{\vec{p}}$, $\hat{v}_{\vec{q}}$ for the momenta $\vec{p}$, $\vec{q}$), and the four-vector $\hat{v}^\mu = (1, \hat{v})$ in the order $(x_4; x_1, x_2, x_3)$; $\mathrm{d}^2\hat{v}$ is the element of solid angle of the directions. Unit: dimensionless. Status: STANDARD.
- $\vec{p}$: The momentum of a neutrino (three components along ordinary space, in the rest frame of the medium); $|\vec{p}| = E$ for a neutrino of negligible mass. Unit: GeV. Status: STANDARD.
- $\varrho_{\vec{p}}$, $\bar{\varrho}_{\vec{p}}$: The 3×3 flavour density matrices of the neutrinos and of the antineutrinos of momentum $\vec{p}$ in the basis e, μ, τ: the diagonal entries are the occupation numbers of the three flavours, the off-diagonal entries the coherence between the flavours; $\bar{\varrho}$ is defined with the flavour indices in the reversed order, so that it obeys an equation of the same form. Unit: dimensionless. Status: STANDARD.
- V_Ψ: The potential energy that dirac16complex gives every neutrino through the current portal of the lepton doublets, $V_\Psi = \mathcal{V}^\ell_{x_4}$ (the same for the three flavours). Unit: GeV. Status: DERIVED.
- $\mathcal{V}^f_\mu$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_\mu = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length ($=$ time, $c = 1$). Status: AUTHOR.

- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- j_X^μ: The convective current of the half X of dirac16complex, $j_X^\mu = \mathrm{i}(\bar\Psi P_X D^\mu \Psi - (D^\mu \bar\Psi) P_X \Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- v_loc: The local value of the electroweak vacuum, $v_\mathrm{loc}^2 = 2\langle \Phi^\dagger \Phi \rangle$: it sets the local masses of the quarks and leptons (each mass times v_loc/v) and the local Fermi constant; $v_\mathrm{loc} = v$ in the Standard-Model vacuum. Unit: GeV. Value: v in the late universe outside dense dirac16complex. Status: DERIVED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\langle \cdots \rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_\mathrm{F})^{-1/2}$ with the Fermi constant $G_\mathrm{F} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_\mathcal{A} \mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.

The potentials and the local electroweak vacuum. Forward scattering on electrons and positrons through W exchange gives the electron flavour the potential $\sqrt{2}G_{\mathrm{F},\mathrm{loc}}(n_e - n_{\bar e})$; neutral currents give all flavours the same potential, here from the neutrons (the neutral-current potentials of electrons and protons cancel in neutral matter for any weak mixing angle); forward scattering on the other neutrinos gives the neutrino-neutrino potential, which depends on the flavour content and the directions of all other neutrinos and makes the equations non-linear. dirac16complex acts on the neutrinos in exactly two ways, both fixed by the action (IN-16):

- through the current portal of the lepton doublets: V_Ψ is the time component of the vector potential $\mathcal{V}_\mu^\ell$ of FE-22, the same for the three flavours because the portals are flavour blind. A multiple of the unit matrix drops out of every commutator: V_Ψ shifts the energies of the neutrinos but does not change their flavour;
- through the Higgs portal: the scalar density of the left-handed half shifts the mass parameter of the Higgs field (FE-15). Where the Higgs field relaxes to the minimum of its potential (a homogeneous, static vacuum with $\mathcal{A} = 0$ in FE-15), its local value is $v_\mathrm{loc}^2 = v^2 - c_\Phi \langle \bar\Psi P_\mathrm{L} \Psi \rangle/\lambda$; in the author's ad hoc vacuum $v_\mathrm{loc}^2 = v^2 + \mathcal{A}^2$ (FE-45). All masses scale with v_loc/v and the Fermi constant with v^2/v_loc^2 (at tree level $G_\mathrm{F} = 1/(\sqrt{2}v^2)$ does not depend on the gauge couplings, so the gauge-kinetic portals do not change it), while the Yukawa matrices, and with them U_PMNS, are unchanged.

So dirac16complex changes neither the mixing angles nor the flavour pattern of the conversion: it rescales the vacuum oscillation frequencies by v_loc^2/v^2 and the potentials by v^2/v_loc^2 (the resonance density of FE-50 by v_loc^4/v^4). In the late universe $\mathcal{A} = 0$, and the dark matter is far too dilute to move v_loc: with FE-47 the shift is $\Delta v^2/v^2 = -\mathfrak{g}_{h\chi}\rho_\chi/(\lambda v^3 \omega_\mathrm{b})$, about $10^{-50}\,\mathfrak{g}_{h\chi}$ for the local dark-matter density 0.3 GeV/cm^3 and $\omega_\mathrm{b} = 100$ GeV. The flavour conversion of UM-3 is therefore that of the Standard Model with three Dirac neutrinos wherever $v_\mathrm{loc} = v$, which includes every supernova and the Sun (DERIVED). The Dirac neutrinos of UM-3 have the magnetic moments of the Standard Model, about 3.2×10^{-19} Bohr magnetons times the mass in eV, too small to turn left-handed into right-handed neutrinos in the magnetic fields of supernovae; the dipole portal does not act on neutrinos directly. Status: STANDARD (the potentials), DERIVED (V_Ψ, v_loc and their consequences; `um3_neutrino.py`, N1).

8.2 Matter-enhanced conversion

$$\tan 2\theta_{\rm m} = \frac{\sin 2\theta}{\cos 2\theta - 2EV_{\rm CC}/\Delta m^2}, \qquad V_{\rm CC} = \sqrt{2}\, G_{\rm F,loc}\, n_e, \qquad n_{e,\rm res} = \frac{\Delta m^2 \cos 2\theta}{2\sqrt{2}\, G_{\rm F,loc}\, E},$$
$$P_{\rm LZ} = \exp\Big(-\frac{\pi}{2}\, \frac{\Delta m^2 \sin^2 2\theta}{2E\cos 2\theta}\, \Big| \frac{{\rm d}\ln n_e}{{\rm d}r}\Big|^{-1}_{\rm res} \Big) \tag{FE-50}$$

Symbols in (FE-50).

- $\theta_{\rm m}$: The mixing angle of two neutrino flavours in matter. Unit: dimensionless. Status: STANDARD.
- θ: The vacuum mixing angle of two neutrino flavours (in the two-flavour description of a resonance, one of the three angles $\theta_{12}, \theta_{13}, \theta_{23}$ of $U_{\rm PMNS}$). Unit: dimensionless. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- $V_{\rm CC}$: The charged-current matter potential of the electron neutrinos (forward scattering on electrons through W exchange). Unit: GeV. Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows ($\Delta\rho$ a perturbation, Δm^2 a difference of squared masses, $\Delta N_{\rm eff}$ an addition to $N_{\rm eff}$). Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; in Δm^2 the squared-mass difference of two neutrino mass states ($m_2^2 - m_1^2$ for the two states of a resonance, multiplied by $v_{\rm loc}^2/v^2$ where this differs from 1). Unit: GeV. Status: STANDARD.
- $G_{\rm F,loc}$: The local Fermi constant, $G_{\rm F,loc} = 1/(\sqrt{2}\, v_{\rm loc}^2)$ ($= G_{\rm F}$ where $v_{\rm loc} = v$). Unit: GeV^{-2}. Status: DERIVED.
- n_e: The number density of electrons (bound and free; in a supernova all are free, in the recombining universe $x_e n_e$ is the density of the free electrons). Unit: GeV^3 (cm^{-3}). Status: STANDARD.
- $n_{e,\rm res}$: The electron density at which the flavour conversion in matter is resonant (the Mikheyev-Smirnov-Wolfenstein resonance). Unit: GeV^3. Status: STANDARD.
- $P_{\rm LZ}$: The probability that a neutrino jumps between the eigenstates of matter at the resonance (Landau-Zener); $P_{\rm LZ} = 0$ is an adiabatic resonance. Unit: dimensionless. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: ${\rm d}X$ is an infinitesimal change of X, ${\rm d}X/{\rm d}Y$ the derivative of X with respect to Y, ${\rm d}^n X$ the volume element of n variables. Status: STANDARD.
- r: The distance from the centre of the star along the path of a neutrino. Unit: length. Status: STANDARD.
- res: The label of a quantity evaluated at the resonance, where $n_e = n_{e,\rm res}$. Status: STANDARD.

The resonant flavour conversion in matter (Mikheyev-Smirnov-Wolfenstein) for two flavours with the vacuum mixing angle θ and the squared-mass difference Δm^2 (which includes the factor $v_{\rm loc}^2/v^2$ of FE-48): the mixing angle in matter $\theta_{\rm m}$ (FE-48 with the diagonal potential $V_{\rm CC}$ in the electron flavour), the electron density of the resonance, where $\theta_{\rm m} = 45°$, and the probability $P_{\rm LZ}$ that the neutrino jumps between the eigenstates of matter at the resonance for the density profile $n_e(r)$ along its path (the Landau-Zener formula, exact for a linear profile). With the measured masses and mixing angles this gives the survival probability of solar neutrinos and the conversion in the outer layers of supernovae (the resonances of Δm^2_{31} and Δm^2_{21}). Status: STANDARD (`um3_neutrino.py`, N2).

8.3 The fast flavour instability of dense neutrino gases

$$\mathcal{G}_{\rm ELN}(\hat v) = \sqrt{2}\, G_{\rm F,loc} \int_0^\infty \frac{E^2\, {\rm d}E}{(2\pi)^3} \Big[\mathcal{F}_e - \bar{\mathcal{F}}_e - \big(\mathcal{F}_x - \bar{\mathcal{F}}_x\big) \Big],$$
$$\det\Big[\eta^{\mu\nu}_{\rm M} + \int {\rm d}^2\hat v\; \mathcal{G}_{\rm ELN}(\hat v)\, \frac{\hat v^\mu \hat v^\nu}{\Omega_{\rm fw} - \vec K\cdot\hat v} \Big] = 0, \qquad \hat v^\mu = (1,\ \hat v) \tag{FE-51}$$

Symbols in (FE-51).

- $\mathcal{G}_{\rm ELN}(\hat v)$: The angular distribution of the electron lepton number carried by the neutrinos minus that carried by the heavy-lepton neutrinos, per unit solid angle of the direction $\hat v$, multiplied by $\sqrt{2} G_{\rm F,loc}$. Unit: GeV. Status: STANDARD.
- $\hat v$, $\hat v^\mu$: The direction of motion of a neutrino: the unit vector $\hat v = \vec p/|\vec p|$ ($\hat v_{\vec p}$, $\hat v_{\vec q}$ for the momenta $\vec p$, $\vec q$), and the four-vector $\hat v^\mu = (1, \hat v)$ in the order $(x_4; x_1, x_2, x_3)$; ${\rm d}^2\hat v$ is the element of solid angle of the directions. Unit: dimensionless. Status: STANDARD.

- $G_{\mathrm{F,loc}}$: The local Fermi constant, $G_{\mathrm{F,loc}} = 1/(\sqrt{2}\, v_{\mathrm{loc}}^2)$ (= G_{F} where $v_{\mathrm{loc}} = v$). Unit: GeV^{-2}. Status: DERIVED.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no bound. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\mathcal{F}_e$, $\bar{\mathcal{F}}_e$, $\mathcal{F}_x$, $\bar{\mathcal{F}}_x$: The occupation numbers of the electron neutrinos, of the electron antineutrinos, and of one heavy-lepton neutrino and antineutrino ($x = \mu$ or τ), as functions of the energy E and the direction $\hat{v}$ (the diagonal entries of $\varrho_{\vec{p}}$ and $\bar{\varrho}_{\vec{p}}$); in a transport code, the same for the number of particles in an energy bin. Unit: dimensionless. Status: STANDARD.
- $\eta_{\mathrm{M}}^{\mu\nu}$: The Minkowski metric of the four-vectors $\hat{v}^\mu$, $\eta_{\mathrm{M}} = \mathrm{diag}(1, -1, -1, -1)$ in the order $(x_4; x_1, x_2, x_3)$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- Ω_{fw}: The (complex) frequency of a flavour wave of the neutrino gas, measured from the shift caused by the potentials; an imaginary part greater than zero is an instability. Unit: GeV. Status: STANDARD.
- $\vec{K}$: The (real) wave vector of a flavour wave of the neutrino gas, measured from the shift caused by the potentials. Unit: GeV. Status: STANDARD.

The fast flavour instability (STANDARD: FE-48 linearised in the flavour coherence, without the vacuum term, which is negligible in the dense gas; Izaguirre, Raffelt and Tamborra 2017). $\mathcal{G}_{\mathrm{ELN}}$ is the angular distribution of the electron lepton number carried by the neutrinos minus that of the heavy-lepton flavours ($\mathcal{F}$ the occupation numbers of ν_e, $\bar{\nu}_e$ and of one heavy flavour ν_x, $\bar{\nu}_x$, as functions of the energy and the direction; $\mathrm{d}^2\hat{v}$ the element of solid angle). A small flavour coherence $\propto \mathrm{e}^{-\mathrm{i}(\Omega_{\mathrm{fw}} x_4 - \vec{K}\cdot\vec{x})}$ grows exponentially if the dispersion relation (the second line, written with the Minkowski metric η_{M} of the four-vectors $\hat{v}^\mu$) has a solution with an imaginary part of Ω_{fw} greater than zero for a real wave vector $\vec{K}$ (the frequency and the wave vector are measured from the shifts that the matter and the neutrino-neutrino potentials cause, which drop out). This happens if and only if $\mathcal{G}_{\mathrm{ELN}}$ changes sign as a function of the direction, an angular crossing (Morinaga 2022). The instability then grows on lengths of the order of $1/(\sqrt{2} G_{\mathrm{F}} n_\nu)$, centimetres in a supernova core, much shorter than the scales of the explosion, and drives the flavours toward equipartition; transport codes therefore model its outcome by a prescription such as FE-52. Status: STANDARD (`um3_neutrino.py`, N3).

$$\begin{aligned} &\mathcal{F}_e \to \mathcal{F}_e - \delta_{\mathrm{FC}}, \qquad \bar{\mathcal{F}}_e \to \bar{\mathcal{F}}_e - \delta_{\mathrm{FC}}, \qquad \mathcal{F}_x = \bar{\mathcal{F}}_x \to \mathcal{F}_x + \frac{\delta_{\mathrm{FC}}}{2}, \\ &\delta_{\mathrm{FC}} = \max\Big(\frac{2}{3}\Big(\min\left(\mathcal{F}_e,\ \bar{\mathcal{F}}_e\right) - \mathcal{F}_x\Big),\ \max\left(\mathcal{F}_e,\ \bar{\mathcal{F}}_e\right) - 1\Big), \\ &\text{applied where}\ \ \rho_{\mathrm{b}} < \rho_{\mathrm{FC}}\ \ \text{and}\ \ x_4 - x_{4,\mathrm{b}} > t_{\mathrm{FC}} \end{aligned} \tag{FE-52}$$

Symbols in (FE-52).

- $\mathcal{F}_e$, $\bar{\mathcal{F}}_e$, $\mathcal{F}_x$, $\bar{\mathcal{F}}_x$: The occupation numbers of the electron neutrinos, of the electron antineutrinos, and of one heavy-lepton neutrino and antineutrino ($x = \mu$ or τ), as functions of the energy E and the direction $\hat{v}$ (the diagonal entries of $\varrho_{\vec{p}}$ and $\bar{\varrho}_{\vec{p}}$); in a transport code, the same for the number of particles in an energy bin. Unit: dimensionless. Status: STANDARD.
- δ_{FC}: The occupation (or number per energy bin) of electron neutrino-antineutrino pairs converted into heavy-lepton pairs by the flavour equipartition FE-52; a negative value converts heavy-lepton pairs into electron pairs. Unit: dimensionless. Status: ASSUMED.
- ρ_{b}: The (rest-mass) energy density of the baryons: in a supernova the local density of the matter (quoted in $\mathrm{g/cm^3}$), in cosmology the mean density of the baryons ($\rho_{\mathrm{b}}^{(0)}$ today). Unit: GeV^4. Status: STANDARD.
- ρ_{FC}: The baryon density below which the flavour equipartition FE-52 is applied (the critical density ρ_{c} of Gogilashvili and Tamborra). Unit: GeV^4 ($\mathrm{g/cm^3}$). Value: 10^9, 10^{10}, 10^{11} or 10^{13} $\mathrm{g/cm^3}$. Status: STANDARD.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.

- $x_{4,\mathrm{b}}$: The time of the bounce of the collapsing stellar core. Unit: length (= time). Status: STANDARD.
- t_{FC}: The time after the bounce from which the flavour equipartition FE-52 is applied. Unit: time. Value: 0.02 s. Status: STANDARD.

The flavour equipartition that models the outcome of the fast instability in the transport of a core-collapse supernova, written in the form that satisfies the constraints stated by Gogilashvili and Tamborra for their scheme: in every energy bin (and in the same way for the flux of every energy bin) electron neutrino-antineutrino pairs are converted into heavy-lepton pairs, which conserves the electron lepton number $\mathcal{F}_e - \bar{\mathcal{F}}_e$, the number of neutrinos $\mathcal{F}_e + 2\mathcal{F}_x$, the number of antineutrinos $\bar{\mathcal{F}}_e + 2\mathcal{F}_x$ and the momentum, with the four heavy-lepton species equal as in the transport ($\mathcal{F}_x = \bar{\mathcal{F}}_x$). The amount δ_{FC} brings the less abundant of ν_e and $\bar{\nu}_e$ into equipartition with the heavy-lepton flavours (complete equipartition of all six species is impossible when the electron lepton number is not zero, because flavour conversion conserves it); the second argument of the maximum keeps every occupation number at most 1 (the Pauli principle) when heavy-lepton pairs are converted into electron pairs, and every occupation number stays between 0 and 1. The map is applied where the baryon density is below ρ_{FC} (their ρ_{c}, varied from 10^9 to 10^{13} g/cm^3) and from $t_{\mathrm{FC}} = 0.02$ s after the bounce of the core at $x_{4,\mathrm{b}}$. In UM-3, FE-48 and FE-49 are those of the Standard Model wherever $v_{\mathrm{loc}} = v$, so the prescription and its consequences carry over unchanged: the supernova physics of UM-3 is that of the Standard Model with three Dirac neutrinos, and the failed-explosion fractions of the paper apply (DERIVED from FE-49). Status: STANDARD (the constraints), ASSUMED (this explicit form), DERIVED (its validity in UM-3; `um3_neutrino.py`, N4).

8.4 Neutrinos in the expanding universe

$$\begin{aligned} &\Big(\partial_{x_4} - a_4' \,|\vec{p}|\, \frac{\partial}{\partial|\vec{p}|}\Big)\varrho_{\vec{p}} = -\mathrm{i}\big[\mathsf{H}_{\vec{p}},\ \varrho_{\vec{p}}\big] + \mathcal{C}_{\vec{p}}, \\ &N_{\mathrm{eff}} = \frac{8}{7}\Big(\frac{11}{4}\Big)^{4/3}\frac{\rho_\nu}{\rho_\gamma}, \qquad \mathsf{T}_\nu^{(0)} = \Big(\frac{4}{11}\Big)^{1/3}\mathsf{T}_\gamma^{(0)}, \qquad n_\nu^{(0)} = \frac{3\zeta(3)}{2\pi^2}\big(\mathsf{T}_\nu^{(0)}\big)^3 \end{aligned} \tag{FE-53}$$

Symbols in (FE-53).

- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\vec{p}$: The momentum of a neutrino (three components along ordinary space, in the rest frame of the medium); $|\vec{p}| = E$ for a neutrino of negligible mass. Unit: GeV. Status: STANDARD.
- $\varrho_{\vec{p}}$, $\bar{\varrho}_{\vec{p}}$: The 3×3 flavour density matrices of the neutrinos and of the antineutrinos of momentum $\vec{p}$ in the basis e, μ, τ: the diagonal entries are the occupation numbers of the three flavours, the off-diagonal entries the coherence between the flavours; $\bar{\varrho}$ is defined with the flavour indices in the reversed order, so that it obeys an equation of the same form. Unit: dimensionless. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\mathsf{H}_{\vec{p}}$, $\bar{\mathsf{H}}_{\vec{p}}$: The flavour Hamiltonians (3×3 Hermitian matrices) of the neutrinos and of the antineutrinos of momentum $\vec{p}$. Unit: GeV. Status: STANDARD.
- $\mathcal{C}_{\vec{p}}$, $\bar{\mathcal{C}}_{\vec{p}}$: The collision terms ($3 \times 3$ matrices) of the neutrinos and the antineutrinos of momentum $\vec{p}$: the rates of change of $\varrho_{\vec{p}}$ and $\bar{\varrho}_{\vec{p}}$ by emission, absorption, scattering and pair processes with the matter. Unit: GeV. Status: STANDARD.
- N_{eff}: The effective number of neutrino species: the energy density of the relativistic particles other than photons in units of that of one neutrino species whose temperature is $(4/11)^{1/3}$ times that of the photons. Unit: dimensionless. Value: 3.044 (Standard Model); measured 2.99 ± 0.17 (Planck 2018). Status: STANDARD.
- ρ_ν: The energy density of the neutrinos and antineutrinos of all flavours. Unit: GeV^4. Status: STANDARD.
- ρ_γ: The energy density of the photons. Unit: GeV^4. Status: STANDARD.
- T_ν: The temperature of the cosmic neutrino background; $\mathsf{T}_\nu^{(0)}$ its value today. Unit: GeV (or K). Value: 1.945 K today. Status: STANDARD.
- T_γ: The temperature of the photons (of the CMB); $\mathsf{T}_\gamma^{(0)}$ its value today. Unit: GeV (or K). Value: 2.7255 K today. Status: STANDARD.

- n_ν: The number density of the cosmic neutrino background per flavour (neutrinos plus antineutrinos, one helicity each); $n_\nu^{(0)}$ its value today. Unit: GeV^3. Value: 112 cm^{-3} today. Status: STANDARD.
- ζ: The Riemann zeta function; $\zeta(3) = 1.2020569$. Unit: dimensionless. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

The neutrinos in the homogeneous, isotropic, expanding fold (the momentum redshifts with the expansion rate a_4') and the cosmic neutrino background (CNB); the superscript (0) marks a value today. Until the temperature falls to about 1 MeV the collisions keep the neutrinos in equilibrium with the plasma; they then decouple, the annihilation of electrons and positrons heats the photons, and the effective number of neutrino species is $N_{\mathrm{eff}} = 3.044$ in the Standard Model (with the small heating of the neutrinos and the flavour conversion during decoupling included). Today the CNB has the temperature $\mathsf{T}_\nu^{(0)} = 1.945$ K and $n_\nu^{(0)} = 112$ neutrinos and antineutrinos per cm^3 and flavour (one helicity each: the right-handed states of the Dirac neutrinos are not populated by the Standard-Model collisions). The right-handed neutrinos add to N_{eff} only if something else brings them into equilibrium (FE-62). Status: STANDARD (`um3_neutrino.py`, N5).

9 The cosmological perturbations

The CMB, the CNB and the BAO are computed on the fold, where four-dimensional gravity is general relativity with G_{N} (FE-59) up to corrections of order ϵ_{r}, the matter is that of the Standard Model, the dark matter is the bound states χ of dirac16complex, and the dark energy is the vacuum energy ρ_Λ of the fold (FE-59). The metric of the fold is the background plus the first-order cosmological term of the power series FE-35 (order ϵ_{c}); the equations are those of Ma and Bertschinger (1995) in the conformal Newtonian gauge, written out here.

$$
\begin{aligned}
&\hat{h}_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu = a^2\Big[-(1+2\psi_{\mathrm{B}})\,\mathrm{d}\tau_{\mathrm{c}}^2 + (1-2\phi_{\mathrm{B}})\big(\mathrm{d}x_1^2+\mathrm{d}x_2^2+\mathrm{d}x_3^2\big)\Big],\\
&a = \mathrm{e}^{a_4}, \qquad \mathrm{d}\tau_{\mathrm{c}} = \frac{\mathrm{d}x_4}{a}, \qquad \mathcal{H} = \frac{\dot{a}}{a},\\
&k^2\phi_{\mathrm{B}} + 3\mathcal{H}\big(\dot{\phi}_{\mathrm{B}} + \mathcal{H}\psi_{\mathrm{B}}\big) = -4\pi G_{\mathrm{N}} a^2\,\Delta\rho, \qquad k^2\big(\dot{\phi}_{\mathrm{B}} + \mathcal{H}\psi_{\mathrm{B}}\big) = 4\pi G_{\mathrm{N}} a^2(\rho+\mathcal{P})\,\vartheta,\\
&\ddot{\phi}_{\mathrm{B}} + \mathcal{H}\big(\dot{\psi}_{\mathrm{B}} + 2\dot{\phi}_{\mathrm{B}}\big) + \Big(2\frac{\ddot{a}}{a} - \mathcal{H}^2\Big)\psi_{\mathrm{B}} + \frac{k^2}{3}\big(\phi_{\mathrm{B}} - \psi_{\mathrm{B}}\big) = 4\pi G_{\mathrm{N}} a^2\,\Delta\mathcal{P},\\
&k^2\big(\phi_{\mathrm{B}} - \psi_{\mathrm{B}}\big) = 12\pi G_{\mathrm{N}} a^2(\rho+\mathcal{P})\,\Sigma_{\mathrm{s}}
\end{aligned}
\qquad \text{(FE-54)}
$$

Symbols in (FE-54).

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}_{\mu\nu}^{(1)}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- x: The coordinates of the fold: x_i ($i = 1, 2, 3$) of ordinary space, x_4 the time, x_r ($r = 5, 6, 7$) the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- a, $\dot{a}$, $\ddot{a}$: The scale factor of the fold, $a = \mathrm{e}^{a_4}$ (normalised to 1 today by the choice of the comoving coordinates x_1, x_2, x_3), and its first and second derivatives with respect to the conformal time τ_{c}. Unit: dimensionless. Status: STANDARD.
- ψ_{B}, $\dot{\psi}_{\mathrm{B}}$: The Bardeen potential of the time-time part of the perturbed metric of the fold (the Newtonian potential) in the conformal Newtonian gauge, a function of τ_{c} and of the wave number k, and its derivative with respect to τ_{c}. Unit: dimensionless. Status: STANDARD.
- τ_{c}: The conformal time of the fold, $\mathrm{d}\tau_{\mathrm{c}} = \mathrm{d}x_4/a$; a dot on a quantity of the cosmological perturbations is $\mathrm{d}/\mathrm{d}\tau_{\mathrm{c}}$. Unit: length (= time). Status: STANDARD.
- ϕ_{B}, $\dot{\phi}_{\mathrm{B}}$, $\ddot{\phi}_{\mathrm{B}}$: The Bardeen potential of the space-space part of the perturbed metric of the fold (the curvature potential) in the conformal Newtonian gauge, and its first and second derivatives with respect to τ_{c}. Unit:

dimensionless. Status: STANDARD.

- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\mathcal{H}$: The conformal expansion rate, $\mathcal{H} = \dot{a}/a = a\,a_4'$. Unit: 1/length. Status: STANDARD.
- k: The comoving wave number of a Fourier mode of the perturbations, $k = |\vec{k}|$. Unit: GeV (1/length). Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_N: Newton's constant of four-dimensional gravity on the fold, $G_\mathrm{N} = 1/M_\mathrm{Pl}^2$. Unit: GeV^{-2}. Value: $6.708{\times}10^{-39}$ GeV^{-2}. Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows ($\Delta\rho$ a perturbation, Δm^2 a difference of squared masses, ΔN_eff an addition to N_eff). Status: STANDARD.
- ρ: The total energy density of the matter of the fold (all species); $\Delta\rho$ its perturbation. Unit: GeV^4. Status: STANDARD.
- $\mathcal{P}$: The total pressure of the matter of the fold; $\Delta\mathcal{P}$ its perturbation. Unit: GeV^4. Status: STANDARD.
- ϑ: The divergence of the velocity of the total matter: $(\rho + \mathcal{P})\vartheta$ is the sum over the species of their energy density plus pressure times the divergence of their velocity. Unit: 1/length. Status: STANDARD.
- Σ_s: The anisotropic stress (shear) of the total matter: $(\rho + \mathcal{P})\Sigma_\mathrm{s}$ is the sum over the species (photons and neutrinos) of their energy density plus pressure times their shear. Unit: dimensionless. Status: STANDARD.

The metric of the fold with the scalar perturbations in the conformal Newtonian gauge (ψ_B and ϕ_B the two Bardeen potentials, functions of the conformal time τ_c and of the comoving wave number k of a Fourier mode; a dot is $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$; the comoving coordinates are chosen so that $a = 1$ today), and the Einstein equations of the fold (FE-59) to first order: the energy constraint, the momentum constraint, the trace of the space-space equations and their traceless part. $\Delta\rho$ and $\Delta\mathcal{P}$ are the perturbations of the total energy density and pressure; $(\rho + \mathcal{P})\vartheta$ and $(\rho + \mathcal{P})\Sigma_\mathrm{s}$ are the sums over the species of (energy density plus pressure) times the divergence of their velocity and times their anisotropic stress. The bulk adds terms of relative order $\epsilon_\mathrm{r} \lesssim 10^{-15}$ (FE-35, FE-59), which are dropped. Status: STANDARD (DERIVED: the fold equations are those of general relativity up to order ϵ_r).

$$
\begin{aligned}
&\dot\delta_\chi = -\vartheta_\chi + 3\dot\phi_\mathrm{B}, \qquad \dot\vartheta_\chi = -\mathcal{H}\vartheta_\chi + k^2\psi_\mathrm{B} + \Gamma_{\chi\mathrm{b}}\big(\vartheta_\mathrm{b} - \vartheta_\chi\big),\\
&\dot\delta_\mathrm{b} = -\vartheta_\mathrm{b} + 3\dot\phi_\mathrm{B}, \qquad \dot\vartheta_\mathrm{b} = -\mathcal{H}\vartheta_\mathrm{b} + c_\mathrm{s}^2k^2\delta_\mathrm{b} + k^2\psi_\mathrm{B} + \frac{a\,x_\mathrm{e}n_e\sigma_\mathrm{T}}{\mathsf{R}}\big(\vartheta_\gamma - \vartheta_\mathrm{b}\big) + \frac{\rho_\chi}{\rho_\mathrm{b}}\,\Gamma_{\chi\mathrm{b}}\big(\vartheta_\chi - \vartheta_\mathrm{b}\big), \qquad \mathsf{R} = \frac{3\rho_\mathrm{b}}{4\rho_\gamma},\\
&\dot\delta_\gamma = -\frac{4}{3}\vartheta_\gamma + 4\dot\phi_\mathrm{B}, \qquad \dot\vartheta_\gamma = k^2\Big(\frac{\delta_\gamma}{4} - \Sigma_\gamma\Big) + k^2\psi_\mathrm{B} + a\,x_\mathrm{e}n_e\sigma_\mathrm{T}\big(\vartheta_\mathrm{b} - \vartheta_\gamma\big),\\
&\dot{\mathsf{F}}_l = \frac{k}{2l+1}\big(l\,\mathsf{F}_{l-1} - (l+1)\,\mathsf{F}_{l+1}\big) - a\,x_\mathrm{e}n_e\sigma_\mathrm{T}\Big(\mathsf{F}_l - \frac{\mathsf{G}_0 + \mathsf{F}_2 + \mathsf{G}_2}{10}\,\delta_{l2}\Big) \quad (l \geq 2),\\
&\dot{\mathsf{G}}_l = \frac{k}{2l+1}\big(l\,\mathsf{G}_{l-1} - (l+1)\,\mathsf{G}_{l+1}\big) - a\,x_\mathrm{e}n_e\sigma_\mathrm{T}\Big(\mathsf{G}_l - \frac{\mathsf{G}_0 + \mathsf{F}_2 + \mathsf{G}_2}{2}\Big(\delta_{l0} + \frac{\delta_{l2}}{5}\Big)\Big) \quad (l \geq 0),\\
&\mathsf{F}_0 = \delta_\gamma, \qquad \mathsf{F}_1 = \frac{4\vartheta_\gamma}{3k}, \qquad \mathsf{F}_2 = 2\Sigma_\gamma,\\
&\dot\delta_\nu = -\frac{4}{3}\vartheta_\nu + 4\dot\phi_\mathrm{B}, \qquad \dot\vartheta_\nu = k^2\Big(\frac{\delta_\nu}{4} - \Sigma_\nu\Big) + k^2\psi_\mathrm{B}, \qquad \dot{\mathsf{N}}_l = \frac{k}{2l+1}\big(l\,\mathsf{N}_{l-1} - (l+1)\,\mathsf{N}_{l+1}\big) \quad (l \geq 2),\\
&\mathsf{N}_1 = \frac{4\vartheta_\nu}{3k}, \qquad \mathsf{N}_2 = 2\Sigma_\nu,\\
&\dot\Upsilon_0 = -\frac{\mathsf{q}k}{\mathsf{E}}\Upsilon_1 - \dot\phi_\mathrm{B}\frac{\mathrm{d}\ln\mathcal{F}_0}{\mathrm{d}\ln\mathsf{q}}, \qquad \dot\Upsilon_1 = \frac{\mathsf{q}k}{3\mathsf{E}}\big(\Upsilon_0 - 2\Upsilon_2\big) - \frac{\mathsf{E}k}{3\mathsf{q}}\,\psi_\mathrm{B}\frac{\mathrm{d}\ln\mathcal{F}_0}{\mathrm{d}\ln\mathsf{q}},\\
&\dot\Upsilon_l = \frac{\mathsf{q}k}{(2l+1)\mathsf{E}}\big(l\,\Upsilon_{l-1} - (l+1)\,\Upsilon_{l+1}\big) \quad (l \geq 2), \qquad \mathsf{E} = \sqrt{\mathsf{q}^2 + a^2m_i^2}, \qquad \mathcal{F}_0 = \frac{1}{\mathrm{e}^{\mathsf{q}/\mathsf{T}_\nu^{(0)}} + 1}
\end{aligned}
\tag{FE-55}
$$

Symbols in (FE-55).

- δ_χ, $\dot\delta_\chi$: The density contrast of the dark matter (the bound states χ of dirac16complex), the perturbation of its energy density divided by its mean, and its derivative with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- ϑ_χ, $\dot\vartheta_\chi$: The divergence of the velocity of the dark matter (for a Fourier mode, ik times its velocity along the wave vector), and its derivative. Unit: 1/length. Status: STANDARD.
- ϕ_B, $\dot\phi_\mathrm{B}$, $\ddot\phi_\mathrm{B}$: The Bardeen potential of the space-space part of the perturbed metric of the fold (the curvature potential) in the conformal Newtonian gauge, and its first and second derivatives with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- $\mathcal{H}$: The conformal expansion rate, $\mathcal{H} = \dot a/a = a\,a_4'$. Unit: 1/length. Status: STANDARD.
- k: The comoving wave number of a Fourier mode of the perturbations, $k = |\vec{k}|$. Unit: GeV (1/length). Status: STANDARD.
- ψ_B, $\dot\psi_\mathrm{B}$: The Bardeen potential of the time-time part of the perturbed metric of the fold (the Newtonian potential) in the conformal Newtonian gauge, a function of τ_c and of the wave number k, and its derivative with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- $\Gamma_{\chi\mathrm{b}}$: The rate (per conformal time) at which the dark matter exchanges momentum with the baryons through the portal couplings of FE-47 (elastic scattering of χ on nuclei and electrons). Unit: 1/length. Value: negligible for the couplings that direct detection allows. Status: DERIVED.
- ϑ_b, $\dot\vartheta_\mathrm{b}$: The divergence of the velocity of the baryons and its derivative. Unit: 1/length. Status: STANDARD.
- δ_b, $\dot\delta_\mathrm{b}$: The density contrast of the baryons and its derivative. Unit: dimensionless. Status: STANDARD.
- c_s: The sound speed of the baryon gas: its square is the temperature of the gas divided by the mean mass per particle, times one minus a third of the logarithmic derivative of that temperature with respect to a. Unit: dimensionless. Status: STANDARD.
- a, $\dot a$, $\ddot a$: The scale factor of the fold, $a = \mathrm{e}^{a_4}$ (normalised to 1 today by the choice of the comoving coordinates x_1, x_2, x_3), and its first and second derivatives with respect to the conformal time τ_c. Unit: dimensionless. Status: STANDARD.
- x_e: The ionisation fraction: the fraction of the electrons that are free. Unit: dimensionless. Status: STANDARD.
- n_e: The number density of electrons (bound and free; in a supernova all are free, in the recombining universe $x_\mathrm{e}n_e$ is the density of the free electrons). Unit: GeV^3 (cm^{-3}). Status: STANDARD.
- σ_T: The Thomson cross-section of the scattering of photons on free electrons. Unit: GeV^{-2}. Value: $6.6524587 \times 10^{-25}$ cm^2. Status: STANDARD.
- R: The ratio of the inertia of the baryons to that of the photons, $\mathsf{R} = 3\rho_\mathrm{b}/(4\rho_\gamma)$. Unit: dimensionless. Status:

STANDARD.

- ϑ_γ, $\dot{\vartheta}_\gamma$: The divergence of the velocity of the photons and its derivative. Unit: 1/length. Status: STANDARD.
- ρ_χ: The energy density of the dark matter, the bound states χ of dirac16complex; $\rho_\chi^{(0)}$ its value today. Unit: GeV^4. Status: DERIVED.
- ρ_b: The (rest-mass) energy density of the baryons: in a supernova the local density of the matter (quoted in $\mathrm{g/cm}^3$), in cosmology the mean density of the baryons ($\rho_\mathrm{b}^{(0)}$ today). Unit: GeV^4. Status: STANDARD.
- ρ_γ: The energy density of the photons. Unit: GeV^4. Status: STANDARD.
- δ_γ, $\dot{\delta}_\gamma$: The density contrast of the photons and its derivative. Unit: dimensionless. Status: STANDARD.
- Σ_γ: The anisotropic stress (shear) of the photons. Unit: dimensionless. Status: STANDARD.
- F_l, $\dot{\mathsf{F}}_l$: The multipoles of the temperature perturbation of the photons ($\mathsf{F}_0 = \delta_\gamma$, $\mathsf{F}_1 = 4\vartheta_\gamma/(3k)$, $\mathsf{F}_2 = 2\Sigma_\gamma$; the $F_{\gamma l}$ of Ma and Bertschinger) and their derivatives with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- l: The multipole index and number ($l = 0, 1, 2, \ldots$) of an expansion in Legendre polynomials of the direction of a photon or a neutrino relative to the wave vector, or of the CMB sky in spherical harmonics. Status: STANDARD.
- G_l, $\dot{\mathsf{G}}_l$: The multipoles of the polarisation perturbation of the photons (the $G_{\gamma l}$ of Ma and Bertschinger) and their derivatives. Unit: dimensionless. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta_N^M = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}} = 1$ for $X = \mathrm{L}$); with upper and lower index strings the generalised Kronecker delta $\delta_{N_1 \cdots N_n}^{M_1 \cdots M_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta_{N_b}^{M_a}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- δ_ν, $\dot{\delta}_\nu$: The density contrast of the relativistic neutrinos and its derivative. Unit: dimensionless. Status: STANDARD.
- ϑ_ν, $\dot{\vartheta}_\nu$: The divergence of the velocity of the relativistic neutrinos and its derivative. Unit: 1/length. Status: STANDARD.
- Σ_ν: The anisotropic stress (shear) of the relativistic neutrinos. Unit: dimensionless. Status: STANDARD.
- N_l, $\dot{\mathsf{N}}_l$: The multipoles of the perturbation of the relativistic neutrinos ($\mathsf{N}_0 = \delta_\nu$, $\mathsf{N}_1 = 4\vartheta_\nu/(3k)$, $\mathsf{N}_2 = 2\Sigma_\nu$) and their derivatives. Unit: dimensionless. Status: STANDARD.
- $\Upsilon_l(\mathsf{q})$, $\dot{\Upsilon}_l$: The multipoles of the relative perturbation of the distribution of a massive neutrino state at the comoving momentum q (the Ψ_l of Ma and Bertschinger) and their derivatives. Unit: dimensionless. Status: STANDARD.
- q: The comoving momentum of a neutrino, $a|\vec{p}|$. Unit: GeV. Status: STANDARD.
- E: The comoving energy of a massive neutrino (a times its energy). Unit: GeV. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $\mathcal{F}_0(\mathsf{q})$: The unperturbed (Fermi-Dirac) occupation number of the cosmic neutrino background at the comoving momentum q. Unit: dimensionless. Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; in Δm^2 the squared-mass difference of two neutrino mass states ($m_2^2 - m_1^2$ for the two states of a resonance, multiplied by v_loc^2/v^2 where this differs from 1). Unit: GeV. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- T_ν: The temperature of the cosmic neutrino background; $\mathsf{T}_\nu^{(0)}$ its value today. Unit: GeV (or K). Value: 1.945 K today. Status: STANDARD.

The Boltzmann equations of the species in the same gauge (Ma and Bertschinger 1995, their equations 42, 66, 63, 49 and 57): the dark matter χ (the bound states of dirac16complex: pressureless, with the drag $\Gamma_{\chi\mathrm{b}}$ on the baryons from its portal couplings, FE-47, which is negligible for the couplings that direct detection allows); the baryons (with the sound speed c_s of the gas and the Thomson coupling to the photons through the free electrons, x_e the ionisation fraction); the photons (the multipoles F_l of the temperature and G_l of the polarisation, l the multipole index; δ_{l0}, δ_{l2} are Kronecker deltas, and the term $l\,\mathsf{G}_{l-1}$ vanishes for $l = 0$); the neutrinos while they are relativistic (the multipoles N_l); and each massive neutrino state i of mass m_i (the multipoles Υ_l of the relative perturbation of its distribution at the comoving momentum q and energy E, with the Fermi-Dirac occupation $\mathcal{F}_0$ of the CNB). The hierarchies are truncated at a large l in a numerical solution. Status: STANDARD (UM-3: the dark matter is χ,

with the drag term).

$$r_{\mathrm{s}} = \int_0^{\tau_{\mathrm{d}}} c_{\gamma\mathrm{b}}\,\mathrm{d}\tau_{\mathrm{c}}, \qquad c_{\gamma\mathrm{b}} = \frac{1}{\sqrt{3(1+\mathsf{R})}},$$
$$D_{\mathrm{M}}(\mathsf{z}) = \int_0^{\mathsf{z}} \frac{\mathrm{d}\mathsf{z}'}{H_4(\mathsf{z}')}, \qquad D_{\mathrm{H}}(\mathsf{z}) = \frac{1}{H_4(\mathsf{z})}, \qquad D_{\mathrm{V}}(\mathsf{z}) = \big(\mathsf{z}\,D_{\mathrm{M}}^2(\mathsf{z})\,D_{\mathrm{H}}(\mathsf{z})\big)^{1/3} \tag{FE-56}$$

Symbols in (FE-56).

- r_{s}: The comoving sound horizon at the drag epoch: the comoving distance that a sound wave of the photon-baryon fluid travels until the baryons are released from the photons. Unit: length. Value: about 147 Mpc (Planck 2018). Status: STANDARD.
- τ_{d}: The conformal time of the drag epoch, when the baryons are released from the photons (redshift about 1060). Unit: length (= time). Status: STANDARD.
- $c_{\gamma\mathrm{b}}$: The sound speed of the tightly coupled photon-baryon fluid. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- τ_{c}: The conformal time of the fold, $\mathrm{d}\tau_{\mathrm{c}} = \mathrm{d}x_4/a$; a dot on a quantity of the cosmological perturbations is $\mathrm{d}/\mathrm{d}\tau_{\mathrm{c}}$. Unit: length (= time). Status: STANDARD.
- R: The ratio of the inertia of the baryons to that of the photons, $\mathsf{R} = 3\rho_{\mathrm{b}}/(4\rho_\gamma)$. Unit: dimensionless. Status: STANDARD.
- $D_{\mathrm{M}}(\mathsf{z})$: The transverse comoving distance (comoving angular-diameter distance) to the redshift z, for a spatially flat fold. Unit: length. Status: STANDARD.
- z: The redshift, $1+\mathsf{z} = 1/a$; z' an integration variable. Unit: dimensionless. Status: STANDARD.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.
- $D_{\mathrm{H}}(\mathsf{z})$: The Hubble distance at the redshift z. Unit: length. Status: STANDARD.
- $D_{\mathrm{V}}(\mathsf{z})$: The volume-averaged distance to the redshift z. Unit: length. Status: STANDARD.

The baryon acoustic oscillations: the sound horizon r_{s} at the drag epoch τ_{d} (the comoving distance that a sound wave of the photon-baryon fluid, of speed $c_{\gamma\mathrm{b}}$, travels before the baryons are released from the photons), and the distances to the redshift z at which galaxy surveys measure it: the transverse comoving distance D_{M} (for a spatially flat fold), the Hubble distance D_{H} and their volume average D_{V} ($H_4 = a_4'$ the expansion rate of the fold as a function of redshift). The surveys measure $D_{\mathrm{M}}/r_{\mathrm{s}}$, $D_{\mathrm{H}}/r_{\mathrm{s}}$ and $D_{\mathrm{V}}/r_{\mathrm{s}}$. Status: STANDARD.

$$C_l^{\mathrm{TT}} = 4\pi \int \frac{\mathrm{d}k}{k}\,\mathcal{P}_{\mathcal{R}}(k)\,\big(\mathcal{I}_l^{\mathrm{T}}(k)\big)^2, \qquad C_l^{\mathrm{TE}} = 4\pi \int \frac{\mathrm{d}k}{k}\,\mathcal{P}_{\mathcal{R}}(k)\,\mathcal{I}_l^{\mathrm{T}}(k)\,\mathcal{I}_l^{\mathrm{E}}(k),$$
$$C_l^{\mathrm{EE}} = 4\pi \int \frac{\mathrm{d}k}{k}\,\mathcal{P}_{\mathcal{R}}(k)\,\big(\mathcal{I}_l^{\mathrm{E}}(k)\big)^2, \qquad \mathcal{P}_{\mathcal{R}}(k) = A_{\mathrm{s}}\Big(\frac{k}{k_*}\Big)^{n_{\mathrm{S}}-1} \tag{FE-57}$$

Symbols in (FE-57).

- C_l^{TT}, C_l^{TE}, C_l^{EE}: The angular power spectra of the CMB at the multipole l: temperature with temperature, temperature with E-mode polarisation, E-mode polarisation with itself. Unit: dimensionless. Status: STANDARD.
- l: The multipole index and number ($l = 0, 1, 2, \ldots$) of an expansion in Legendre polynomials of the direction of a photon or a neutrino relative to the wave vector, or of the CMB sky in spherical harmonics. Status: STANDARD.
- T, E: The labels of the CMB observables: T the temperature anisotropy, E the E-mode (gradient) polarisation; TT, TE, EE label the spectra of their products. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

- k: The comoving wave number of a Fourier mode of the perturbations, $k = |\vec{k}|$. Unit: GeV (1/length). Status: STANDARD.
- $\mathcal{P}_\mathcal{R}(k)$: The primordial power spectrum of the comoving curvature perturbation, per logarithmic interval of k. Unit: dimensionless. Status: ASSUMED.
- $\mathcal{I}_l^{\mathrm{T}}(k)$, $\mathcal{I}_l^{\mathrm{E}}(k)$: The transfer functions of the CMB: the multipole l of the temperature or E-mode anisotropy today produced by a Fourier mode k of unit primordial curvature perturbation, computed from FE-54 and FE-55 (the line-of-sight integral). Unit: dimensionless. Status: STANDARD.
- A_s: The amplitude of the primordial curvature spectrum at the pivot k_*. Unit: dimensionless. Value: 2.10×10^{-9} (Planck 2018). Status: ASSUMED.
- k_*: The pivot wave number of the primordial spectrum. Unit: GeV (1/length). Value: 0.05 Mpc^{-1}. Status: STANDARD.
- n_S: The spectral index of the primordial curvature spectrum. Unit: dimensionless. Value: 0.965 (Planck 2018). Status: ASSUMED.

The angular power spectra of the CMB temperature (T) and E-mode polarisation (E) at the multipole l: the transfer functions $\mathcal{I}_l^{\mathrm{T}}$, $\mathcal{I}_l^{\mathrm{E}}$ follow from FE-54 and FE-55, integrated from the early universe to today (the line-of-sight integral), and are weighted with the primordial spectrum $\mathcal{P}_\mathcal{R}$ of the comoving curvature perturbation (amplitude A_s, spectral index n_S, pivot k_*). The primordial spectrum is an input of UM-3 (ASSUMED: the author's primordial era is not computed to the precision this needs). Status: STANDARD.

$$\Omega_\chi = \frac{\rho_\chi^{(0)}}{\rho_\mathrm{c}^{(0)}}, \qquad \Omega_\mathrm{b} = \frac{\rho_\mathrm{b}^{(0)}}{\rho_\mathrm{c}^{(0)}}, \qquad \Omega_\Lambda = \frac{\rho_\Lambda}{\rho_\mathrm{c}^{(0)}}, \qquad \Omega_\nu h_{100}^2 = \frac{\sum_i m_i}{93.14\ \mathrm{eV}},$$
$$\rho_\mathrm{c}^{(0)} = \frac{3\big(H_4^{(0)}\big)^2}{8\pi G_\mathrm{N}}, \qquad H_4^{(0)} = 100\, h_{100}\ \mathrm{km/s/Mpc} \tag{FE-58}$$

Symbols in (FE-58).

- Ω_χ: The density parameter of the dark matter today. Unit: dimensionless. Value: $\Omega_\chi h_{100}^2 = 0.120$ (Planck 2018). Status: DERIVED.
- ρ_χ: The energy density of the dark matter, the bound states χ of dirac16complex; $\rho_\chi^{(0)}$ its value today. Unit: GeV^4. Status: DERIVED.
- ρ_c: The critical density, the energy density of a spatially flat fold at the expansion rate H_4; $\rho_\mathrm{c}^{(0)}$ its value today. Unit: GeV^4. Value: $\rho_\mathrm{c}^{(0)} = 1.0537 \times 10^{-5}\, h_{100}^2\ \mathrm{GeV/cm}^3$. Status: STANDARD.
- Ω_b: The density parameter of the baryons today. Unit: dimensionless. Value: $\Omega_\mathrm{b} h_{100}^2 = 0.0224$ (Planck 2018). Status: STANDARD.
- ρ_b: The (rest-mass) energy density of the baryons: in a supernova the local density of the matter (quoted in $\mathrm{g/cm}^3$), in cosmology the mean density of the baryons ($\rho_\mathrm{b}^{(0)}$ today). Unit: GeV^4. Status: STANDARD.
- Ω_Λ: The density parameter of the dark energy (the vacuum energy of the fold) today. Unit: dimensionless. Value: 0.685 (Planck 2018). Status: DERIVED.
- ρ_Λ: The vacuum energy density of the fold, the dark energy of UM-3: $\rho_\Lambda = \rho_\mathrm{ff} - 31 H V_*/(4\kappa)$ (FE-59). Unit: GeV^4. Status: DERIVED.
- Ω_ν, $\Omega_\nu^{(i)}$: The density parameter of the neutrinos of the CNB: Ω_ν that of the massive states today; $\Omega_\nu^{(i)}(\mathsf{z})$ the energy density of the mass state i at the redshift z in units of the critical density today. Unit: dimensionless. Status: STANDARD.
- h_{100}: The present expansion rate of the fold in units of 100 km/s/Mpc. Unit: dimensionless. Value: 0.674 (Planck 2018). Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; in Δm^2 the squared-mass difference of two neutrino mass states ($m_2^2 - m_1^2$ for the two states of a resonance, multiplied by v_loc^2/v^2 where this differs from 1). Unit: GeV. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\, h_{100}$ km/s/Mpc. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

- $G_{\rm N}$: Newton's constant of four-dimensional gravity on the fold, $G_{\rm N} = 1/M_{\rm Pl}^2$. Unit: GeV^{-2}. Value: $6.708{\times}10^{-39}$ GeV^{-2}. Status: STANDARD.

The density parameters today: the dark matter (the relic density of χ, FE-61), the baryons, the dark energy (the vacuum energy of the fold, FE-59) and the massive neutrinos of the CNB (for the three left-handed states with the temperature and the small non-thermal distortion of the Standard Model; the right-handed states are not populated). Together with $A_{\rm s}$, $n_{\rm S}$ and the optical depth of reionisation they fix the CMB, the CNB and the BAO. Status: STANDARD.

10 The thermal history

$$\frac{1}{\kappa_4}\,\hat{G}^{\mu}{}_{\nu} = \mathcal{T}_{\rm SM}{}^{\mu}{}_{\nu} + \mathcal{T}_{\rm P}{}^{\mu}{}_{\nu} + \mathcal{T}_{\rm ff}{}^{\mu}{}_{\nu} - \frac{V_*}{\kappa}\,\mathcal{J}^{\mu}{}_{\nu}, \qquad \mathcal{J}^{a}{}_{b} = \kappa\,\mathcal{S}^{a}{}_{b} \text{ (the junction condition)},$$
$$H_4^2 = \frac{8\pi G_{\rm N}}{3}\big(\rho_{\rm SM} + \rho_\chi + \rho_\Lambda\big), \qquad 8\pi G_{\rm N} = \frac{\kappa_4}{1+5\epsilon_{\rm r}/6}, \qquad \epsilon_{\rm r} = \frac{\kappa_4 V_*}{\kappa H}, \qquad \rho_\Lambda = \rho_{\rm ff} - \frac{31 H V_*}{4\kappa}, \tag{FE-59}$$
$$\frac{\hat{R}}{2\kappa_4} + q_{\rm SM} + q_\chi + q_{\rm P} + q_{\rm ff} = \frac{V_*}{\kappa}\,\mathcal{J}^{(r)}{}_{(r)}, \qquad q_{\rm SM} = \big\langle \mathcal{L}_{\rm H} + \mathcal{L}_{\rm gauge} + \mathcal{L}_{\rm f} + \mathcal{L}_{\rm Y}\big\rangle, \qquad q_{\rm P} = \langle \mathcal{L}_{\rm P}\rangle$$

Symbols in (FE-59).

- κ_4: The coupling of the induced gravity of the fold, $\kappa_4 = 8\pi G_{\rm ind}$; four-dimensional gravity on the fold has $8\pi G_{\rm N} = \kappa_4/(1+5\epsilon_{\rm r}/6)$ (the Friedmann equation of the fold to first order in $\epsilon_{\rm r}$). Unit: GeV^{-2}. Value: $\kappa_4 \simeq 8\pi G_{\rm N} = 8\pi/M_{\rm Pl}^2$. Status: ASSUMED.
- $\hat{G}^{\mu}{}_{\nu}$: The Einstein tensor of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold. Unit: 1/length^2. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- $\mathcal{T}_{\rm SM}$: The energy-momentum tensor of the Standard Model on the fold, $\mathcal{T}_{\rm H} + \mathcal{T}_{\rm gauge} + \mathcal{T}_{\rm f}$. Unit: GeV^4. Status: STANDARD.
- $\mathcal{T}_{\rm P}$: The energy-momentum tensor of the portals of dirac16complex. Unit: GeV^4. Status: DERIVED.
- $\mathcal{T}_{\rm ff}{}^{a}{}_{b}$: The energy-momentum tensor of the fold fluid (frame components of the fold, four-dimensional normalisation). Unit: GeV^4. Status: ASSUMED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\rm t}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{J}^{a}{}_{b}$: The jump of the junction potential across the fold computed from the metric: the geometric side of the junction condition $\mathcal{J} = \kappa\mathcal{S}$, which equates it to κ times the energy-momentum tensor $\mathcal{S}$ of everything on the fold. Unit: GeV. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- $\mathcal{S}^{a}{}_{b}$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_{\rm N}$: Newton's constant of four-dimensional gravity on the fold, $G_{\rm N} = 1/M_{\rm Pl}^2$. Unit: GeV^{-2}. Value: $6.708{\times}10^{-39}$ GeV^{-2}. Status: STANDARD.
- $\rho_{\rm SM}$: The energy density of the Standard-Model particles. Unit: GeV^4. Status: STANDARD.
- ρ_χ: The energy density of the dark matter, the bound states χ of dirac16complex; $\rho_\chi^{(0)}$ its value today. Unit: GeV^4. Status: DERIVED.

- ρ_Λ: The vacuum energy density of the fold, the dark energy of UM-3: $\rho_\Lambda = \rho_{\mathrm{ff}} - 31HV_*/(4\kappa)$ (FE-59). Unit: GeV^4. Status: DERIVED.
- ϵ_r: The bookkeeping parameter of the response of the bulk to the matter of the fold: the ratio of the bulk to the fold gravity, $\epsilon_\mathrm{r} = \kappa_4 V_*/(\kappa H)$. Unit: dimensionless. Value: $\lesssim 10^{-15}$ (an estimate: if a fraction ϵ_r of the gravitational waves of the fold travelled with the bulk gravitons, whose speed squared is 15/2 times that of light, their speed would differ from that of light by about $3\epsilon_\mathrm{r}$, and the gravitational waves and the light of the neutron-star merger GW170817 arrived together to about 10^{-15}). Status: DERIVED (the definition), ESTIMATE (the bound).
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- ρ_{ff}: The energy density of the fold fluid. Unit: GeV^4. Value: $31HV_*/(4\kappa)$ at the static block. Status: ASSUMED.
- $\hat{R}$: The scalar curvature of the metric $\hat{h}_{\mu\nu}$ of the fold. Unit: 1/length^2. Status: STANDARD.
- q_SM: The pressure of the Standard Model along the extra times at the fold: the mean on-shell Lagrangian density of the Standard-Model fields (the extra-time component of the energy-momentum tensor of every term of the fold is its Lagrangian, because the fields of the fold do not depend on the extra times). Unit: GeV^4. Status: DERIVED.
- q_χ: The pressure of the bound states of dirac16complex along the extra times, integrated across the fold (their on-shell Lagrangian, the self-interaction energy per volume; zero for a dilute gas). Unit: GeV^4. Status: DERIVED.
- q_P: The pressure of the portals along the extra times: their mean on-shell Lagrangian density. Unit: GeV^4. Status: DERIVED.
- q_{ff}: The pressure of the fold fluid along the extra times. Unit: GeV^4. Value: $-61HV_*/(8\kappa)$ at the static block. Status: ASSUMED.
- r: An extra-time direction of the fold: $r \in \{5,6,7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\mathcal{L}_\mathrm{H}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\mathcal{L}_\mathrm{gauge}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{f}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{Y}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{P}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

The Einstein equations of the fold: FE-4 and FE-5 combined. The left-hand side is the induced gravity; the sources are the Standard Model ($\mathcal{T}_\mathrm{SM} = \mathcal{T}_\mathrm{H} + \mathcal{T}_\mathrm{gauge} + \mathcal{T}_\mathrm{f}$), the portals, the fold fluid and the bulk, which acts through the jump $\mathcal{J}$ of the junction potential computed from the metric (FE-30, or FE-33 with frozen extra times). For a homogeneous fold with frozen extra times ${\mathcal{J}^{(4)}}_{(4)} = -(31 + 10\mathsf{A}^2)H/4$ (FE-33): its constant part cancels against the fold fluid except for the remainder ρ_Λ, the dark energy of UM-3 (a vacuum energy of the fold), and its part $-\frac{5}{2}\,a_4'^2/H$ adds to the left-hand side, so that the Friedmann equation of the fold has Newton's constant reduced by the factor $1 + 5\epsilon_\mathrm{r}/6$ (ϵ_r the ratio of bulk to fold gravity; DERIVED from FE-33 with $\mathsf{A} = a_4'/H$ and ${\hat{G}^{(4)}}_{(4)} = -3a_4'^2$). The space-space components give, in the same way, the acceleration equation of general relativity up to terms of order ϵ_r. The component along the extra times (the third line; q is a pressure along the extra times, for every term of the fold its on-shell Lagrangian density by FE-5, and the induced gravity contributes $\hat{R}/(2\kappa_4)$, $\hat{R}$ the scalar curvature of the fold) is an additional equation: the equation of the size of the extra times at the fold (the radion). To leading order in ϵ_r it requires that the sum over the matter of the fold of $q + (\rho - 3p)/2$ vanishes. Radiation satisfies it; non-relativistic matter ($q = 0$ for the bound states of dirac16complex, $q = -\rho$ for classical particles on the fold) and vacuum energy ($q = -\rho$) do not. The first-order terms of the power series FE-35 must therefore carry the difference: if the bulk medium does not respond, the only available bulk mode is the difference mode of FE-35, which changes the acceleration equation to $5H_4' + 9H_4^2 = -\kappa_4(p + q)$ (dust would expand as $x_4^{5/6}$, which observation excludes); UM-3 therefore requires (ASSUMED) that the bulk medium responds with the radion mode of FE-35, which changes only the extra-time component and keeps the four-dimensional equations those of general relativity (DERIVED, `um3_series.py`, SR1 to SR6). The radion mode with the jumps $c\,(1, -1, 5)$ of $(\mathfrak{a}_1', \mathfrak{b}_1', \mathfrak{n}_1')$ at the fold changes the extra-time component of the junction by $\kappa\,\delta{\mathcal{S}^{(r)}}_{(r)} = 5c/8$; today it must carry the mismatch

$0.793\,\rho_{\mathrm{c}}^{(0)}$ of the dark matter ($q=0$), the baryons and the vacuum energy ($q=-\rho$), which needs the amplitude $c/H = 3.806\,\mathsf{A}^2/\epsilon_{\mathrm{r}}$ ($\mathsf{A} = a_4'/H$, $\rho_{\mathrm{c}}^{(0)}$ the critical density of FE-58), a small term of the power series when $\mathsf{A}^2 \ll \epsilon_{\mathrm{r}}$, that is when the bulk curvature scale H exceeds the Hubble rate by far more than $\epsilon_{\mathrm{r}}^{-1/2}$ (DERIVED, SR6). With $\epsilon_{\mathrm{r}} \lesssim 10^{-15}$ the fold then expands as in general relativity. Status: DERIVED (with the ASSUMED induced gravity and the ASSUMED response of the bulk medium).

$$\rho_{\mathrm{SM}} = \frac{\pi^2}{30}\,g_\rho(\mathsf{T})\,\mathsf{T}^4, \qquad s = \frac{2\pi^2}{45}\,g_{\mathrm{s}}(\mathsf{T})\,\mathsf{T}^3, \qquad H_4 = \sqrt{\frac{8\pi^3 g_\rho}{90}}\,\frac{\mathsf{T}^2}{M_{\mathrm{Pl}}}, \qquad M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2} \tag{FE-60}$$

Symbols in (FE-60).

- ρ_{SM}: The energy density of the Standard-Model particles. Unit: GeV^4. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- g_ρ: The effective number of relativistic degrees of freedom of the energy density. Unit: dimensionless. Value: 106.75 in the Standard Model above the electroweak scale. Status: STANDARD.
- T: The temperature of the Standard-Model plasma (of the photons after the annihilation of electrons and positrons). Unit: GeV. Status: STANDARD.
- s: The entropy density of the Standard-Model plasma. Unit: GeV^3. Status: STANDARD.
- g_{s}: The effective number of relativistic degrees of freedom of the entropy density. Unit: dimensionless. Value: 3.909 today. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.
- M_{Pl}: The Planck mass, $M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2}$. Unit: GeV. Value: 1.221×10^{19} GeV. Status: STANDARD.
- G_{N}: Newton's constant of four-dimensional gravity on the fold, $G_{\mathrm{N}} = 1/M_{\mathrm{Pl}}^2$. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.

The radiation of the Standard Model in thermal equilibrium at the temperature T: the energy density and the entropy density with the effective numbers of relativistic degrees of freedom g_ρ and g_{s} (106.75 above the electroweak scale in the Standard Model; three right-handed neutrinos in equilibrium would add $\frac{7}{8}\times 6 = 5.25$; the bound states of dirac16complex add theirs while they are relativistic), and the expansion rate of the radiation era. Status: STANDARD.

$$\frac{\mathrm{d}n_\chi}{\mathrm{d}x_4} + 3H_4\,n_\chi = -\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle\big(n_\chi^2 - n_{\chi,\mathrm{eq}}^2\big), \qquad n_{\chi,\mathrm{eq}} = g_\chi\Big(\frac{\omega_{\mathrm{b}}\mathsf{T}}{2\pi}\Big)^{3/2}\mathrm{e}^{-\omega_{\mathrm{b}}/\mathsf{T}}, \qquad \Omega_\chi h_{100}^2 \simeq \frac{1.07\times10^9\ \mathrm{GeV}^{-1}\,x_{\mathrm{f}}}{\sqrt{g_*}\,M_{\mathrm{Pl}}\,\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle} \tag{FE-61}$$

Symbols in (FE-61).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- n_χ: The number density of the dark-matter particles χ (particles plus antiparticles). Unit: GeV^3. Status: DERIVED.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c=1$). Status: AUTHOR.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.
- σ_{ann}: The annihilation cross-section of the particles χ into Standard-Model particles (through the portals) in the normalisation of FE-61, where n_χ counts the four particle and the four antiparticle states of the bound level: $\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle$ is a quarter of the annihilation rate $(\sigma\mathsf{v})_{\chi\bar\chi}$ of one particle χ_a with one antiparticle $\bar\chi_a$. Unit: GeV^{-2}. Status: DERIVED.
- v: The relative velocity of two annihilating particles; $\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle$ is the thermal average of the product. Unit: dimensionless. Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.

- $n_{\chi,\mathrm{eq}}$: The number density of χ in thermal equilibrium with the plasma (non-relativistic). Unit: GeV^3. Status: STANDARD.
- g_χ: The number of internal states of the bound level of χ that are in thermal equilibrium. Unit: dimensionless. Status: DERIVED.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- T: The temperature of the Standard-Model plasma (of the photons after the annihilation of electrons and positrons). Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- Ω_χ: The density parameter of the dark matter today. Unit: dimensionless. Value: $\Omega_\chi h_{100}^2 = 0.120$ (Planck 2018). Status: DERIVED.
- h_{100}: The present expansion rate of the fold in units of 100 km/s/Mpc. Unit: dimensionless. Value: 0.674 (Planck 2018). Status: STANDARD.
- x_f: The ratio of the mass of χ to the temperature at which its annihilations stop (freeze-out). Unit: dimensionless. Value: 23 to 30 (for $\omega_\mathrm{b} = 10$ GeV to 10 TeV). Status: STANDARD.
- g_*: The effective number of relativistic degrees of freedom at the freeze-out (g_ρ there). Unit: dimensionless. Status: STANDARD.
- M_Pl: The Planck mass, $M_\mathrm{Pl} = G_\mathrm{N}^{-1/2}$. Unit: GeV. Value: 1.221×10^{19} GeV. Status: STANDARD.

The relic density of the bound states of dirac16complex: their number density n_χ (particles plus antiparticles) follows the Boltzmann equation with the thermally averaged annihilation cross-section $\langle\sigma_\mathrm{ann}\mathsf{v}\rangle$ into Standard-Model particles through the portal couplings of FE-47 (Higgs exchange, pairs of gluons and photons, pairs of fermions); g_χ is the number of states of the bound level that are in equilibrium. The freeze-out at $x_\mathrm{f} = \omega_\mathrm{b}/\mathsf{T}_\mathrm{f} \approx 20$ to 25 (T_f the temperature at which the annihilations stop) gives the standard estimate of the relic density for an annihilation that does not depend on the velocity (g_* the degrees of freedom at freeze-out). Status: STANDARD (UM-3: the cross-section from the portals).

$$\frac{\mathrm{d}\rho_{\nu_\mathrm{R}}}{\mathrm{d}x_4} + 4H_4\,\rho_{\nu_\mathrm{R}} = \mathcal{C}_{\nu_\mathrm{R}}, \qquad \Delta N_\mathrm{eff} = 3\Big(\frac{10.75}{g_\mathrm{s}(\mathsf{T}_\mathrm{dec})}\Big)^{4/3} \quad \text{(three right-handed neutrinos decoupling at } \mathsf{T}_\mathrm{dec}) \quad \text{(FE-62)}$$

Symbols in (FE-62).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- ρ_{ν_R}: The energy density of the three right-handed neutrinos. Unit: GeV^4. Status: STANDARD.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.
- $\mathcal{C}_{\nu_\mathrm{R}}$: The collision term of the right-handed neutrinos: the rate at which collisions produce (or remove) their energy density. Unit: GeV^5. Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows ($\Delta\rho$ a perturbation, Δm^2 a difference of squared masses, ΔN_eff an addition to N_eff). Status: STANDARD.
- N_eff: The effective number of neutrino species: the energy density of the relativistic particles other than photons in units of that of one neutrino species whose temperature is $(4/11)^{1/3}$ times that of the photons. Unit: dimensionless. Value: 3.044 (Standard Model); measured 2.99 ± 0.17 (Planck 2018). Status: STANDARD.
- g_s: The effective number of relativistic degrees of freedom of the entropy density. Unit: dimensionless. Value: 3.909 today. Status: STANDARD.
- T_dec: The temperature at which the right-handed neutrinos decouple from the plasma. Unit: GeV. Status: STANDARD.

The right-handed neutrinos in the early universe: their energy density grows by the collision term $\mathcal{C}_{\nu_\mathrm{R}}$ (production by the neutrino Yukawa couplings, negligible for Dirac masses below 0.1 eV because it is suppressed by the square of the

mass over the temperature, and by the current portal c_X^ν with dirac16complex), and if they were in equilibrium until the temperature $\mathsf{T}_{\rm dec}$ they add $\Delta N_{\rm eff}$ to the effective number of neutrinos (from the conservation of entropy after their decoupling). The measured $N_{\rm eff} = 2.99 \pm 0.17$ (Planck 2018) and 2.89 ± 0.11 (ACT DR6, the sixth data release of the Atacama Cosmology Telescope, with Planck, lensing, BAO and nucleosynthesis, 2025) bound $\Delta N_{\rm eff}$ and with it the portal couplings c_X^ν: if the portal keeps the right-handed neutrinos in equilibrium with χ they decouple at $\mathsf{T}_{\rm dec} \approx \omega_{\rm b}/13$ to $\omega_{\rm b}/17$ and add $\Delta N_{\rm eff} = 0.14$ to 0.24 for $\omega_{\rm b} = 10$ TeV to 10 GeV, 2.7 to 3.5 standard deviations above the ACT value; the right-handed neutrinos of UM-3 must therefore never thermalise (c_X^ν much smaller than the lepton-doublet couplings), and then $N_{\rm eff} = 3.044$ (DERIVED, `um3_thermal.py`, T5). Status: STANDARD (the equations), DERIVED (the bound).

11 The coupled system

11.1 Which field appears in which equation

equation	the fields in it	how they enter
bulk gravity FE-1	metric, dirac16complex, stabiliser	the Lovelock tensors of the metric; $\mathcal{T}_\Psi$; $\mathcal{T}_{\rm stab}$
fold junction FE-4, FE-5, FE-59	metric, every field of the fold, dirac16complex at the fold	the jump of the junction potential; $\mathcal{T}_{\rm H}$, $\mathcal{T}_{\rm gauge}$, $\mathcal{T}_{\rm f}$, $\mathcal{T}_{\rm P}$, $\mathcal{T}_{\rm ff}$; the induced gravity $\hat{G}/\kappa_4$
dirac16complex FE-12, FE-13, FE-41	metric, itself, every Standard-Model field	the spin connection; the self-interaction; the fold operator $\mathbb{M}$ of the portals (the Higgs portal for $\Psi_{\rm L}$ only)
Higgs FE-15	metric, dirac16complex, W, B, quarks and leptons	$\hat{h}$ and $\mathcal{Z}$; $c_\Phi \bar{\Psi} P_{\rm L} \Psi$; D_μ; the Yukawa source Θ
gauge fields FE-16 to FE-18	metric, dirac16complex, Higgs, quarks and leptons	$\hat{h}$ and $\mathcal{Z}$; the portal factors $\mathcal{K}$ and the dipole density $\mathcal{D}_X$; the currents
quarks and leptons FE-19 to FE-24	metric, gauge fields, Higgs, dirac16complex	$\hat{h}$; D_μ; the Yukawa matrices; the potentials $\mathcal{V}^f$
neutrino flavours FE-48 to FE-53	expansion, Higgs, dirac16complex, matter, other neutrinos	a_4'; $v_{\rm loc}$ and $G_{\rm F,loc}$; V_Ψ; $\mathsf{V}_{\rm m}$ and $\mathsf{V}_{\nu\nu}$
cosmology FE-54 to FE-58	metric perturbations, dark matter χ, baryons, photons, neutrinos	$\phi_{\rm B}$ and $\psi_{\rm B}$; $\Gamma_{\chi{\rm b}}$; Thomson scattering; ρ_Λ
thermal history FE-59 to FE-62	metric, plasma, dark matter χ, right-handed neutrinos	H_4, $G_{\rm N}$ and ρ_Λ; $\langle\sigma_{\rm ann}\mathsf{v}\rangle$; $\mathcal{C}_{\nu_{\rm R}}$

11.2 How the system is solved

The equations are coupled in a chain that UM-3 solves in steps, each with the results of the earlier ones:

1. The background: the author's vacuum in the primordial era (FE-31: exact in the bulk, with its fold source) and the static block with frozen extra times in the late universe (FE-32, FE-33: exact with the stabiliser and the fold fluid).
2. The modes of dirac16complex on the background: the bulk continuum and the bound states (FE-36 to FE-40) with the mean fields of the Standard-Model vacuum (FE-39).
3. The Kohn-Sham states (FE-41, FE-42): the bound states filled to a given density, with the self-interaction and the mean fields, to self-consistency; their energy-momentum tensor enters the bulk equations and the fold junction (the next order of FE-35).
4. The Standard Model on the fold (FE-44 to FE-47): the masses with the ad hoc vacuum, the residuals FE-26, the couplings of the bound states.
5. The neutrino flavours (FE-48 to FE-53), the thermal history (FE-59 to FE-62) and the relic density of χ.
6. The cosmological perturbations (FE-54 to FE-58): the CMB, the CNB and the BAO.

12 What was verified

Every statement labelled DERIVED in this document rests on a computation in the folder `universe_model_3/code/` of the repository, which checks itself and writes a report to `universe_model_3/reports/` (the equations and results themselves are all written out above). The checks used here:

- `um3_dirac_matrices.py` (55 checks): the author's matrices, their reality, the Clifford relations, the chiral projectors, Pin(4,4).
- `um3_geometry.py` (26 checks): the exact curvature of the author's metric, the Lovelock tensors by two routes, the trace and Bianchi identities, the background equations FE-29, the author's vacuum, the degeneracy, the frozen extra times, the spin connection FE-14 and the vielbein postulate.
- `um3_fold.py` (13 checks): the proper chart, the general warped metric FE-34, quasi-linearity, the junction potential, the fold sources FE-30 to FE-33 (also by smoothing the kink), the conservation law FE-11 at the fold.
- `um3_bulk.py` (10 checks): the zero modes and the speeds of bulk scalars, photons, fermions and gravitons (with the Lovelock terms), the dispersion of dirac16complex with chiral masses.
- `um3_fields.py` (16 checks): the reality and chirality of the bilinears of dirac16complex, the Krein operator, the four-dimensional chirality, the electroweak masses and the photon of the ad hoc vacuum, its isospin charge and residual, the Yukawa invariance, the anomalies with three right-handed neutrinos, the electric charges.
- `um3_spectra.py` (5 checks): the transfer matrices, the fold transfer, the bound states FE-40 with the exact 16×16 matrices, their Krein signature.
- `um3_neutrino.py` (5 checks): the flavour blindness of V_Ψ and the invariance of the mixing under v_{loc} (N1), the resonance formulas FE-50 against an exact integration (N2), the crossing criterion of FE-51 (N3), the conservation laws of FE-52 (N4), the CNB numbers of FE-53 (N5).
- `um3_background.py` (6 checks): the histories of FE-29 with constant rates (only $(\mathsf{A}, \mathsf{U}) = (\pm 1, 0)$, a degenerate double root), the violation of the null energy condition by the stabiliser and by the bulk medium of the frozen branch, the bulk sources that a varying b_4 needs in the primordial era (of second order in U, without flux at $\mathsf{U} = 2/3$), the factor $\mathcal{Z}$ of the couplings of the fold, the gauge equivalence of the ad hoc vacuum (FE-45) and its radial residual, with numbers for three values of a_{40}.
- `um3_series.py` (6 checks): the linear operator of the first-order terms of FE-35 around the static block, its only static solution (the difference mode $\propto \mathrm{e}^{6H|\sigma|}$), the linear junction conditions and the radion mode, the radion condition of FE-59, the acceleration law with a rigid bulk medium, the amplitude of the response today.
- `um3_kohnsham.py` (9 checks): the Kohn-Sham equations FE-41 for a homogeneous gas of the bound particles χ, solved self-consistently with the author's matrices (reduced exactly by the rotations of ordinary space and of the extra times, and lifted back to sixteen components for the checks): the free level FE-40 (K1), first-order perturbation theory in the dilute limit (K2), the residuals of FE-41 and of the matching at the fold in three regimes of H, with the dimensionless strength $\lambda_\Psi n_\chi / V_*$ of the self-interaction (n_χ the number density of the particles on the fold) up to 1 in units of m_Ψ (K3), the exponents of the dilute limit (K4), the saturation of the proper density far from the fold and the attractive case (K5), the Higgs-portal loop through FE-49 (K6), the equation of state from FE-42 (K7), the level raised through the free threshold m_Ψ without unbinding (K8), and a weakly bound point that direct detection allows (K9).
- `um3_flavour.py` (4 checks): the survival probability of solar neutrinos by exact integration of FE-48 and FE-49, against the adiabatic formula of FE-50 and the measurements of Borexino; the three-flavour conversion in the envelope of a supernova for both mass orderings; the fast instability near a neutrinosphere and the dispersion relation FE-51 at its solutions; the equipartition FE-52 (its conservation laws, its fixed point, the bound $0 \leq \mathcal{F} \leq 1$) and its effect on the heating behind the shock.
- `um3_cosmology.py` (6 checks): the expansion rate of FE-59 and the ruler FE-56 computed independently and by the Boltzmann code CAMB; the derived quantities of the CMB against Planck 2018; the CMB spectra (FE-57) against the Planck 2018 high-l data, with a refit for thermal right-handed neutrinos; the BAO distances against DESI DR2 (the second data release of the Dark Energy Spectroscopic Instrument); the CNB numbers of FE-53 and FE-58.
- `um3_thermal.py` (6 checks): g_ρ and g_{s} of FE-60, the relic density FE-61 by numerical solution, the current portals against the limits of direct detection, the Higgs-portal coupling of FE-47 and the binding of χ, the right-handed neutrinos and N_{eff} (FE-62), the ad hoc vacuum before nucleosynthesis.